

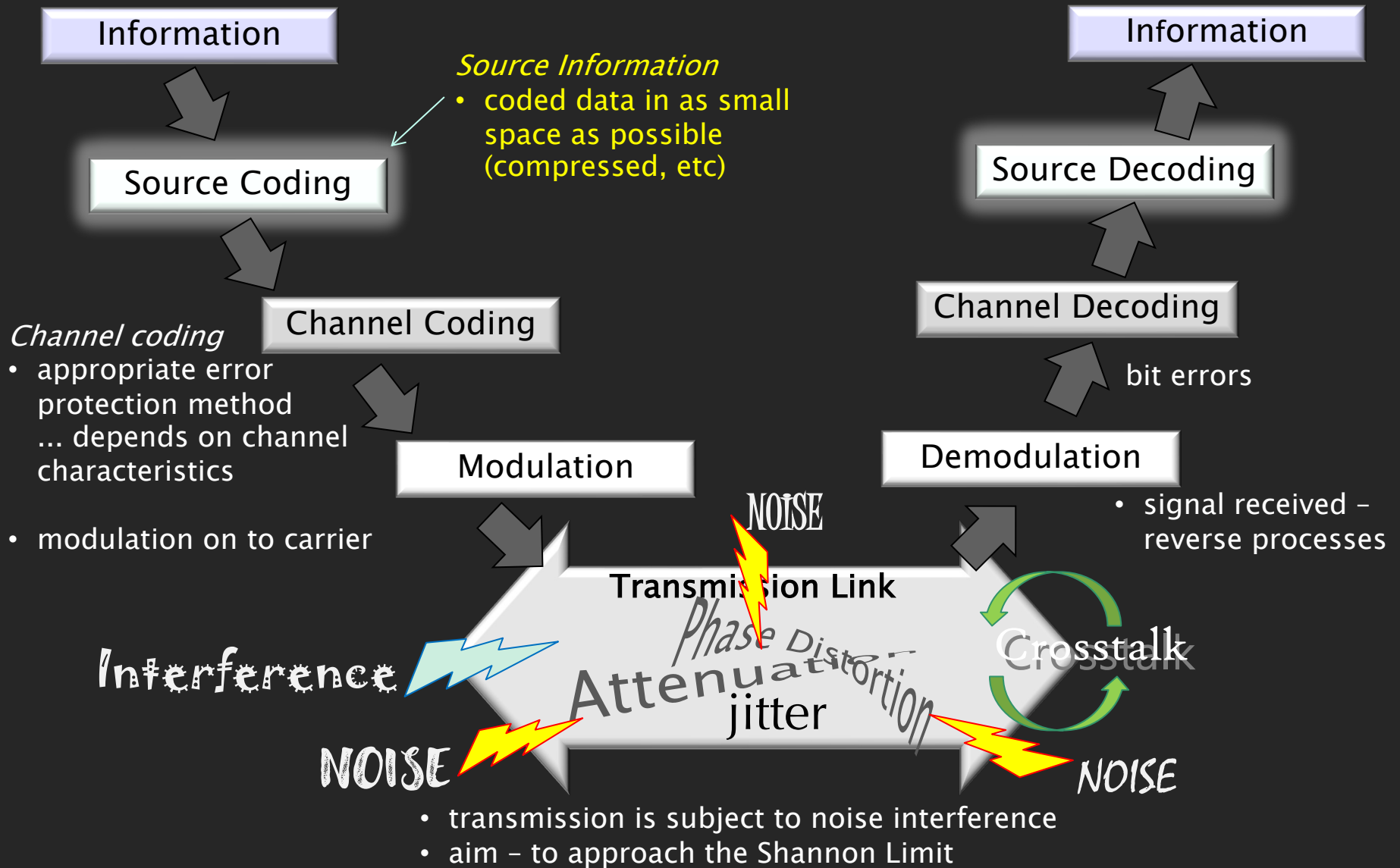
Broadcast Technology UG1

Modulation, Multiplexing

The Playout System

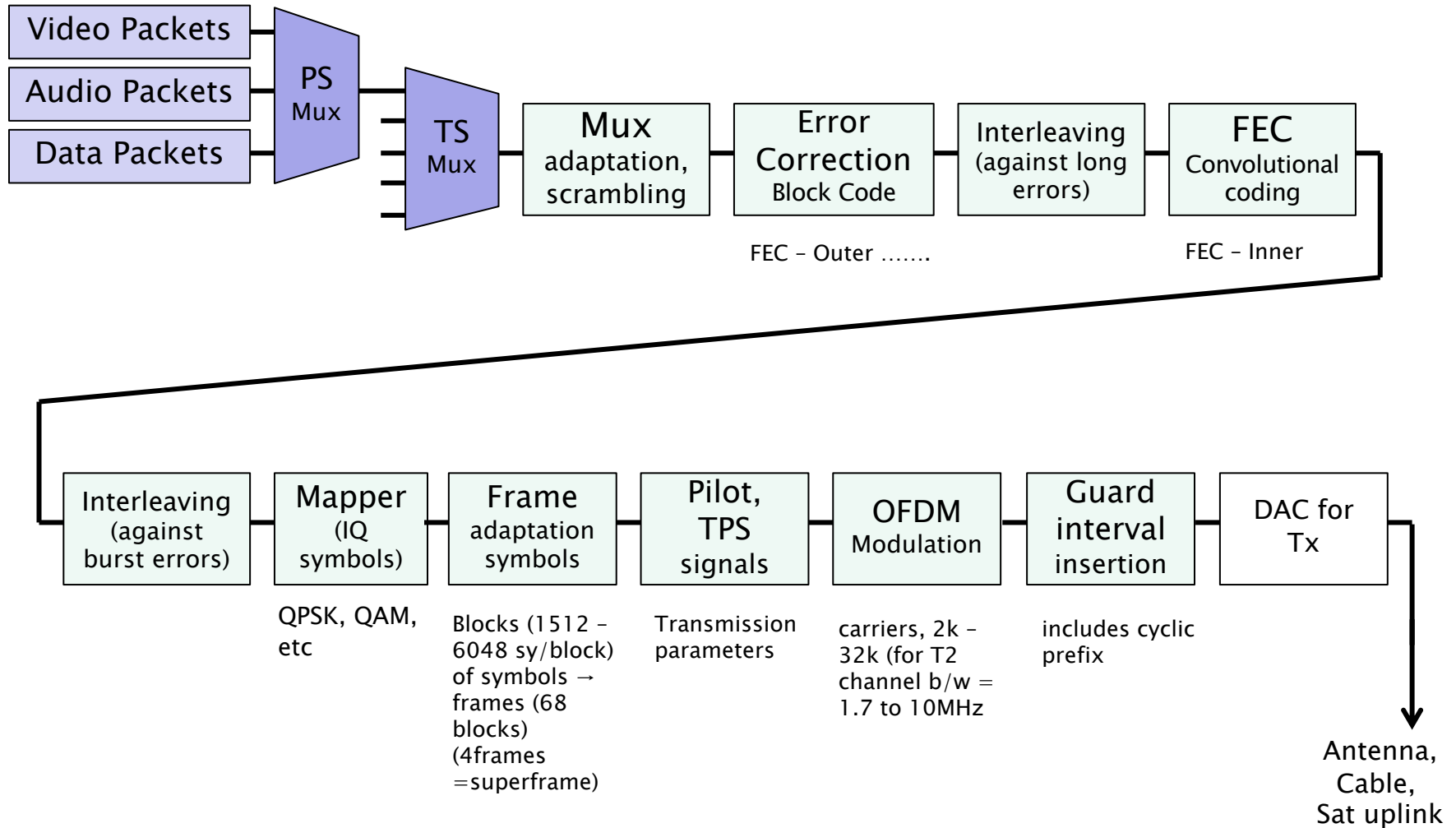
(a quick revision from the previous session)

Information Transmission, Signal Encoding



DVB Transmission Payout System

(Based on DVB-T, but -C, -S and -H are similar)



Inter-Symbol Interference for Modulated Data

Causes

In radio:

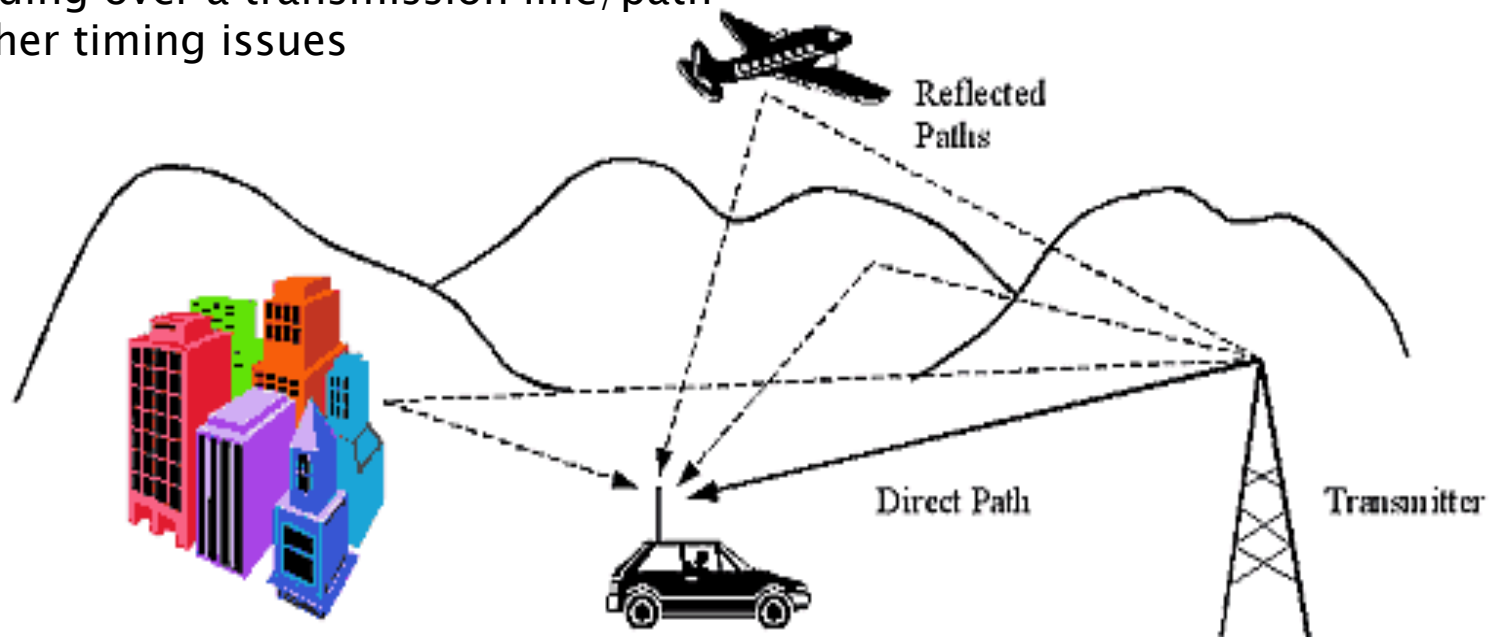
- multipath interference, including atmospherics

On cables:

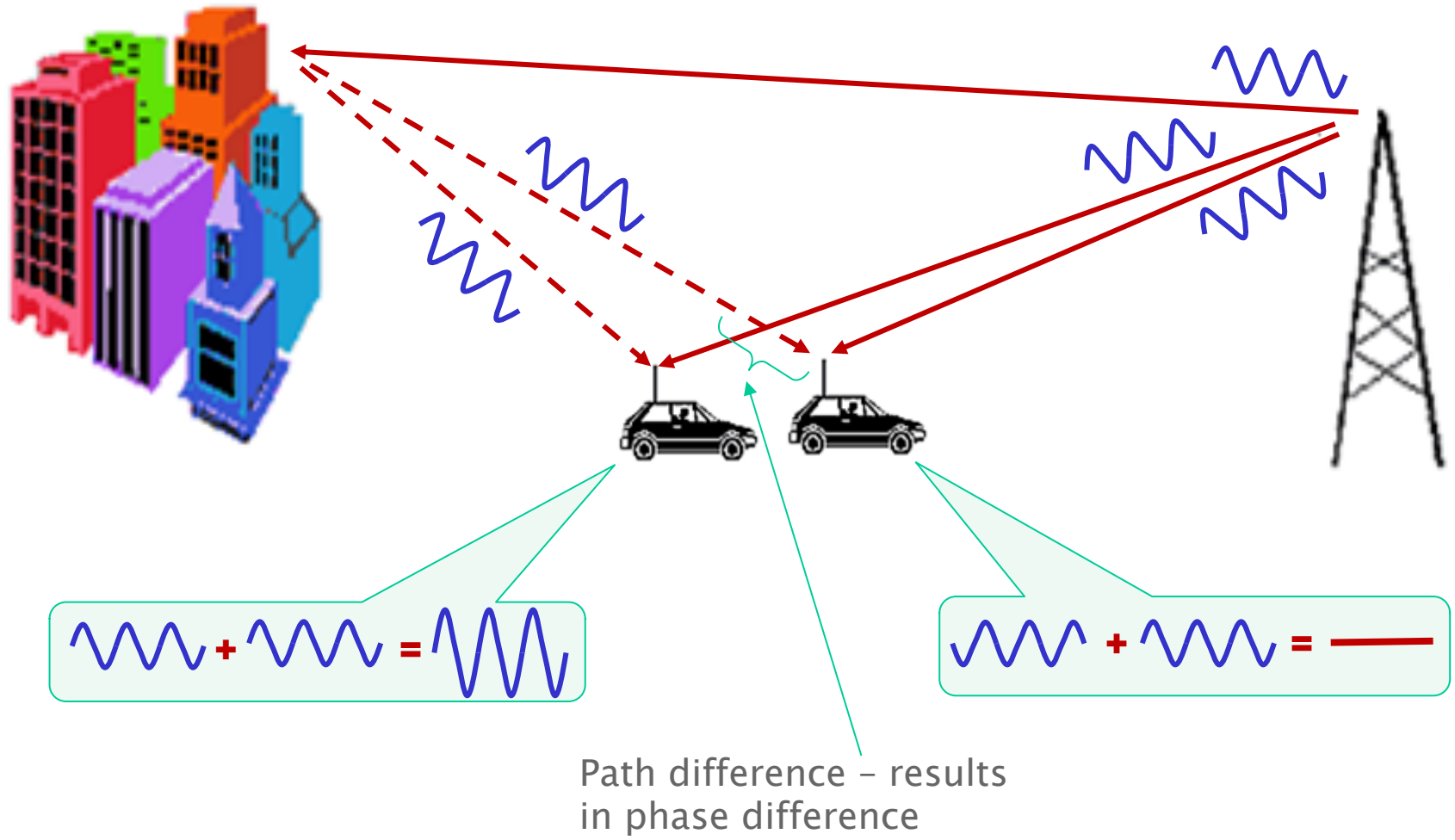
- transmission line propagation, phase delays, electrical characteristics

Implications on radio and cable:

- channel gain variations
- signal spreading over a transmission line/path
- jitter and other timing issues

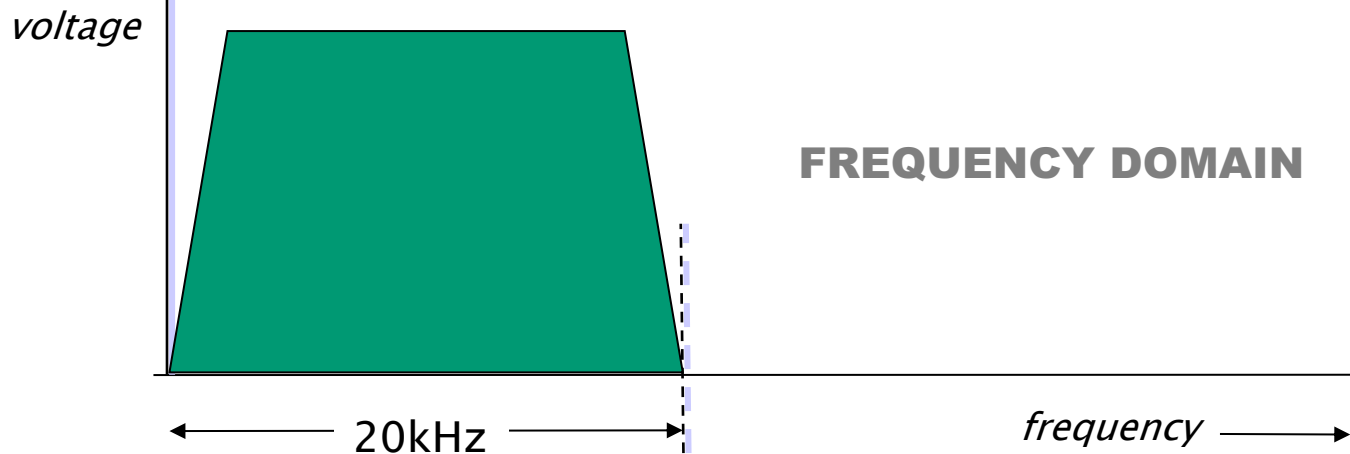
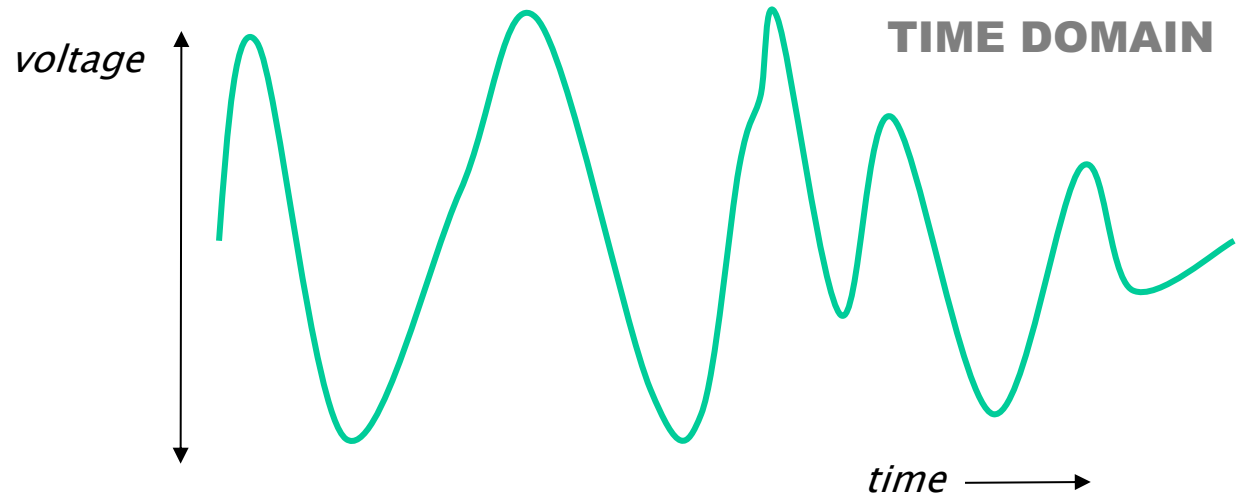


Multipath Interference in Radio

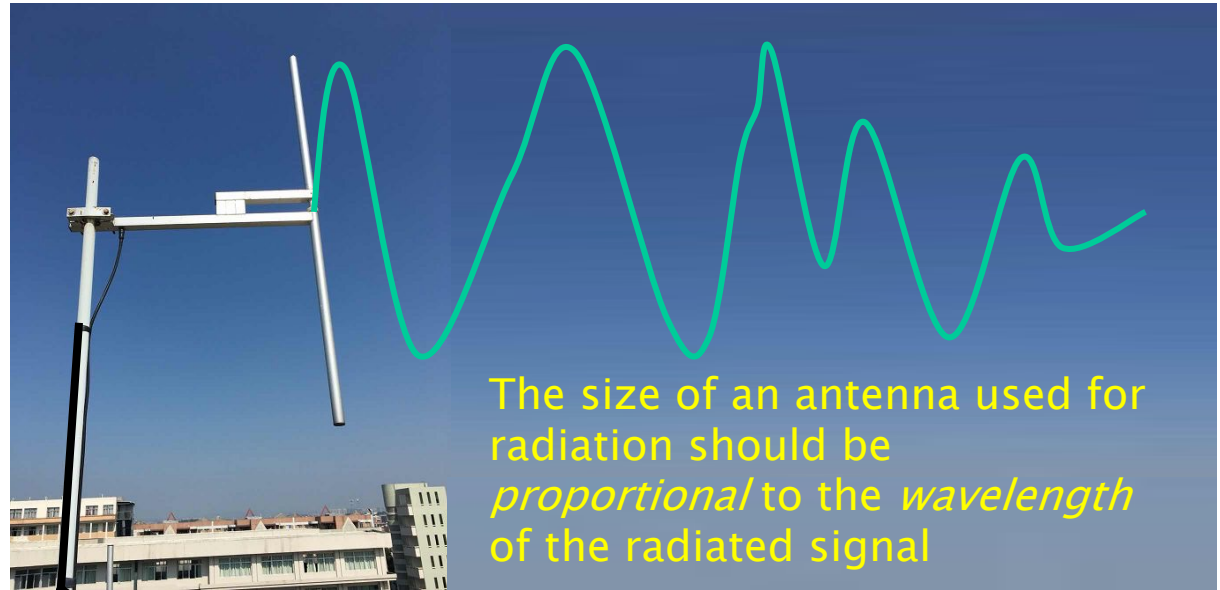


Modulation, AM, FM, Phase, (QAM ... later on ...)

Signals – *baseband*

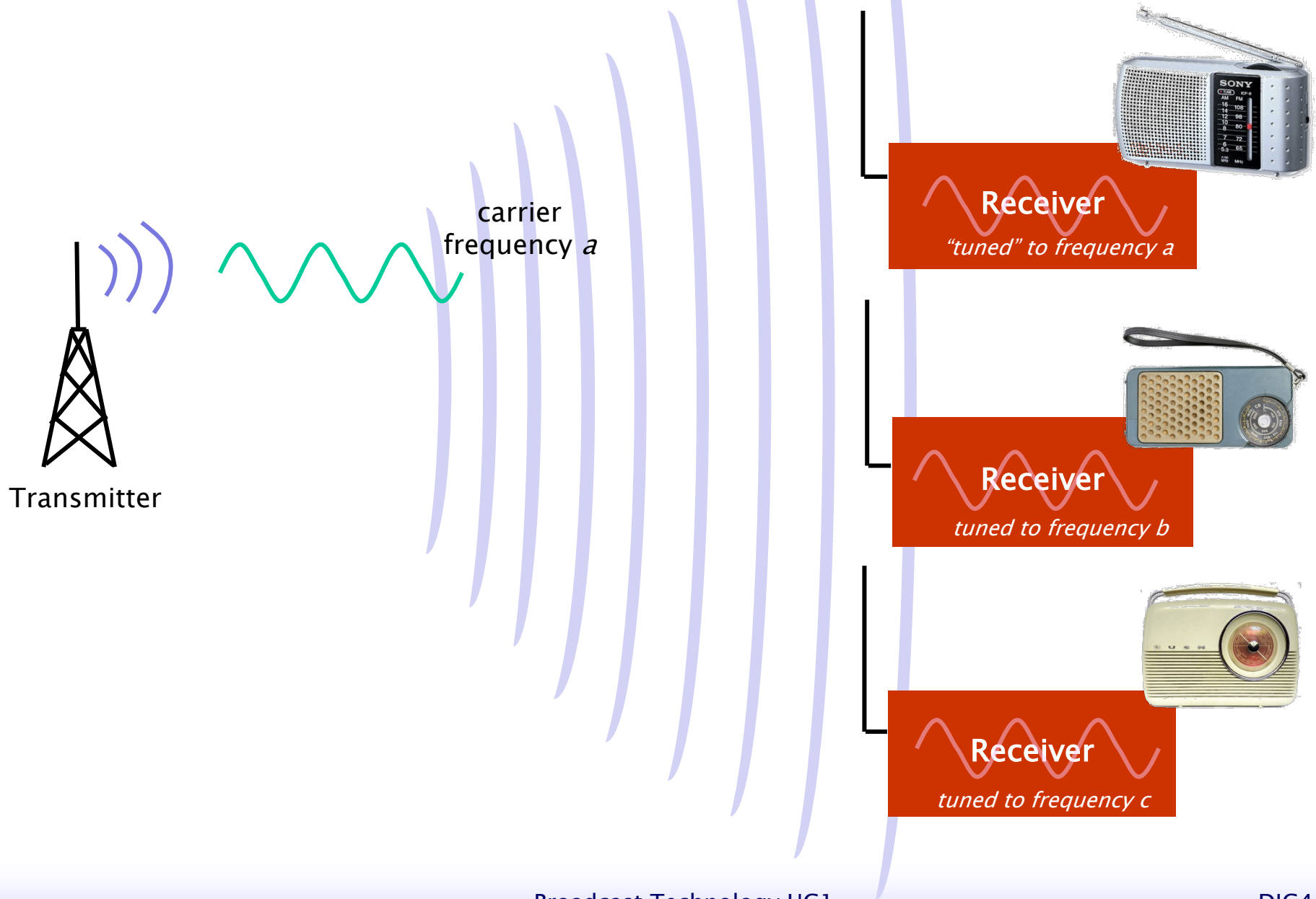


Transmission Characteristics; Modulation and Carriers

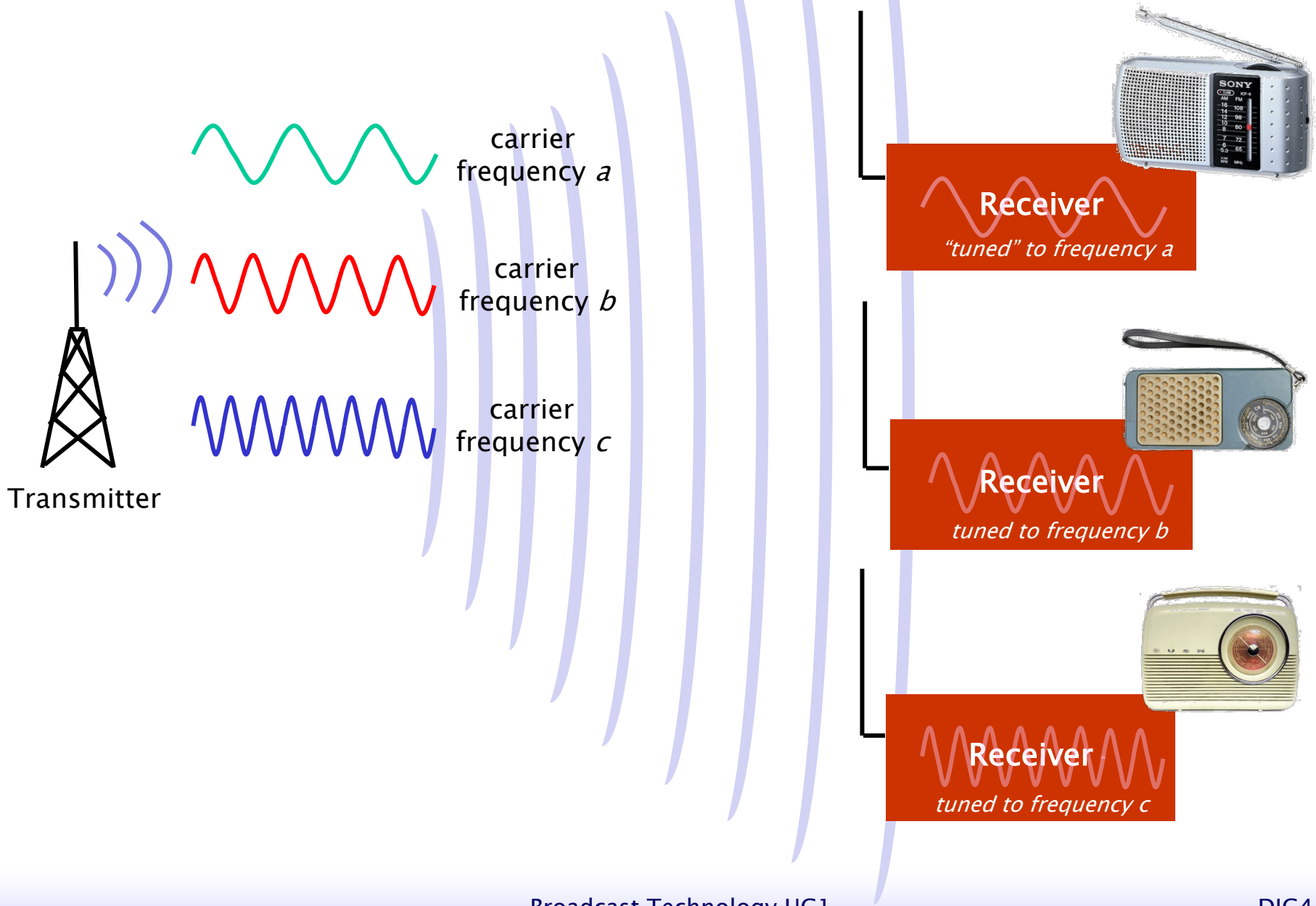


- λ for a very low frequency [baseband] signal will be large
... so the size of the antenna won't be practical
- Instead, the signal can be shifted to a higher frequency
... using a high frequency carrier
- Frequency, amplitude and/or phase of the carrier will be modulated by the message signal ... then radiated through a smaller antenna

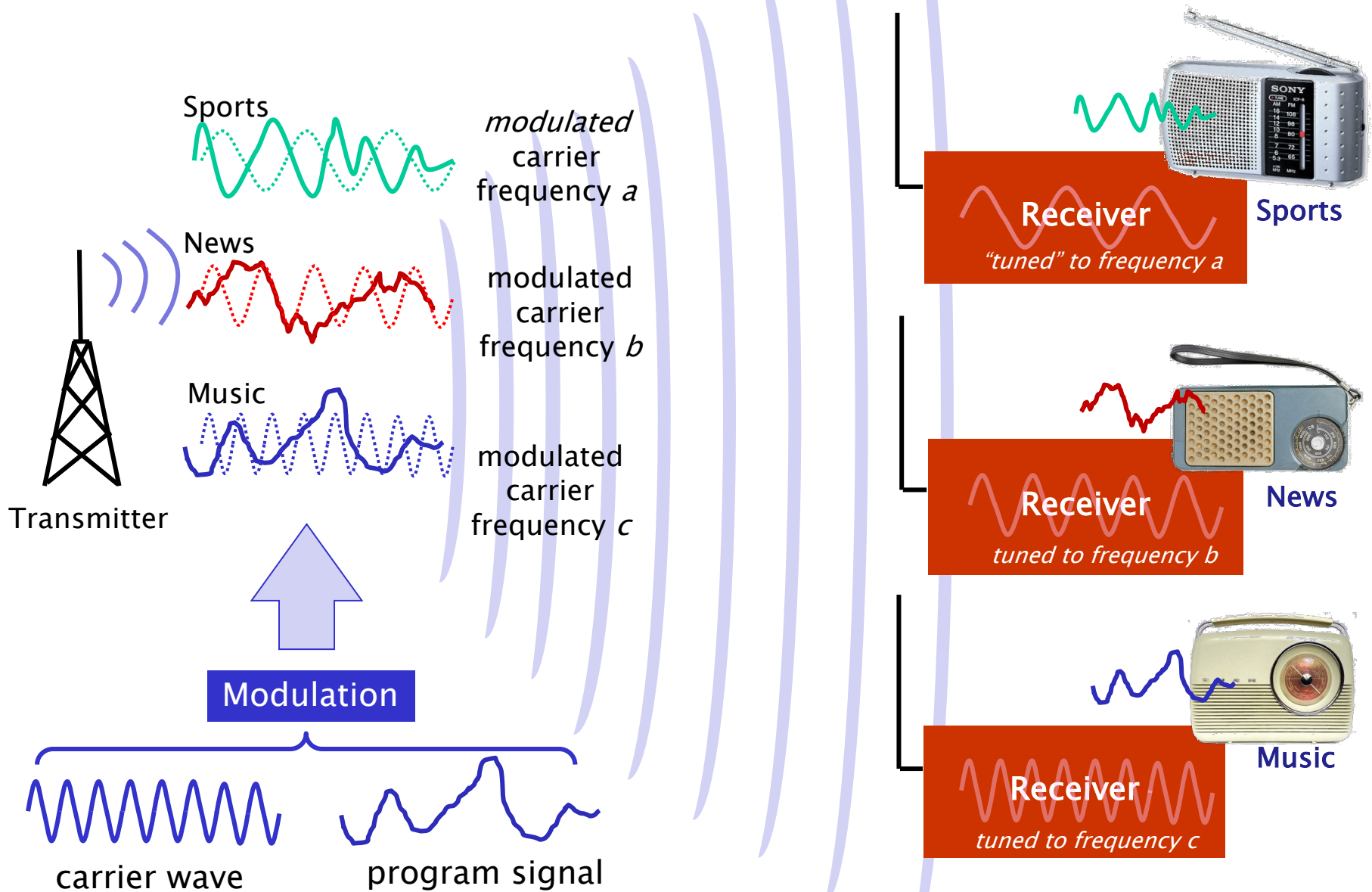
Carrier Waves



Carrier Waves

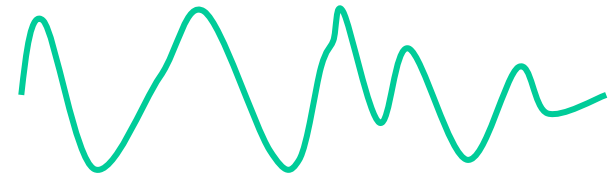


Modulation

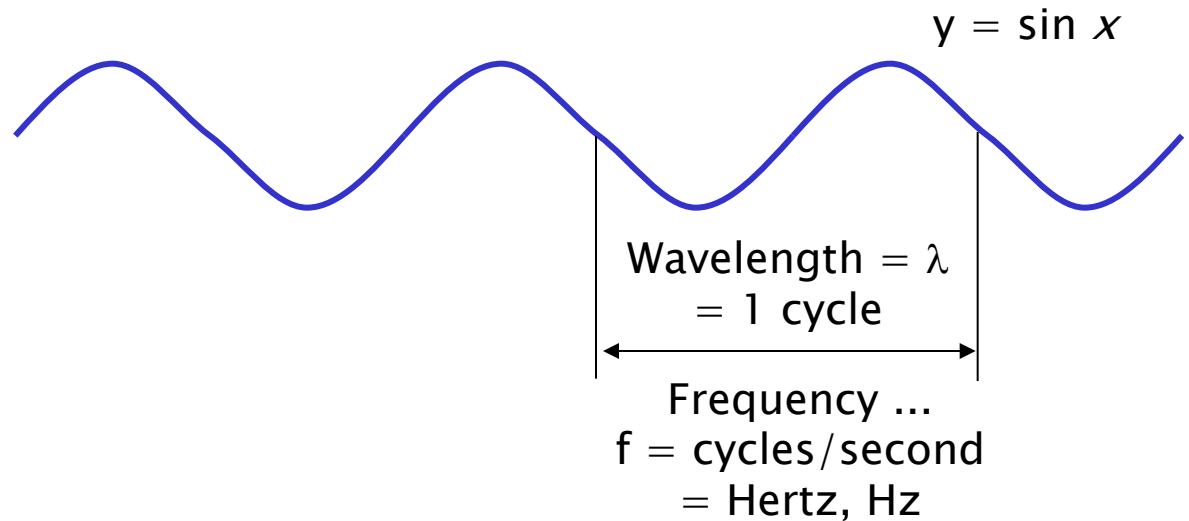


Modulation

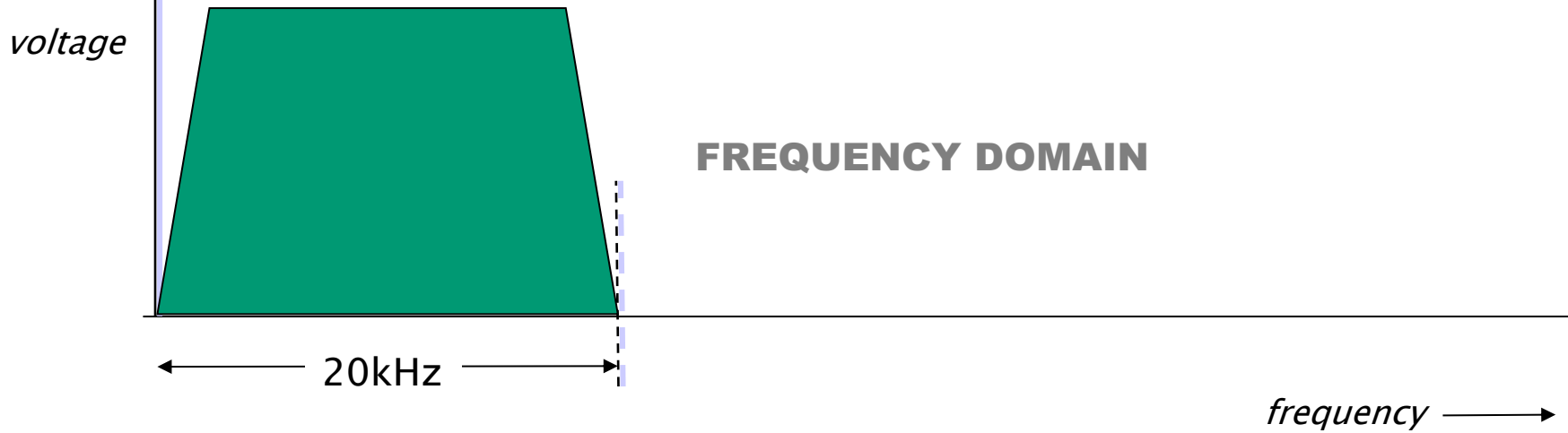
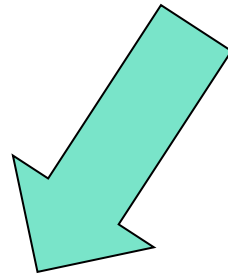
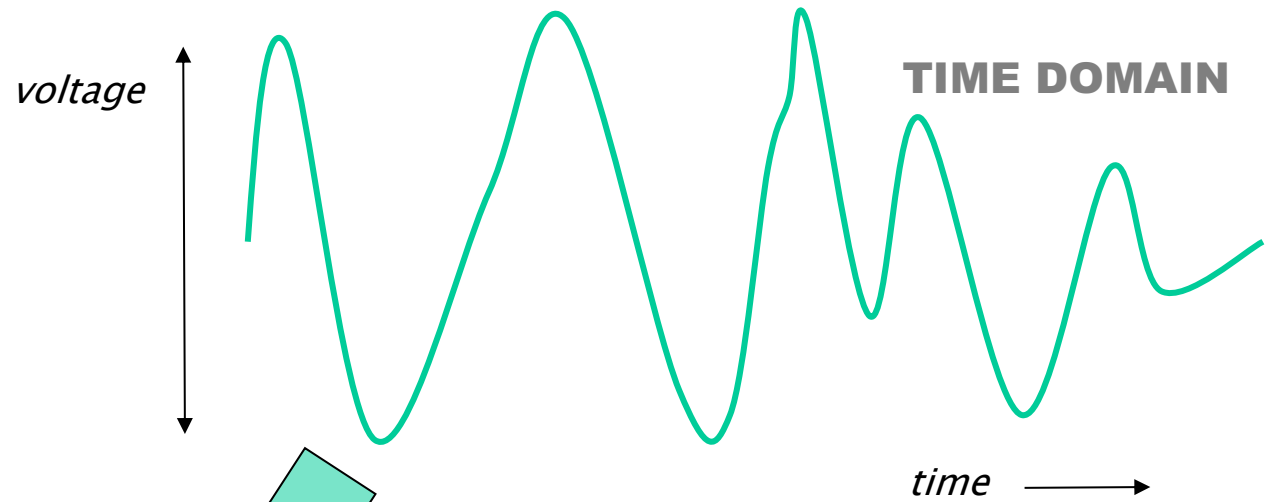
- To convey a *Baseband* signal on a *Carrier* signal
- Necessary to allow a multitude of baseband signals to simultaneously exist



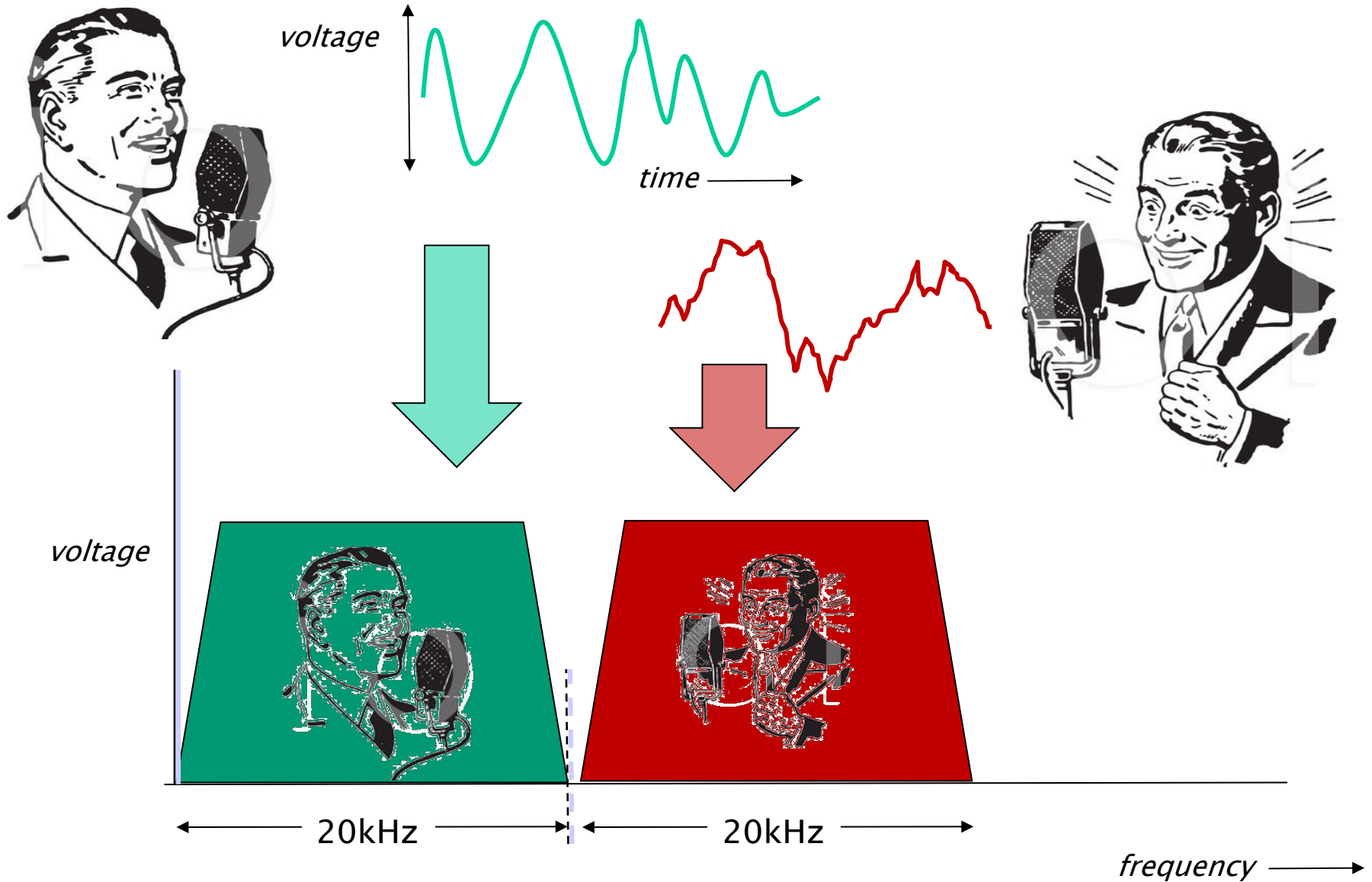
- Carrier – sinusoidal



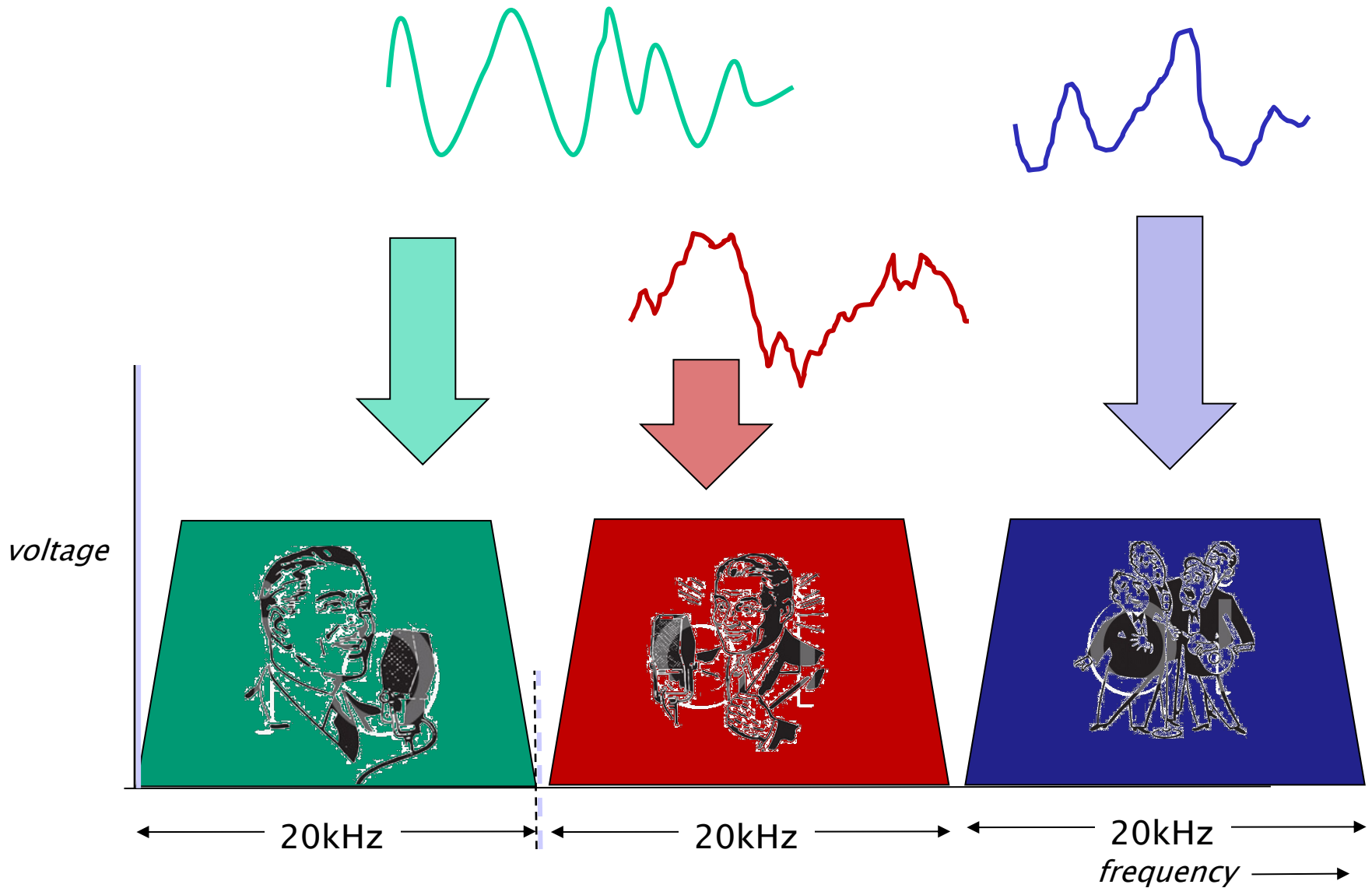
Signals



Signals – transmitting more than one signal on a channel – multiplexing

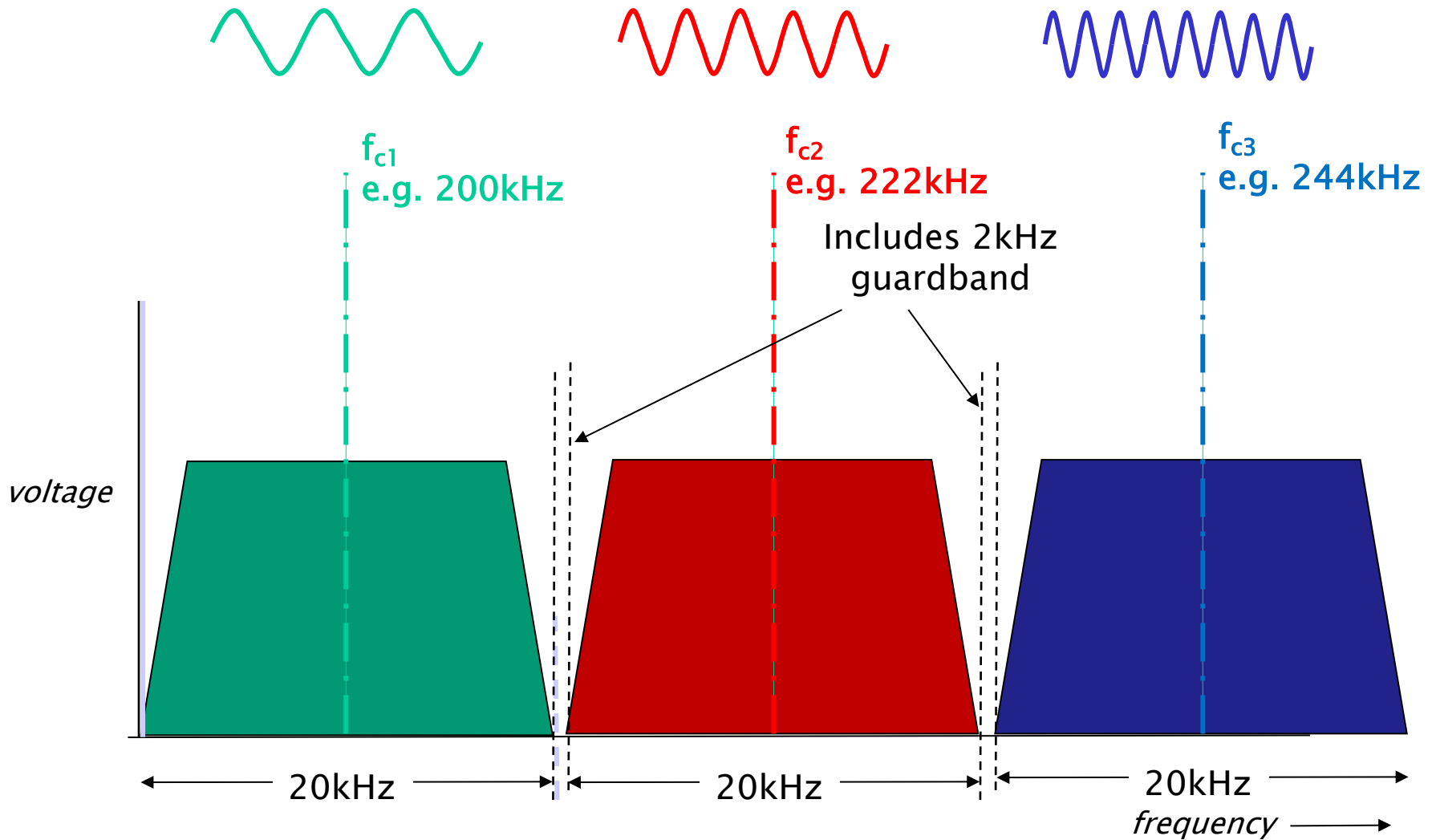


Signals – transmitting more than one signal on a channel – multiplexing

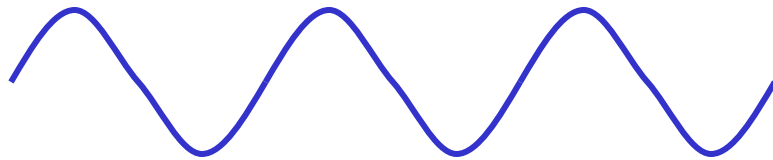


Signals – multiplexing

Each channel needs a separate carrier



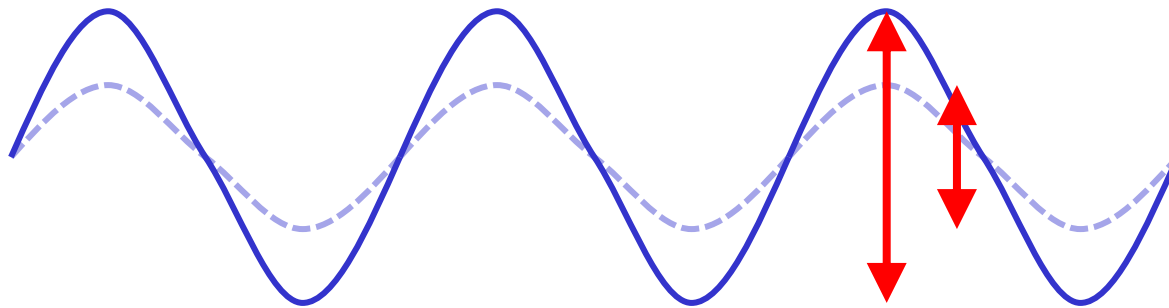
Modulation



$$y = \sin x$$

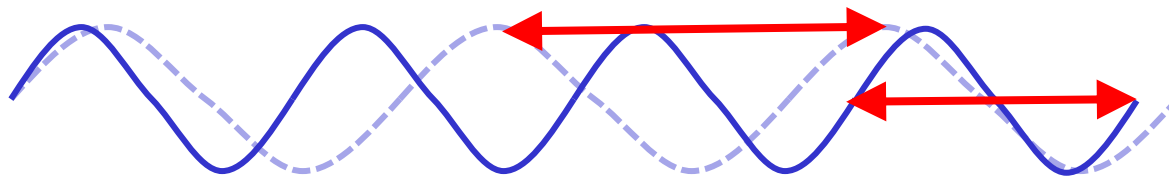
- Carrier – sinusoidal

- Carrier parameters can change to convey the baseband signal:



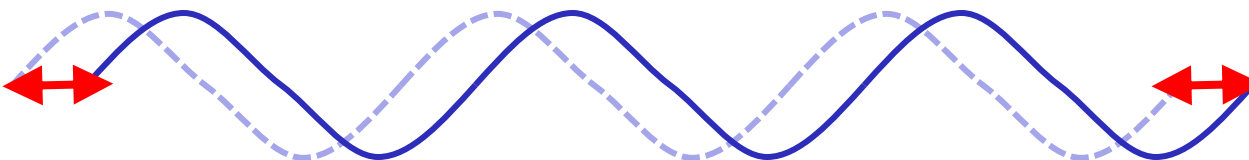
Amplitude

- ... Amplitude Modulation – AM



Frequency

- ... Frequency Modulation – FM



Phase

- ... Phase Modulation (PM and many other types)

What Is The Determining Factor For A Carrier Frequency?

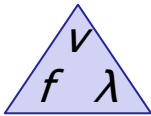
Basics

- The size of an antenna used for radiation should be proportional to the wavelength of the radiated signal

Wavelength λ = speed of wave in free space $\times$ time cycle = $c \times \tau$

$$d = \frac{d}{t} \cdot t$$

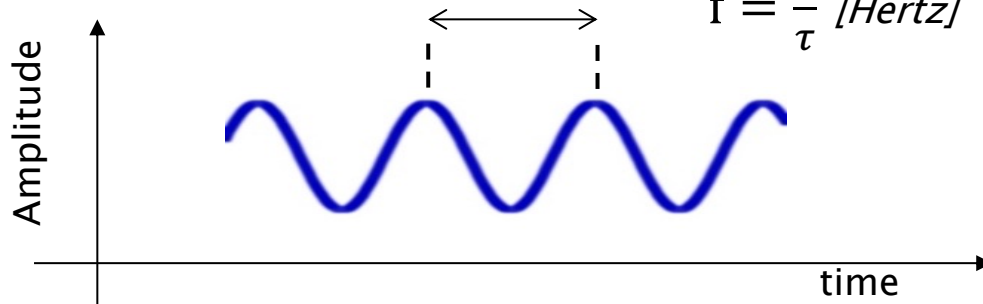
$$f = \frac{1}{\tau} \quad \dots \quad \lambda = \frac{\text{speed of wave in free space}}{\text{frequency of the signal}} = \frac{c}{f}$$



(c is essentially speed of light = $3 \times 10^8 \text{ ms}^{-1}$)

One cycle = τ [seconds]

$$f = \frac{1}{\tau} \text{ [Hertz]}$$



What Is The Determining Factor For A Carrier Frequency?

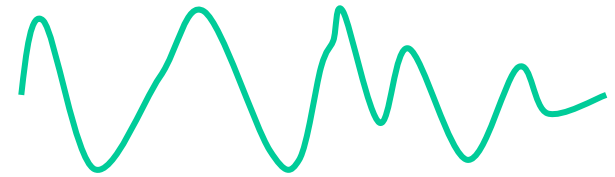
- λ for a very low frequency signal will be large
... so the size of the antenna won't be practical
- Instead, the signal can be shifted to a higher frequency
... using a high frequency carrier
- Frequency, amplitude and/or phase of the carrier will be modulated by the message signal
... then radiated through a smaller antenna

Receiver

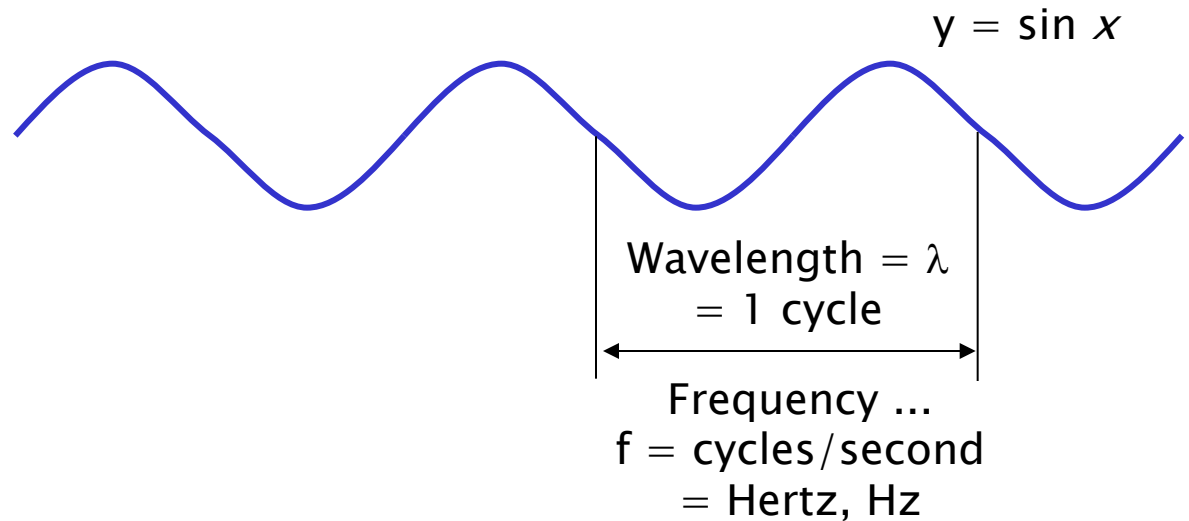
- Received signal needs to be demodulated to achieve the original message signal

Modulation

- To convey a *Baseband* signal on a *Carrier* signal
- Necessary to allow a multitude of baseband signals to simultaneously exist

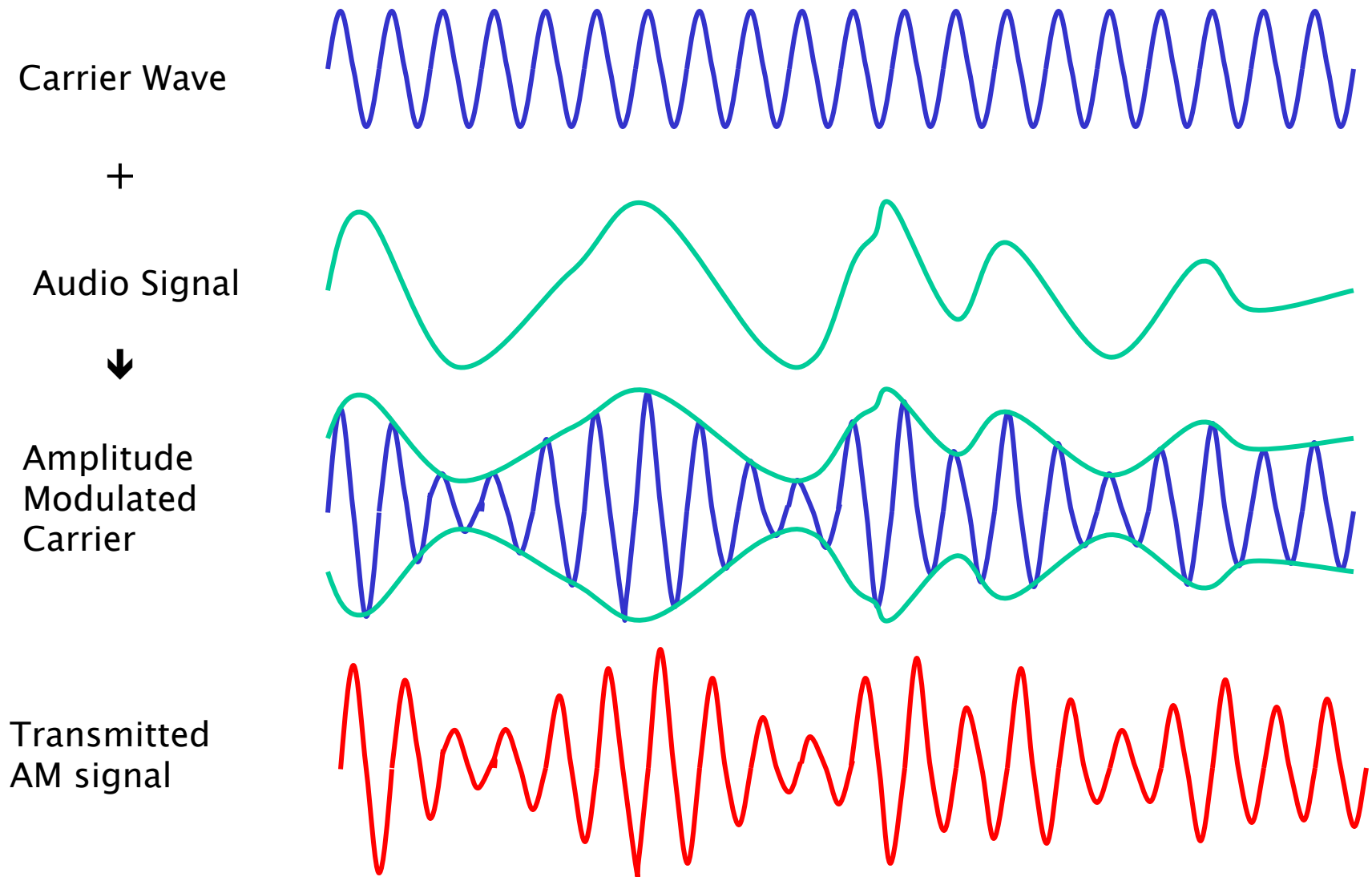


- Carrier – sinusoidal

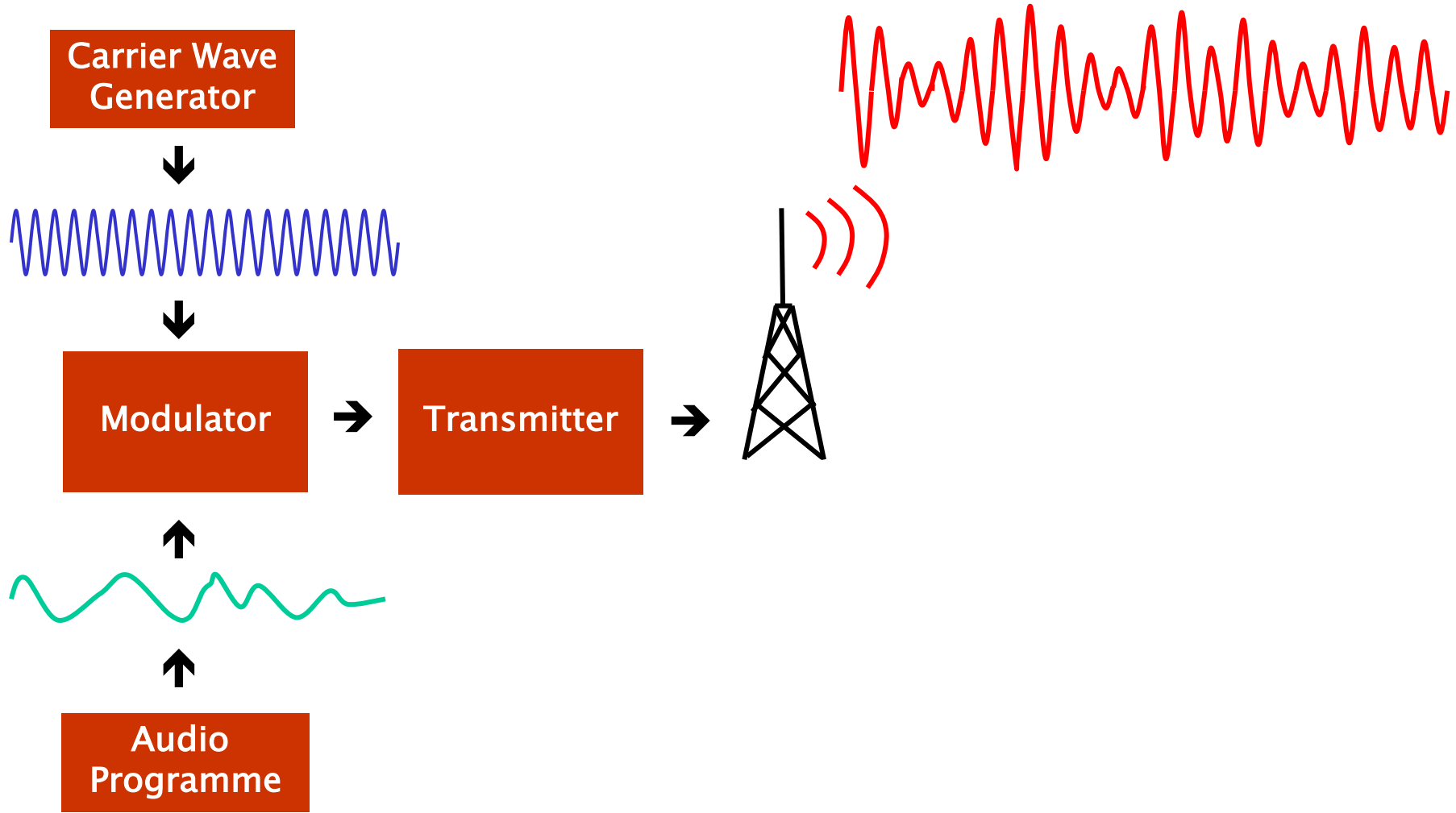


Analogue Modulation – AM and FM

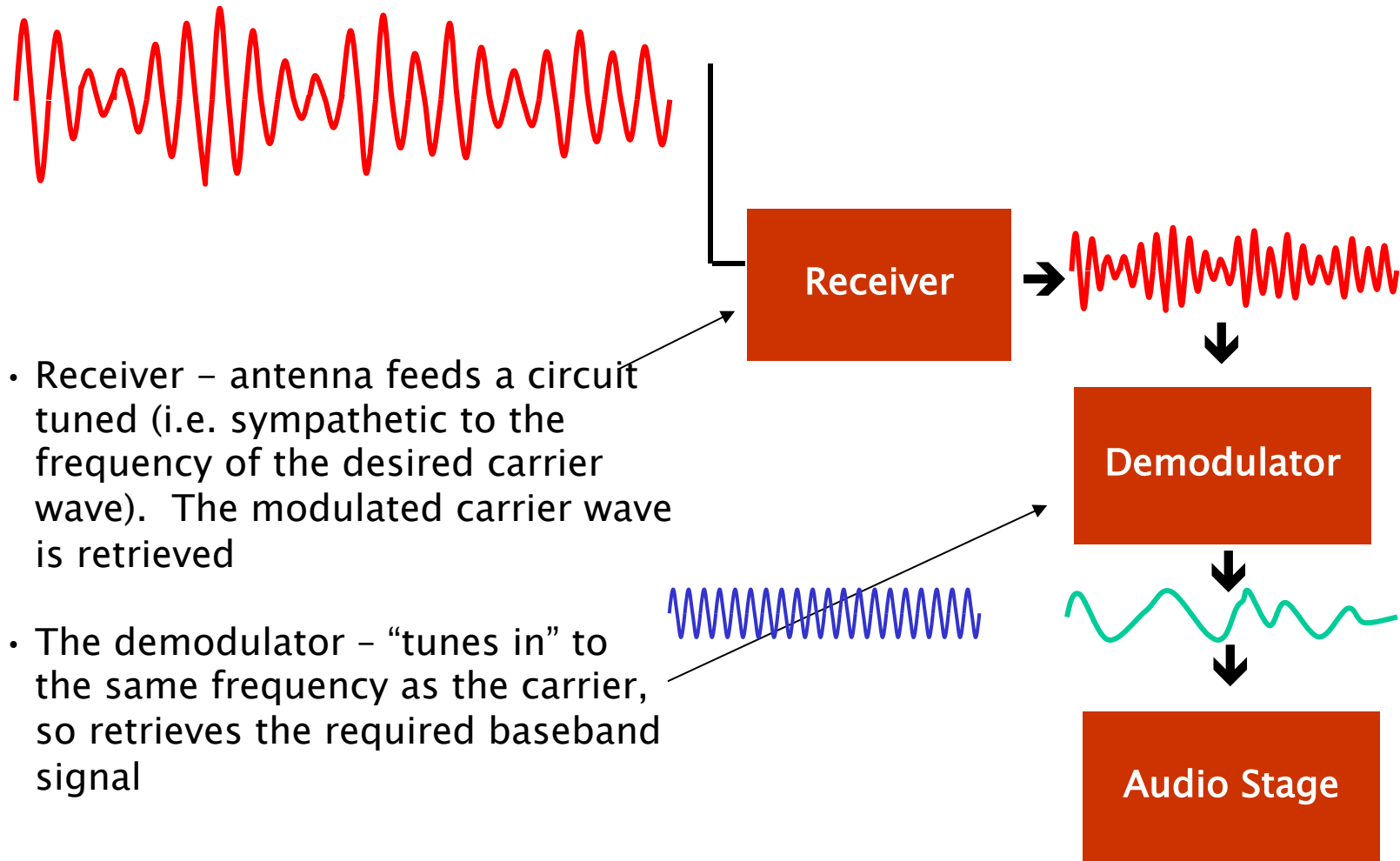
Amplitude Modulation – AM



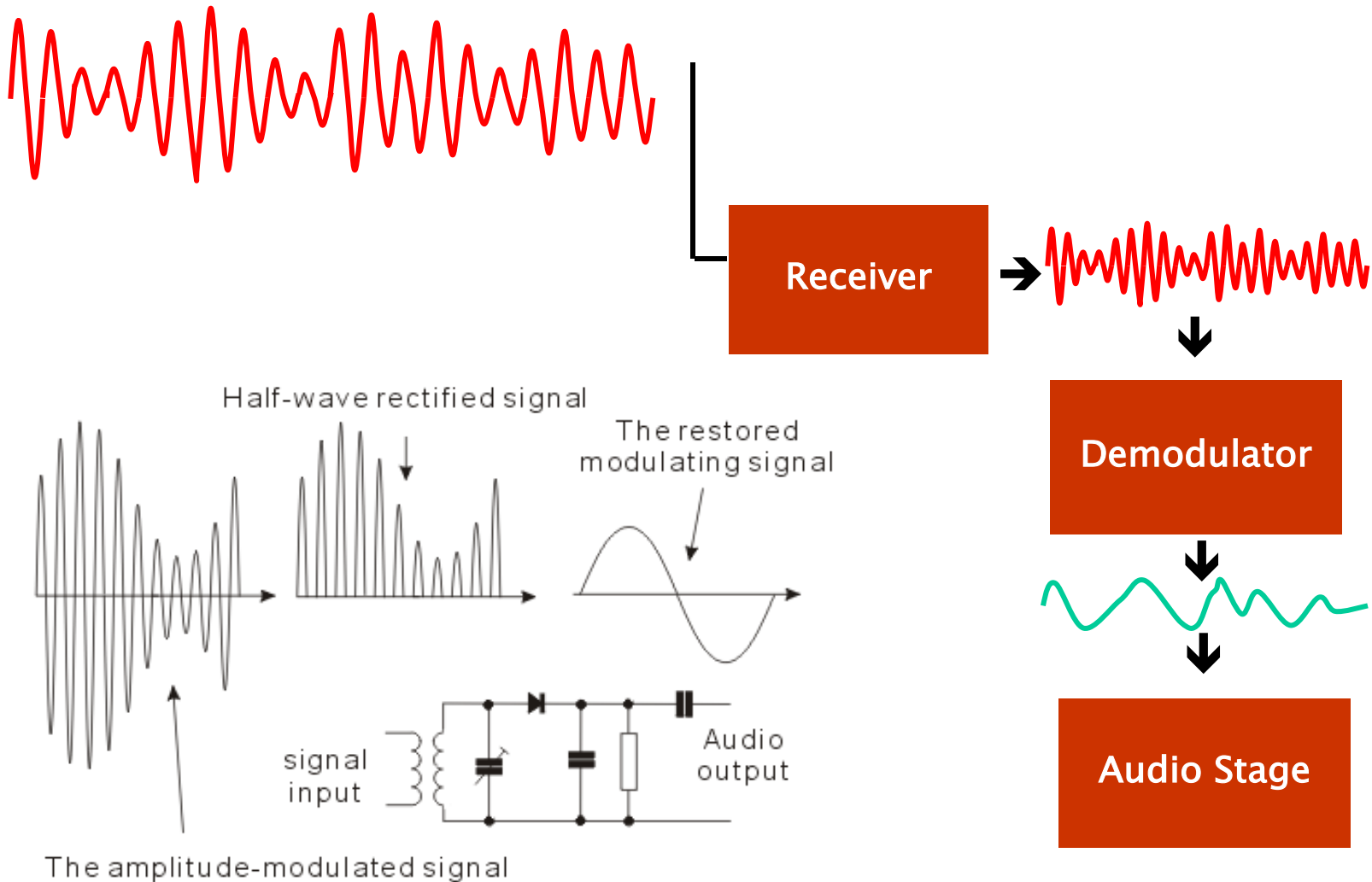
Amplitude Modulation



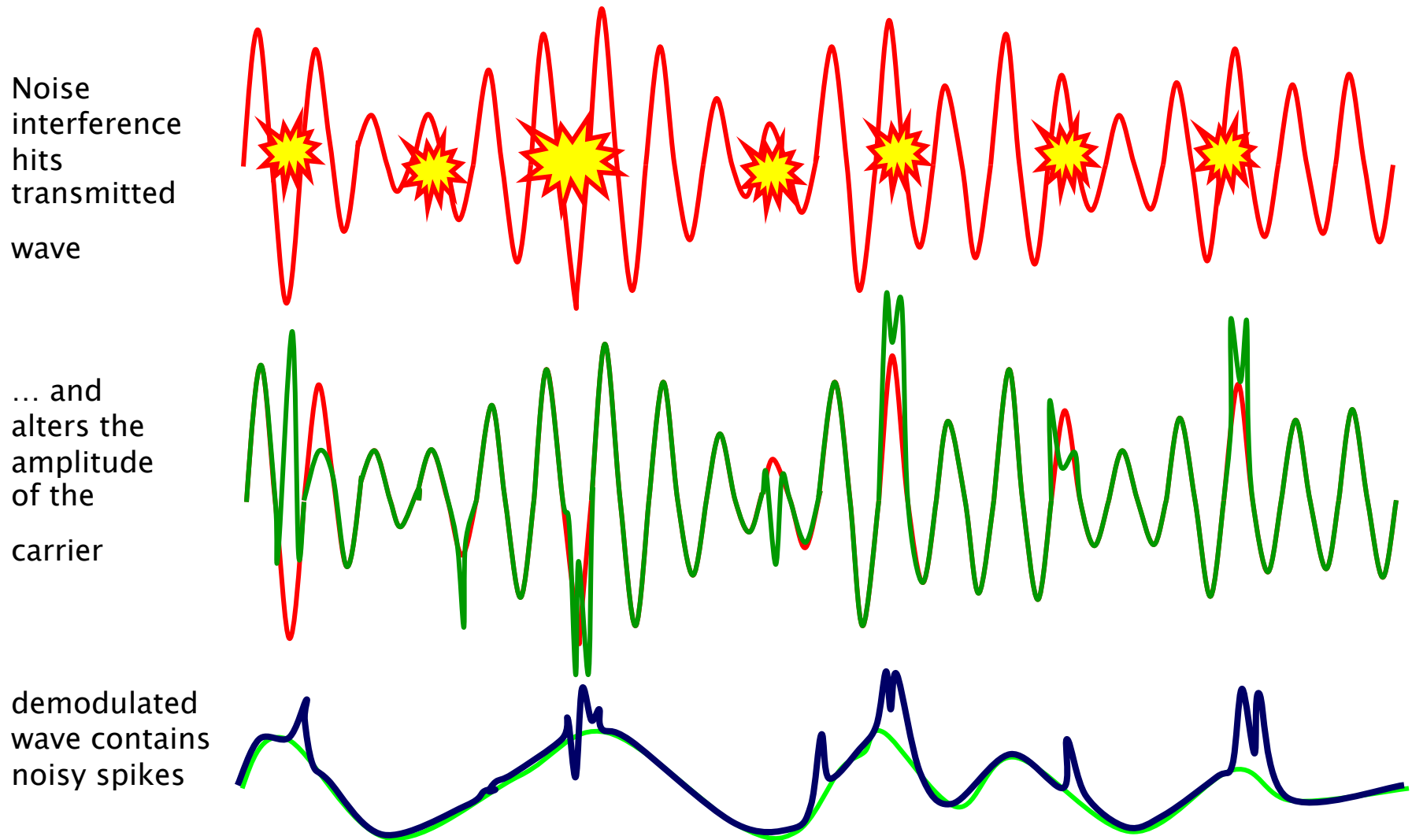
Amplitude Demodulation



Amplitude Demodulation

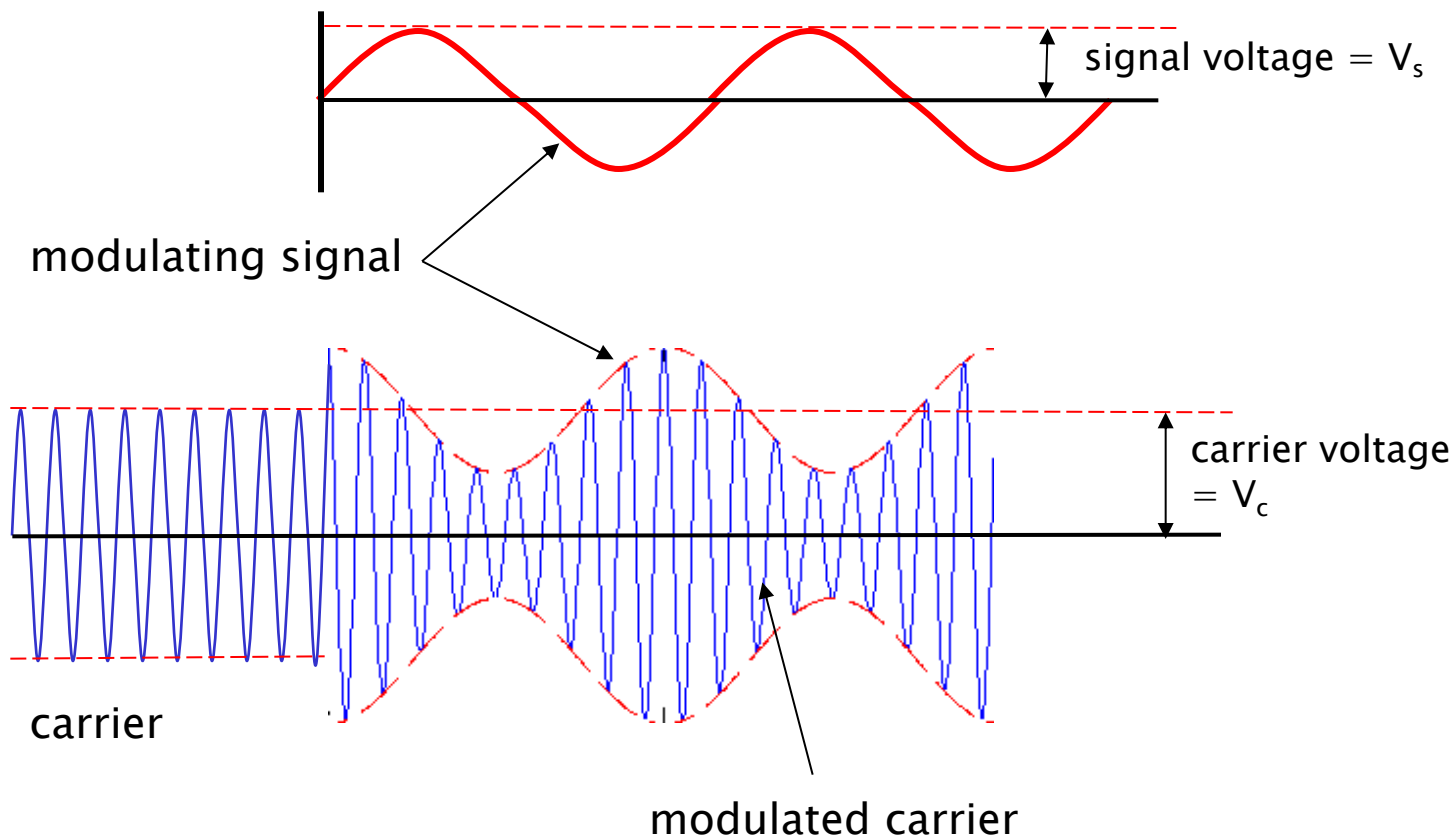


AM Prone to Noise Interference



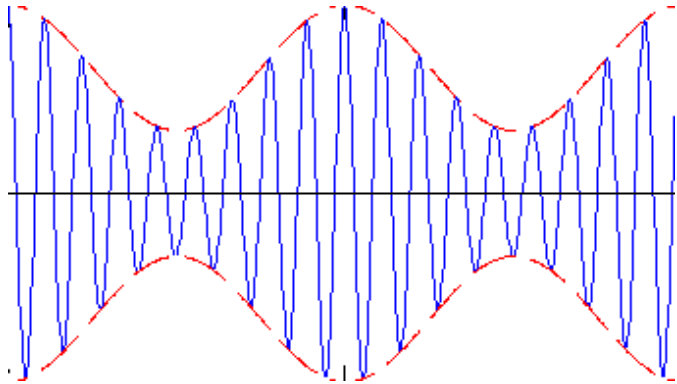
Modulation Index

The ratio of the depth of modulation



$$\text{Modulation Index} = m = \frac{V_s}{V_c}$$

Modulation Index



Here, modulating signal V_s is half the magnitude of carrier voltage V_c

i.e. Carrier Voltage = V_c
... and $V_s = 0.5 \times V_c$

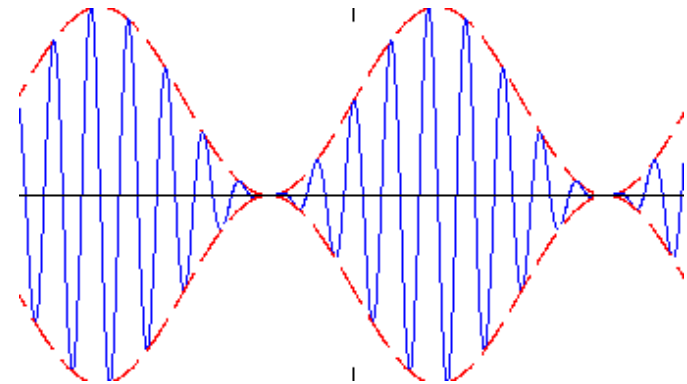
$$\begin{aligned}\therefore m &= V_s / V_c \\ &= 0.5 \nu / \nu \\ &= 0.5 \text{ (or 50\%)}\end{aligned}$$

Here, the modulating signal V_s is the same magnitude as the carrier V_c ;

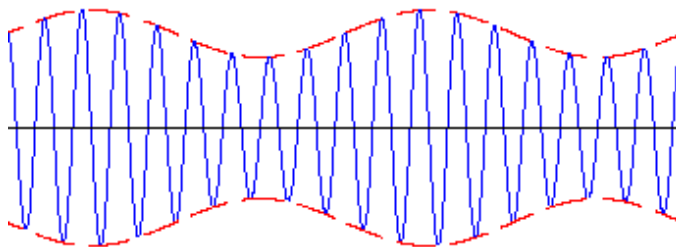
i.e. $V_s = V_c$

$$\therefore m = V_s / V_c = 1 \text{ (or 100\%)}$$

note – the modulated wave reduces to zero



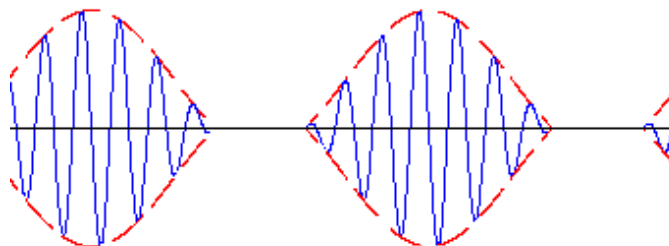
Modulation Index



Here the modulation index is 0.25

- giving a fairly shallow modulated carrier wave
- signal strength is low compared to noise (S/N ratio)

Here the modulation index is 1.5
– giving an over modulated carrier



Amplitude Modulation

Carrier Signal = $v_c = V_c \sin \omega_c t$;
 $\omega_c = 2\pi f_c$ where $f_c = \text{carrier freq}$

Modulating Signal = $v_s = V_s \sin \omega_s t$;
 $\omega_s = 2\pi f_s$; where $f_s = \text{modulation freq}$

So the voltage of the carrier varies between
 $(V_c + V_m)$ *and* $(V_c - V_m)$

Modulation Index = $m = V_s / V_c$; so $V_s = m V_c$

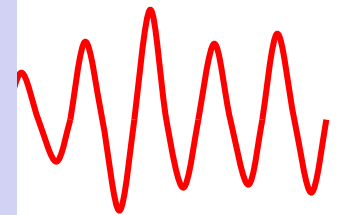
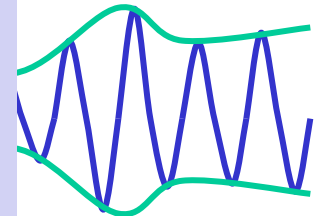
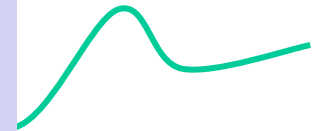
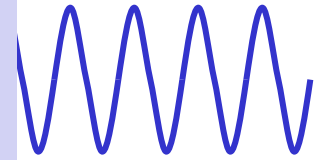
$$\begin{aligned} v_c &= (V_c + V_s \sin \omega_s t) \sin \omega_c t \\ &= V_c \sin \omega_c t + V_s \sin \omega_s t \cdot \sin \omega_c t \end{aligned}$$

since

$$\sin \omega_s t \cdot \sin \omega_c t = \frac{1}{2} [\cos(\omega_c - \omega_s)t - \cos(\omega_c + \omega_s)t]$$

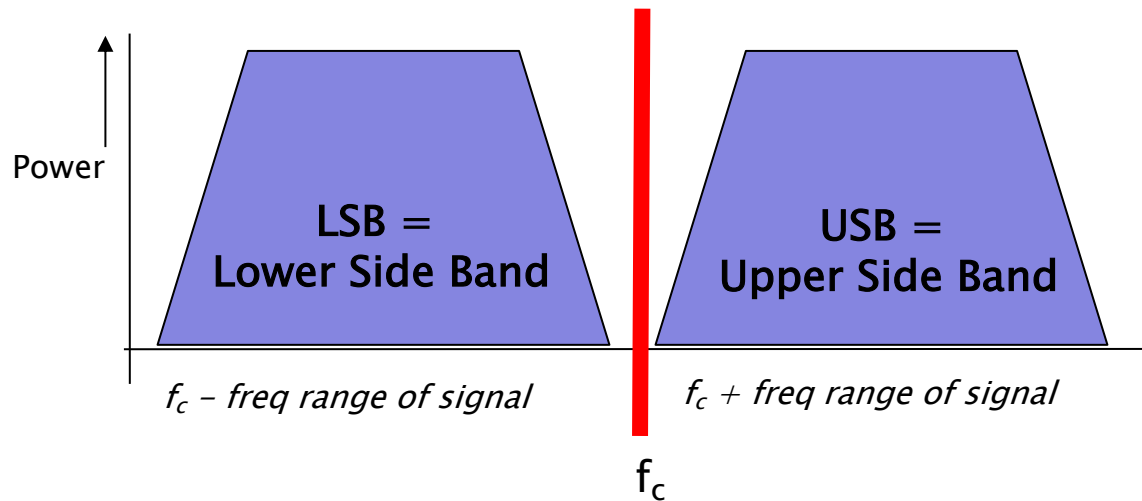
then

$$v_c = (V_c \sin \omega_c t) + \frac{1}{2} m V_c \cos(\omega_c - \omega_s)t - \frac{1}{2} m V_c \cos(\omega_c + \omega_s)t$$



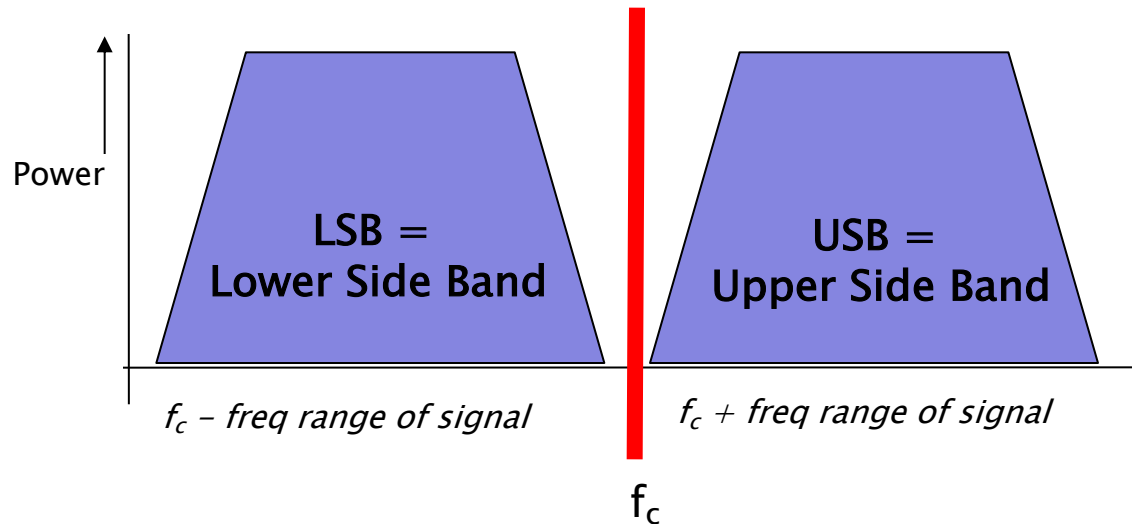
Amplitude Modulation

$$v_c = \underbrace{(V_c \sin \omega_c t)}_{\text{Carrier}} + \underbrace{\frac{1}{2} m V_c \cos(\omega_c - \omega_s) t}_{\text{USB}} - \underbrace{\frac{1}{2} m V_c \cos(\omega_c + \omega_s) t}_{\text{LSB}}$$



AM Frequency Spectrum – Sidebands

AM process produces a spectrum containing the carrier and two sidebands of frequencies related to the modulating signal
e.g.:



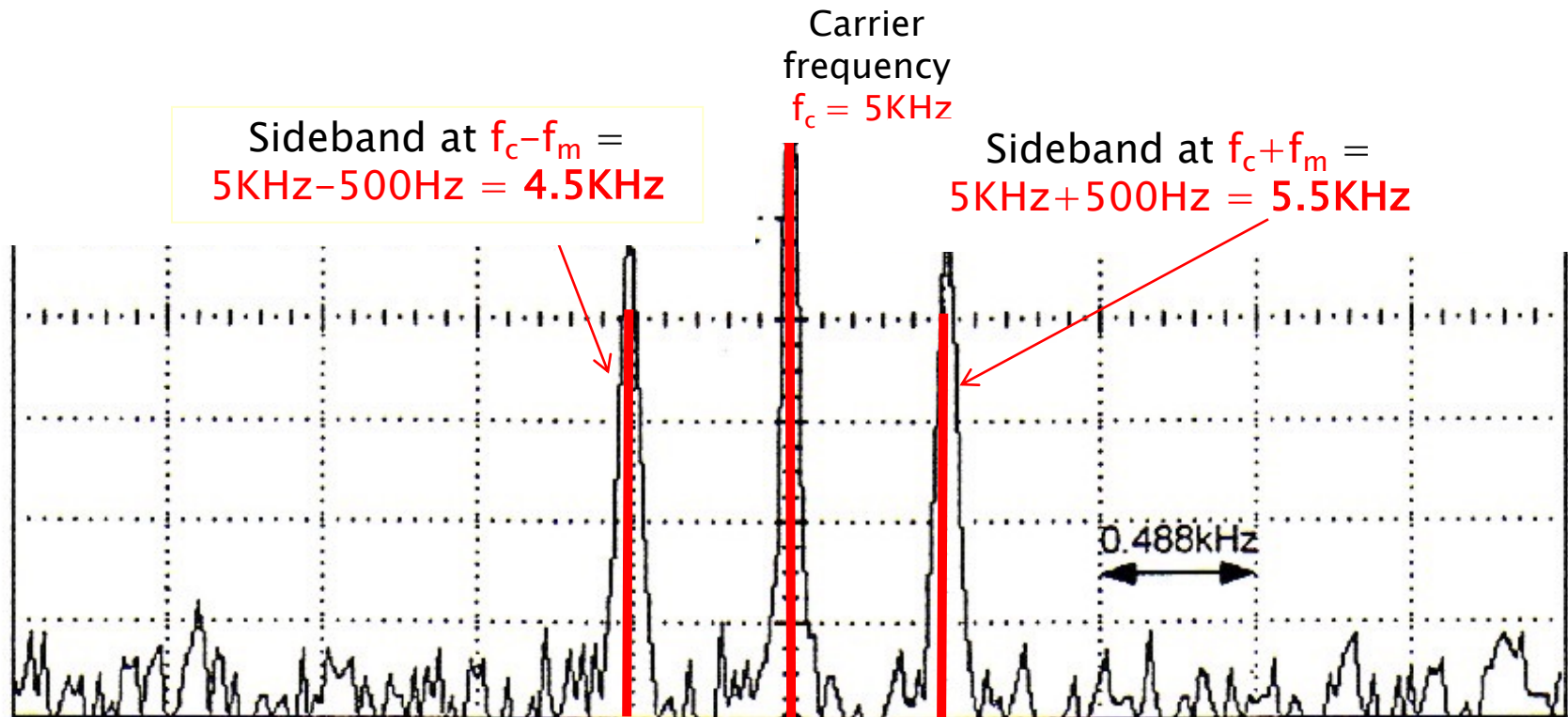
e.g. modulating signal is between 100Hz and 8KHz,
and carrier freq = 600KHz (=600,000Hz)

Then – USB extends *up* from **$[600,000 + 100]\text{Hz}$** to **$[600,000 + 8,000]\text{Hz}$**
= **600.1KHz up to 608KHz**

LSB extends *down* from **$[600,000 - 100]\text{Hz}$** to **$[600,000 - 8,000]\text{Hz}$**
= **599.9KHz down to 592KHz**

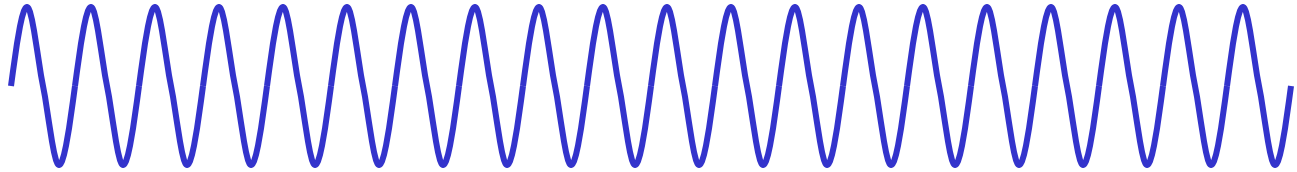
AM Frequency Spectrum – Sidebands

Signal frequency, $f_m = 500\text{Hz}$

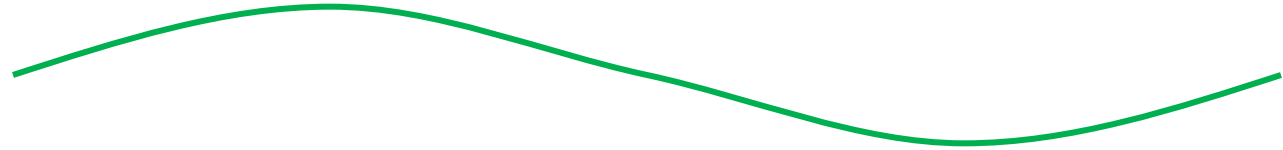


Frequency Modulation

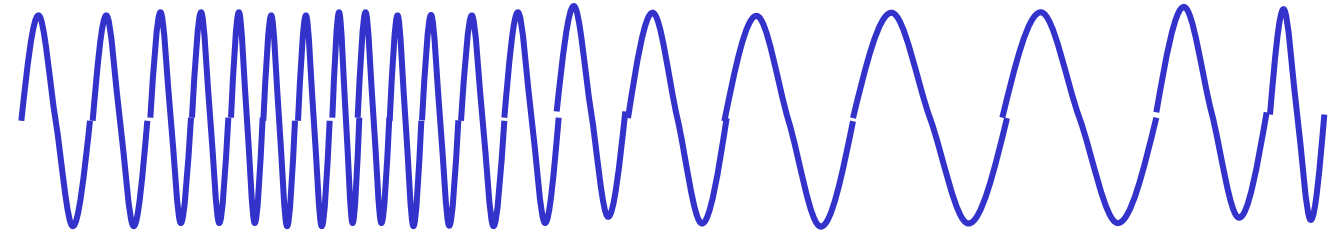
Unmodulated
Carrier Wave



Audio Signal



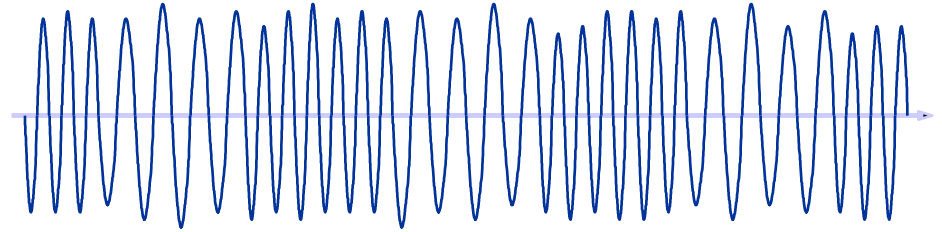
Frequency
Modulated
Wave



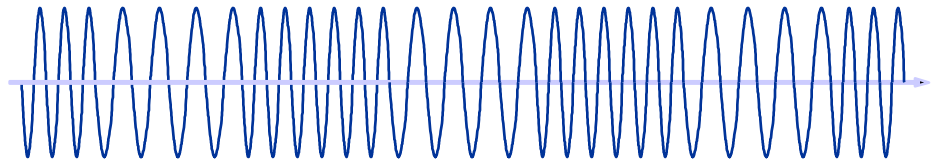
i.e. carrier freq increases with audio signal amplitude

FM Demodulation

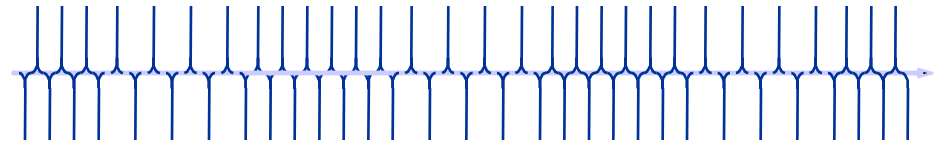
Received signal $S(t)$



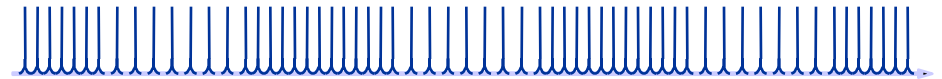
Limited and filtered signal



Zero Crossing Detection



Fully rectified signal



Pulse Generator



Low Pass Filter
Regenerator Threshold



Regenerated baseband
signal $S_m(t)$



Frequency Modulation – Modulation Index

Often denoted as β

indicates how much the frequency of the carrier will deviate in relation to the frequency of the modulating signal

i.e.

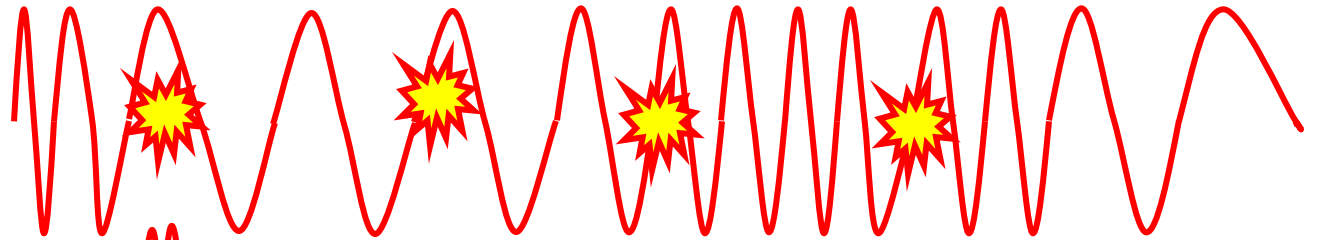
$$\beta = \frac{\Delta f}{f_m} = \text{the change of frequency} / \text{frequency of the carrier}$$

e.g. $\beta=5$ means the frequency of the modulated carrier will deviate by 5x the modulating signal

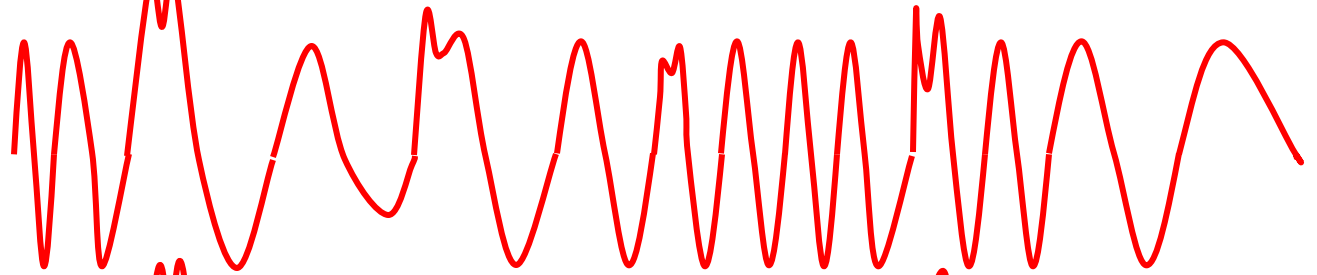
e.g. if freq range of the signal is 20Hz to 15KHz (i.e. typical FM spec) then the frequency of the carrier will deviate by 100Hz up to 75KHz

Frequency Modulation – Immunity to noise

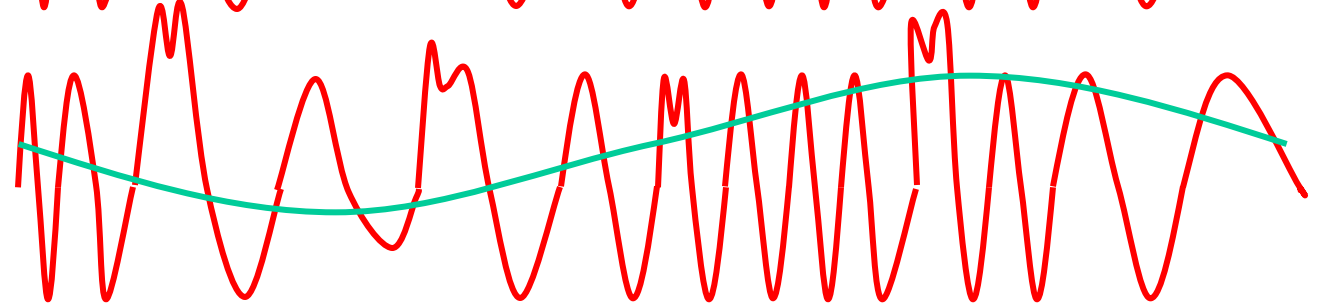
Transmitted FM signal



Noise-hit FM signal



Amplitude of signal is corrupted...

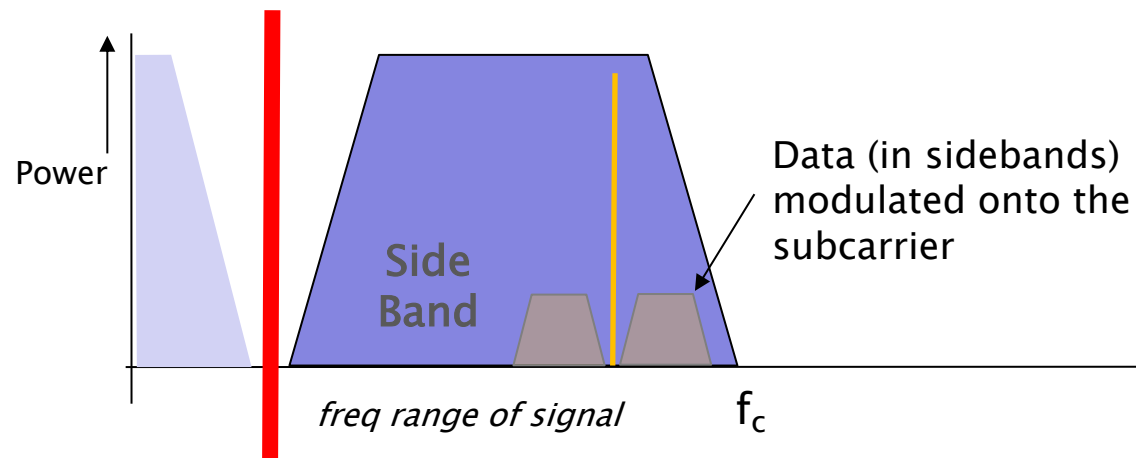
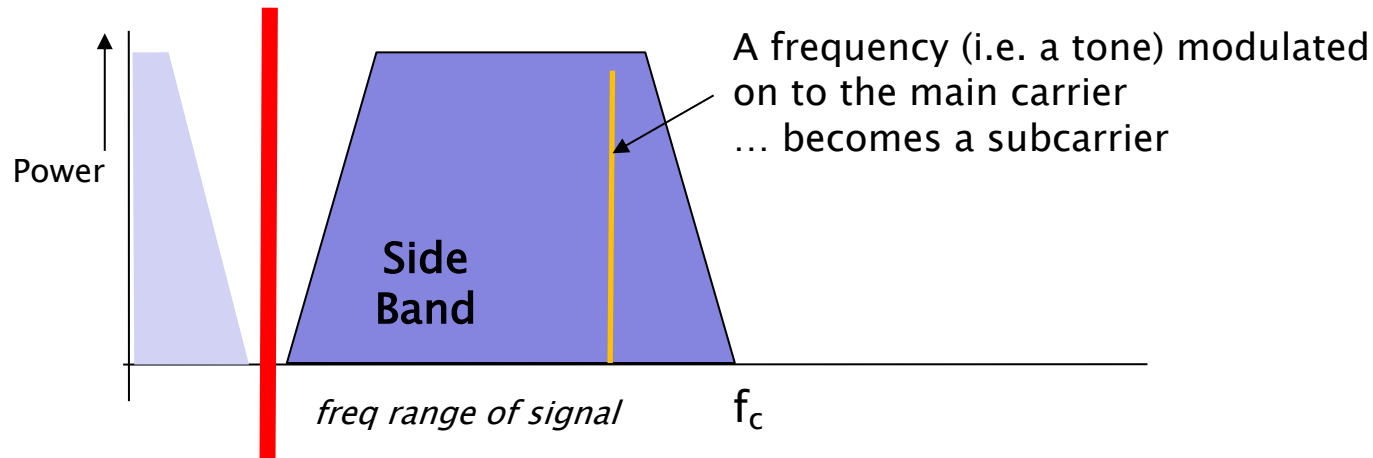


... however – frequency of signal is still intact:

Since we are only interested in the frequency of the carrier, the original audio signal can still be recovered

Subcarriers

An example of subcarrier – consider an AM transmission:



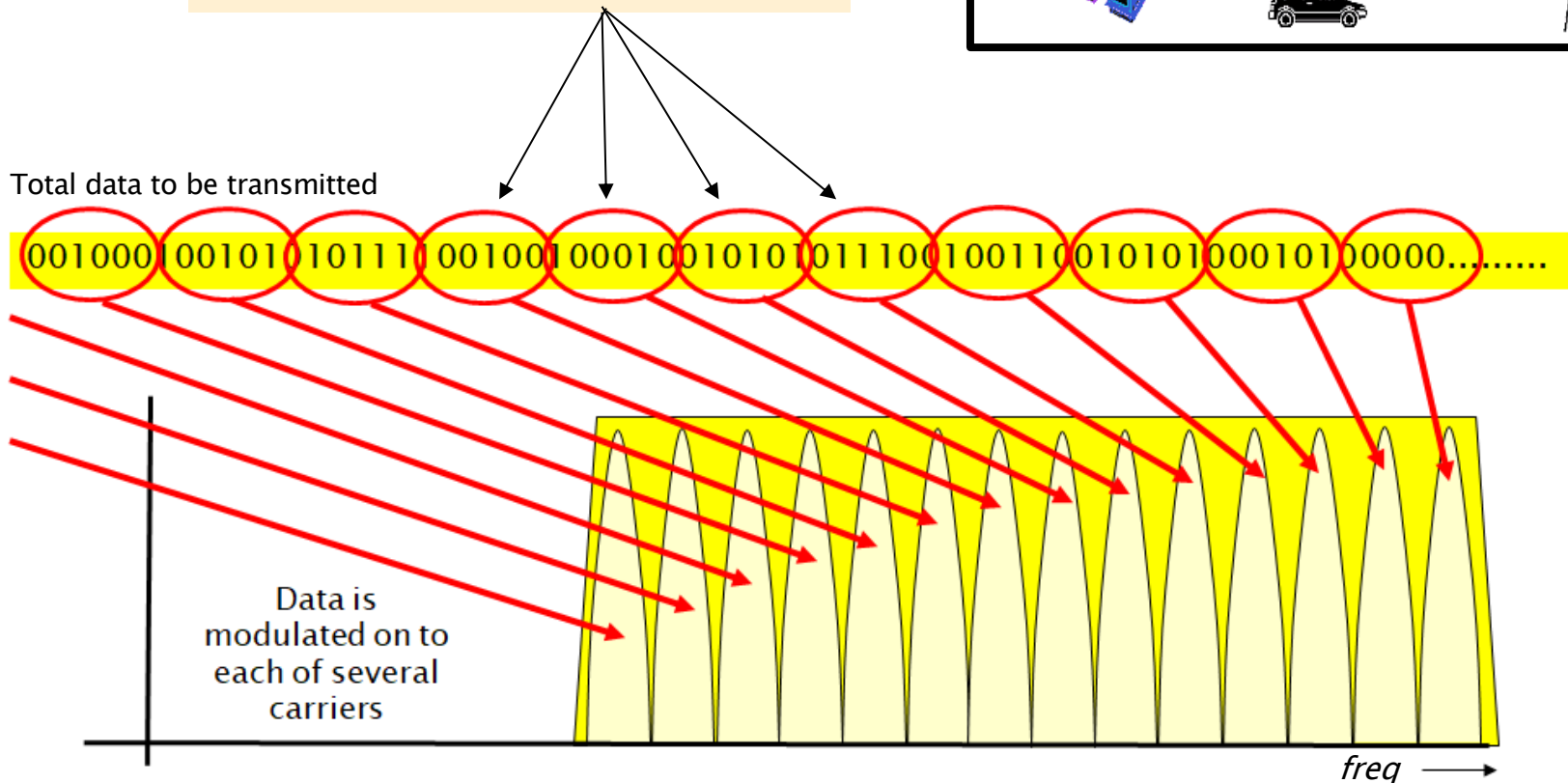
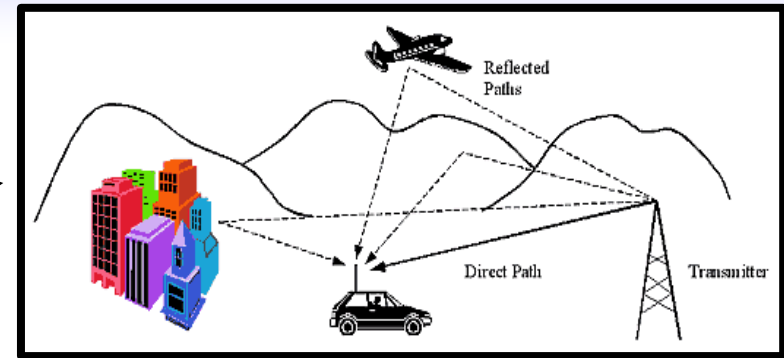
This example shows AM modulation on the subcarrier ... but could be other forms of modulation

- phase (PM)
- frequency (FM)
- etc

so, in effect we have two levels of modulation (at least)

Subcarriers ...??

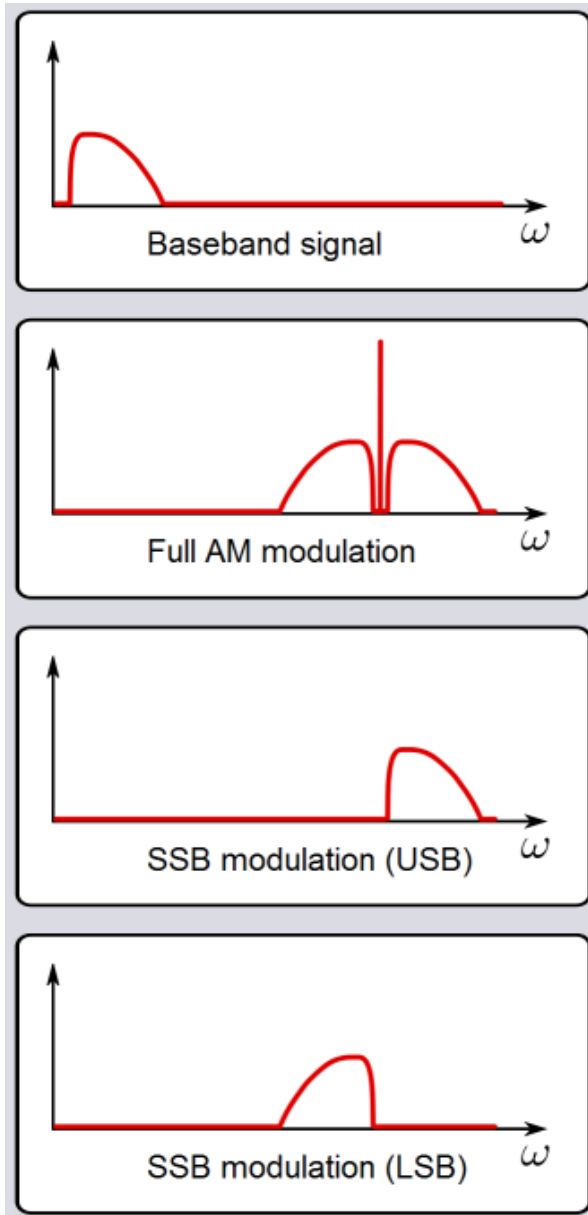
Intersymbol interference due to multipath reflections
Solution: multiple carriers



But these are separate carriers – not subcarriers modulated onto one carrier
... but they are often referred to as “subcarriers”

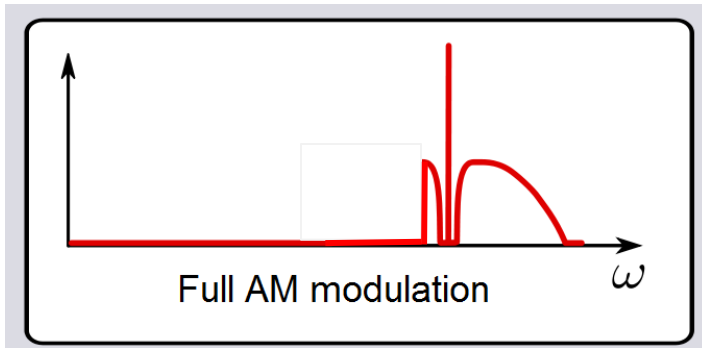
Spectrum of an Analogue PAL TV Transmission

Sidebands – Single, Vestigial



- SSB produced by filtering one of the sidebands, usually the lower (LSB)
- Carrier is also often reduced or removed entirely (suppressed)
= single sideband suppressed carrier (SSBSC)
- No information is lost in the process
- Radio frequency transmission is now concentrated in a single sideband
 $\therefore$ effective power output is greater than in normal AM
(since the carrier and redundant sideband account for well over half of the power output of an AM transmitter)
- Requires special reception to demodulate

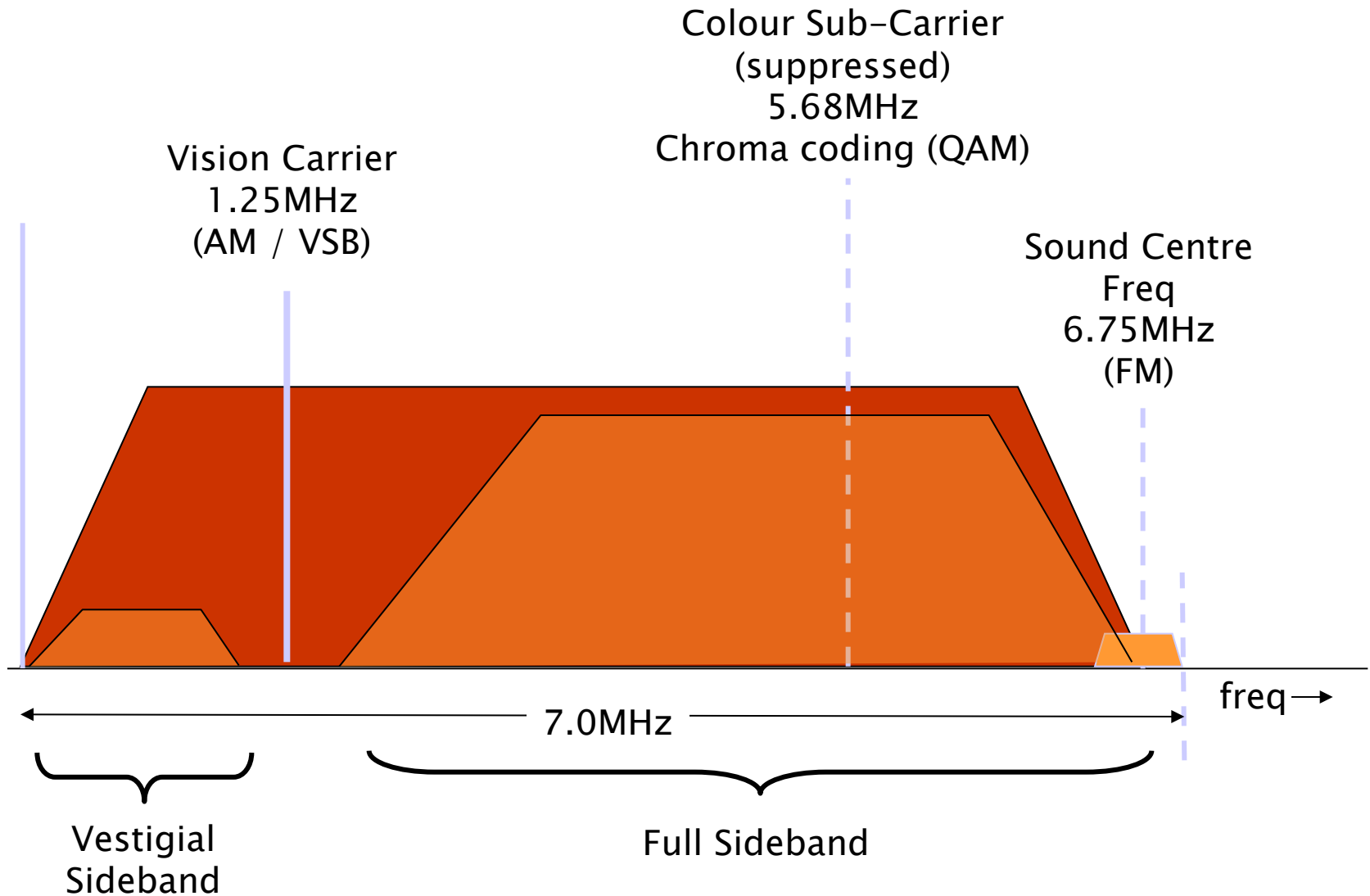
Vestigial Sideband (VSB)



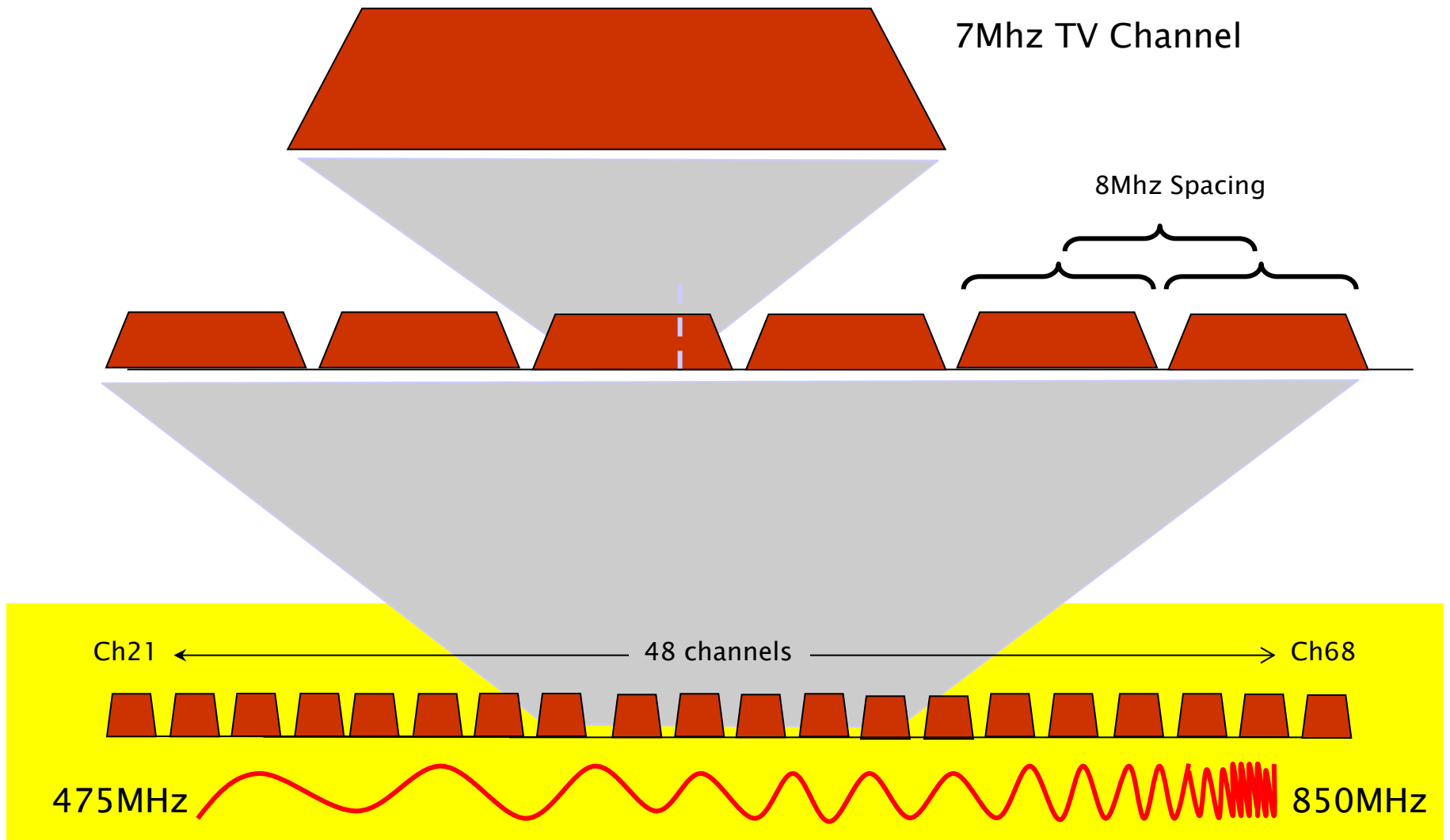
- sideband that has been only partly cut off or suppressed
- to reduce the large bandwidth of the original video signal
- also be used in digital transmission, such as the ATSC standardized 8-VSB
- by the RF and IF filters

- For video – SSB would be desirable
... but the video signal has significant low frequency content, plus square sync pulses
... so some of the lower sideband is retained
- VSB – full upper sideband of bandwidth transmitted = 4MHz
... but only 1.25MHz of the lower sideband is transmitted, plus the carrier
- effectively makes the system AM at low modulation frequencies and SSB at high modulation frequencies
- absence of the lower sideband components at high frequencies must be compensated for, and this is done by the RF and IF filters

PAL Signal Spectrum



Broadcast Channels Frequency Spacing



UK FREE-TV

Freeview: Sutton Coldfield Birmingham

Which Freeview channels does the Sutton Coldfield transmitter broadcast?

If you have any kind of Freeview fault, follow this [Freeview reset procedure](#) first.

Digital television services are broadcast on a multiplex (or Mux) where many stations occupy a single broadcast frequency, as shown below.

Mux	Aerial position	Frequency	Height Mode	Watts
PSB1 BBCA	horizontal max	C43 (650.0MHz)	64QAM 8K 2/3 24.1Mb/s DVB-T MPEG2	200,000W
PSB2 D3+4	horizontal max	C46 (674.0MHz)	64QAM 8K 2/3 24.1Mb/s DVB-T MPEG2	200,000W
PSB3 BBCB	horizontal max	C40+ (626.2MHz)	256QAM 32KE 2/3 40.2Mb/s DVB-T2 MPEG4	200,000W
COM4 SDN	horizontal max	C42 (642.0MHz)	64QAM 8K 3/4 27.1Mb/s DVB-T MPEG2	200,000W
COM5 ArqA	horizontal max	C45 (695.0MHz)	64QAM 8K 3/4 27.1Mb/s DVB-T MPEG2	200,000W
COM6 ArqB	horizontal max	C39+ (618.2MHz)	64QAM 8K 3/4 27.1Mb/s DVB-T MPEG2	200,000W

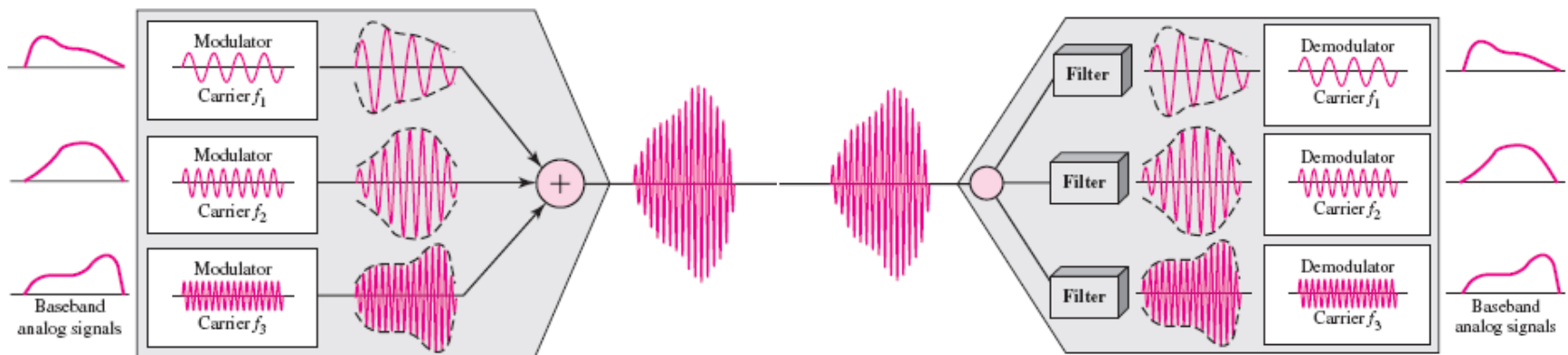
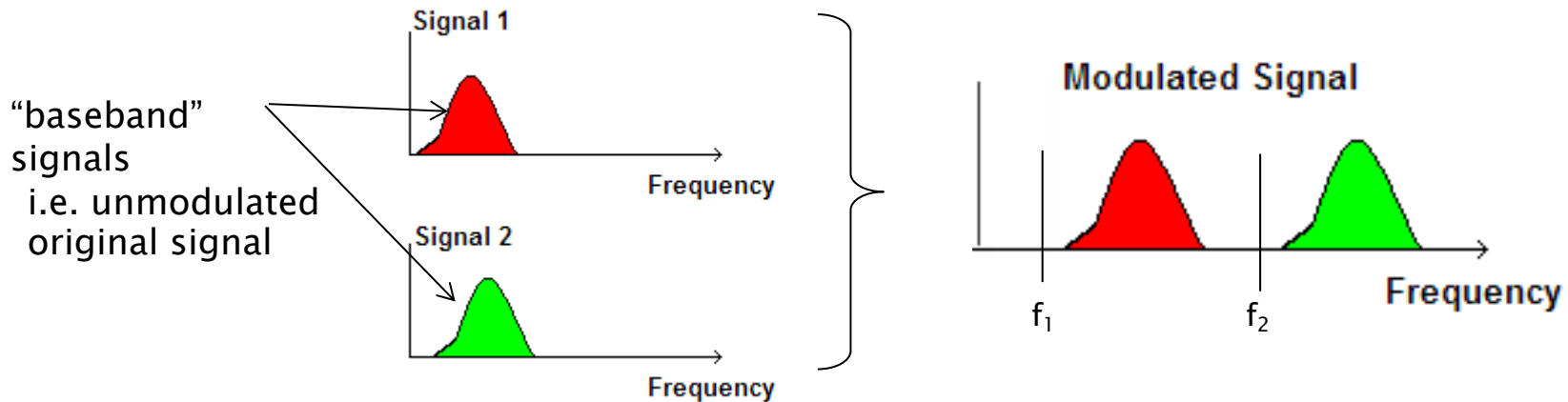
Mux	Aerial position	Frequency	Height Mode	Watts
PSB1 BBCA	horizontal max	C43 (650.0MHz)	64QAM 8K 2/3 24.1Mb/s DVB-T MPEG2	200,000W
 1 BBC One West Midlands, 2 BBC Two England, 7 BBC Three, 9 BBC Four, 70 CBBC Channel, 71 CBeebies, 80 BBC News, 81 BBC Parliament, 301 BBC Red Button 1, plus 12 others				
PSB2 D3+4	horizontal max	C46 (674.0MHz)	64QAM 8K 2/3 24.1Mb/s DVB-T MPEG2	200,000W
 3 ITV (Central (West micro region)), 4 Channel 4 Midlands ads, 5 Channel 5 Part Network ads, 6 ITV 2, 13 Channel 4+1 Midlands ads, 14 More 4, 15 Film 4, 28 E4, 33 ITV +1 (Central west),				
PSB3 BBCB	horizontal max	C40+ (626.2MHz)	256QAM 32KE 2/3 40.2Mb/s DVB-T2 MPEG4	200,000W
 73 CBBC HD, 101 BBC One HD (England no regional news), 102 BBC Two HD (England), 103 ITV HD (ITV Central West), 104 Channel 4 HD Midlands ads, 105 BBC Three HD, 303 BBC Red Button HD, plus 1 others				

Multiplexing: FDM, TDM Statistical

Multiplexing: combination of signals to transmit on a common bearer / channel

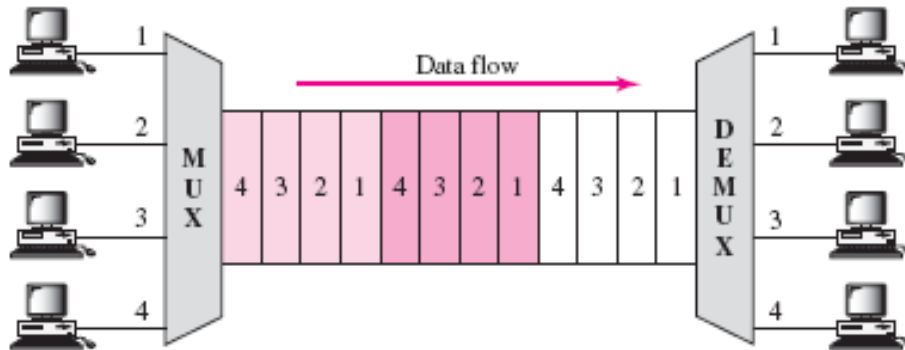
Multiplexing – Frequency Division Multiplexing (FDM)

- Each signal is modulated onto a different frequency (*subcarrier*)



Multiplexing – Time Division Multiplexing (TDM)

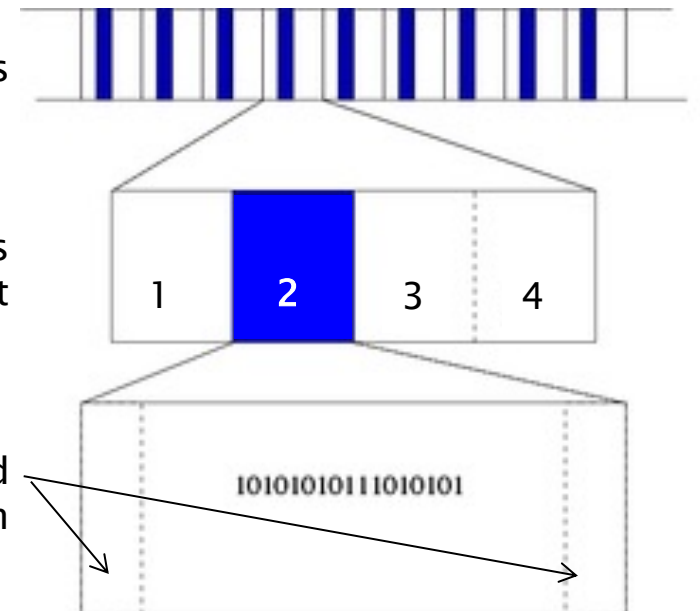
- Multiple signals are interleaved (in bits or blocks) over time



Data streams divided into frames

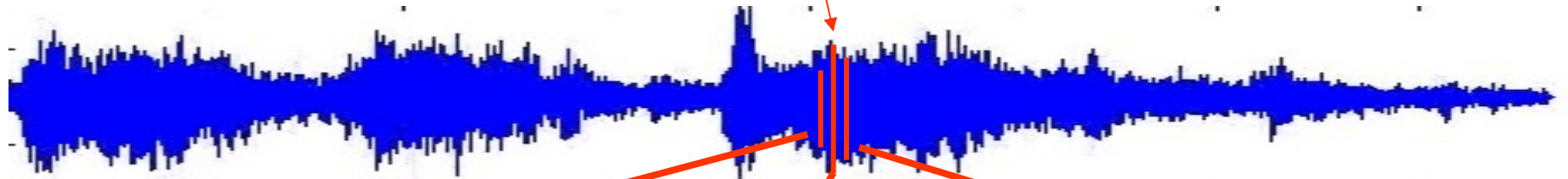
Frames divided into time slots
Each user is allocated one time slot

Time slots contain data with guard period
if needed for synchronisation



Time Division Multiplex

4kHz voice sampled at 8kHz (f_s) into 8 bits binary

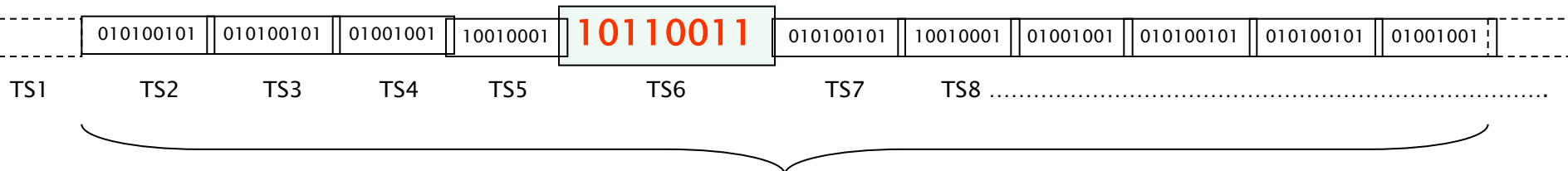


Analogue – Digital: Samples are dropped into allocated timeslot for each frame

TS6 previous frame

TS6 this frame

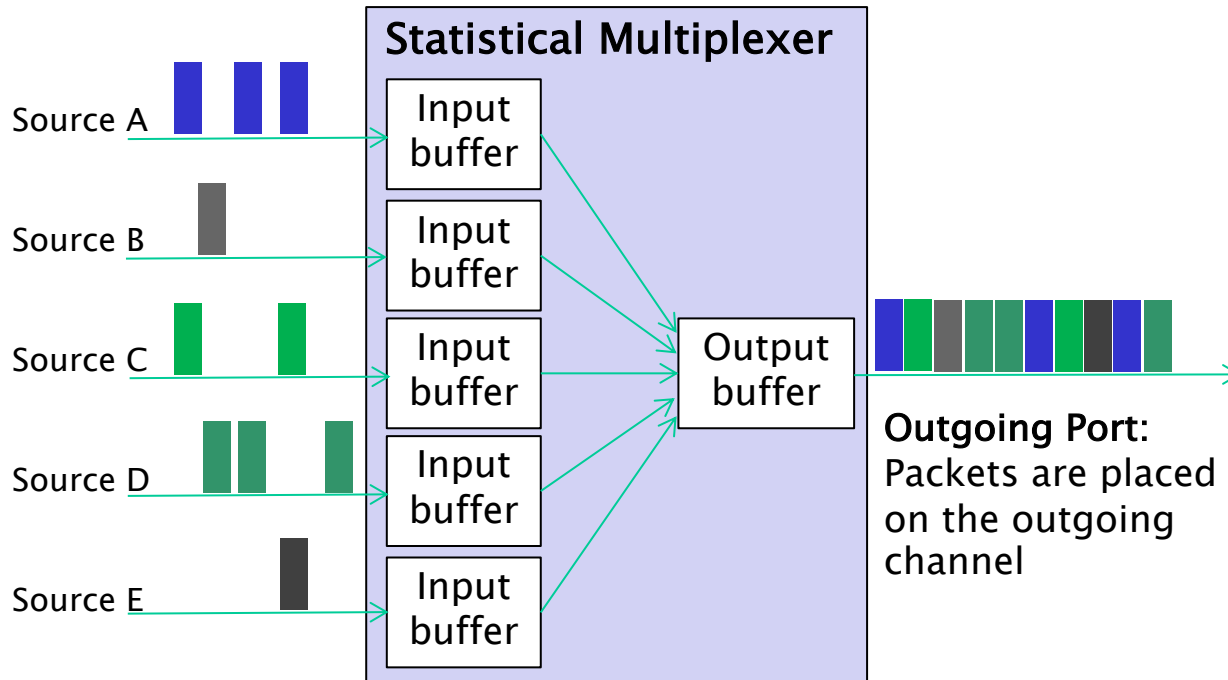
TS6 next frame



e.g. for UK / European E1 Multiplexes:

- 30 channels of speech into 32 channel frame “Multiplex”
- 32 channels x 8b x 8kHz = 2.048Mbps-1
 - (includes 1 channel signalling and 1 channel synchronisation)

Multiplexing – Statistical Multiplexing

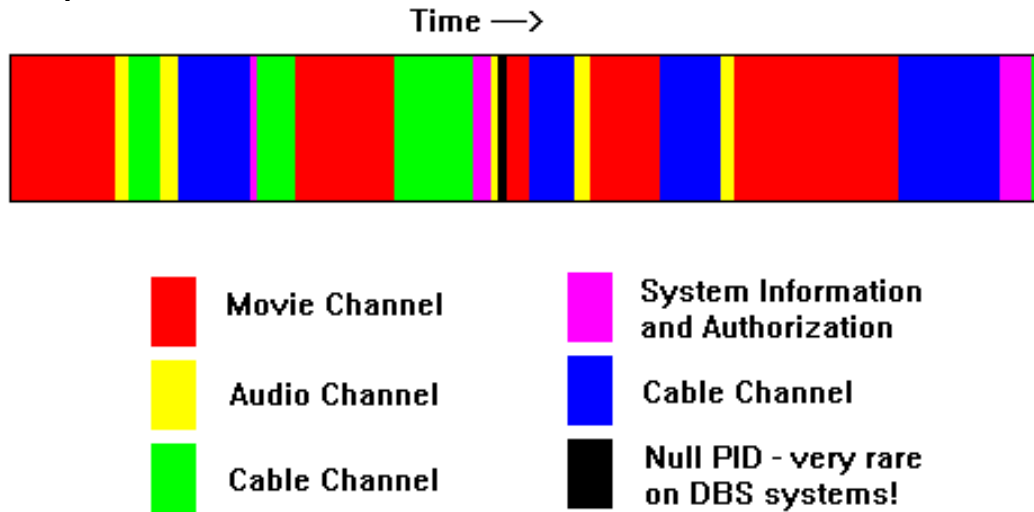


Incoming ports:
Data blocks arrive asynchronously from variable-rate sources

- Input buffers smooth fluctuation in input data rates
- Headers containing addresses and other information are added to the data blocks – to form packets
- Packets are then fed to the output buffer

Multiplexing – Statistical Multiplexing

e.g. DVB multiplex:



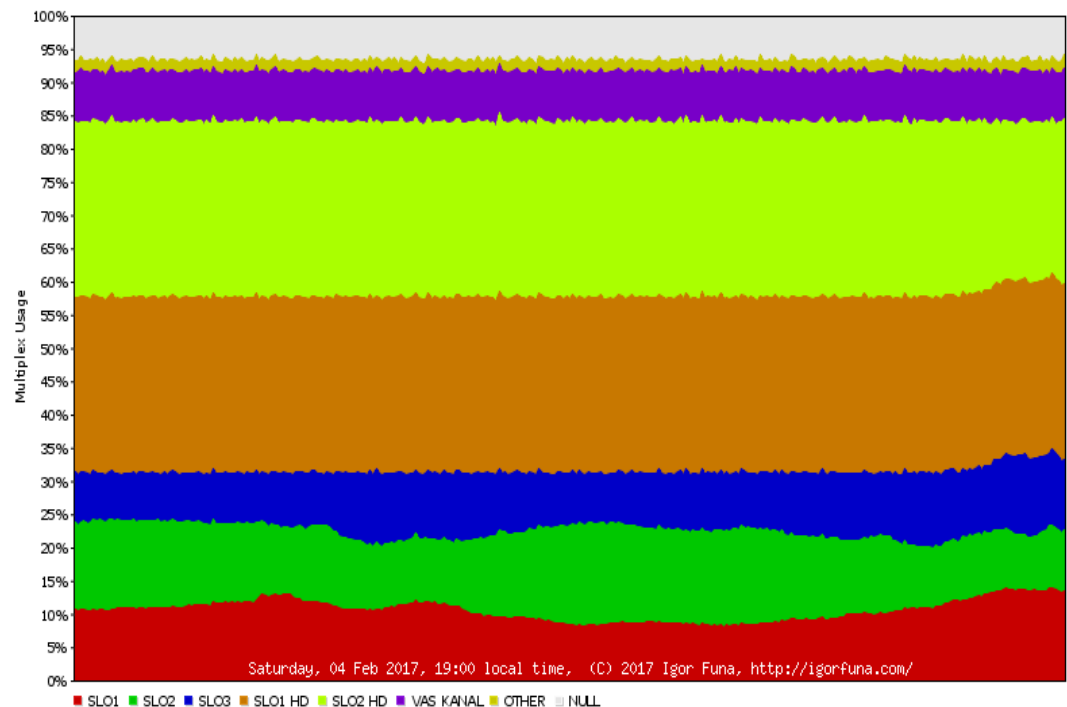
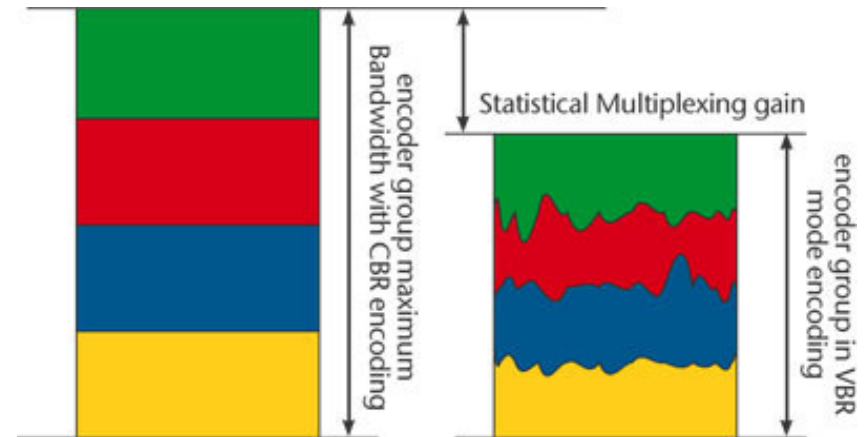
Packet format: Control, management and content assigned into fields



Statistical Multiplexing

Dynamic capacity allocation

- Variable Bit Rate – VBR
- one channel capacity reallocated to meet immediate service needs of other channels
- allow additional elementary bit streams to be added at the input of the multiplexer
- ... or allow elementary bit streams to be multiplexed at a *second stage*

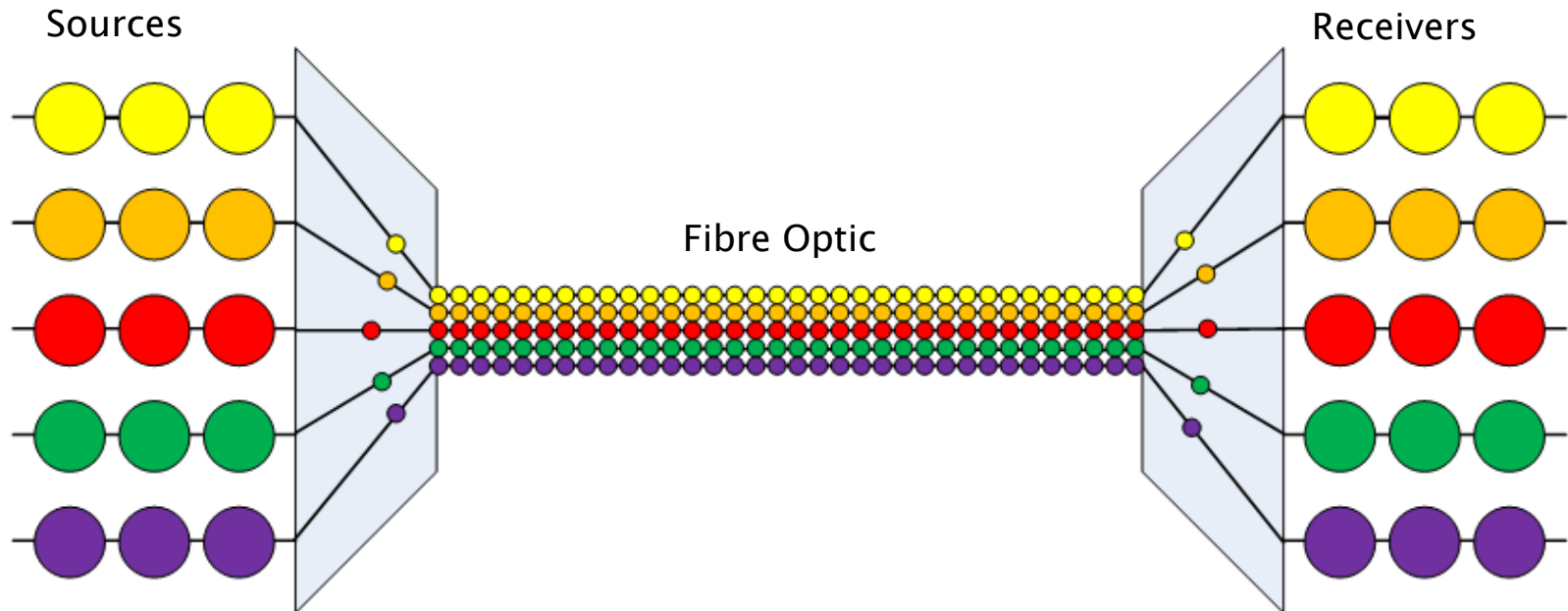


<http://igorfuna.com/dvb-t/slovenia/multiplex-a-usage-chart>

WDM – Wave Division Multiplexing

A number of multiplexes (each of many channels) are carried on one fibre

- Each multiplex is a string of data
 - each on it's own wavelength
- Uses single mode fibre



Modulation for Digital Data

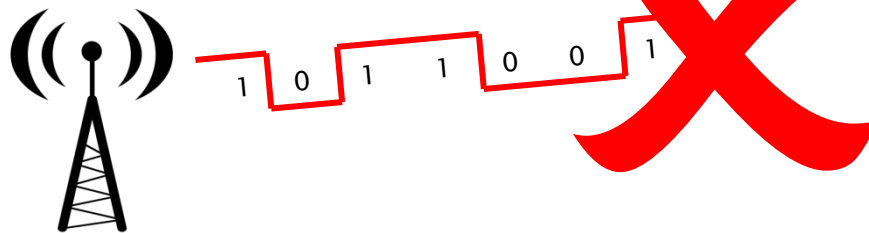
Modulation for Digital Data

objective:

We need to transmit a digital file of

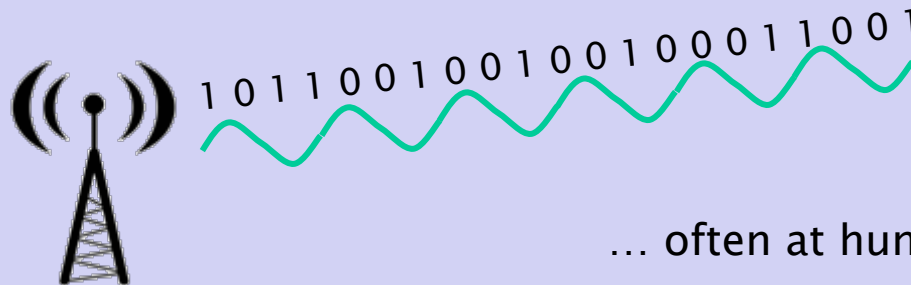
00010101100110101000101010100101100110101...

- can't transmit “neat” digital 1's and 0's from a transmitter
 - (or receive them at a receiver)



- transmission technologies operate only on analogue signals
 - we also need to retain the carrier frequency concept to allow for many stations

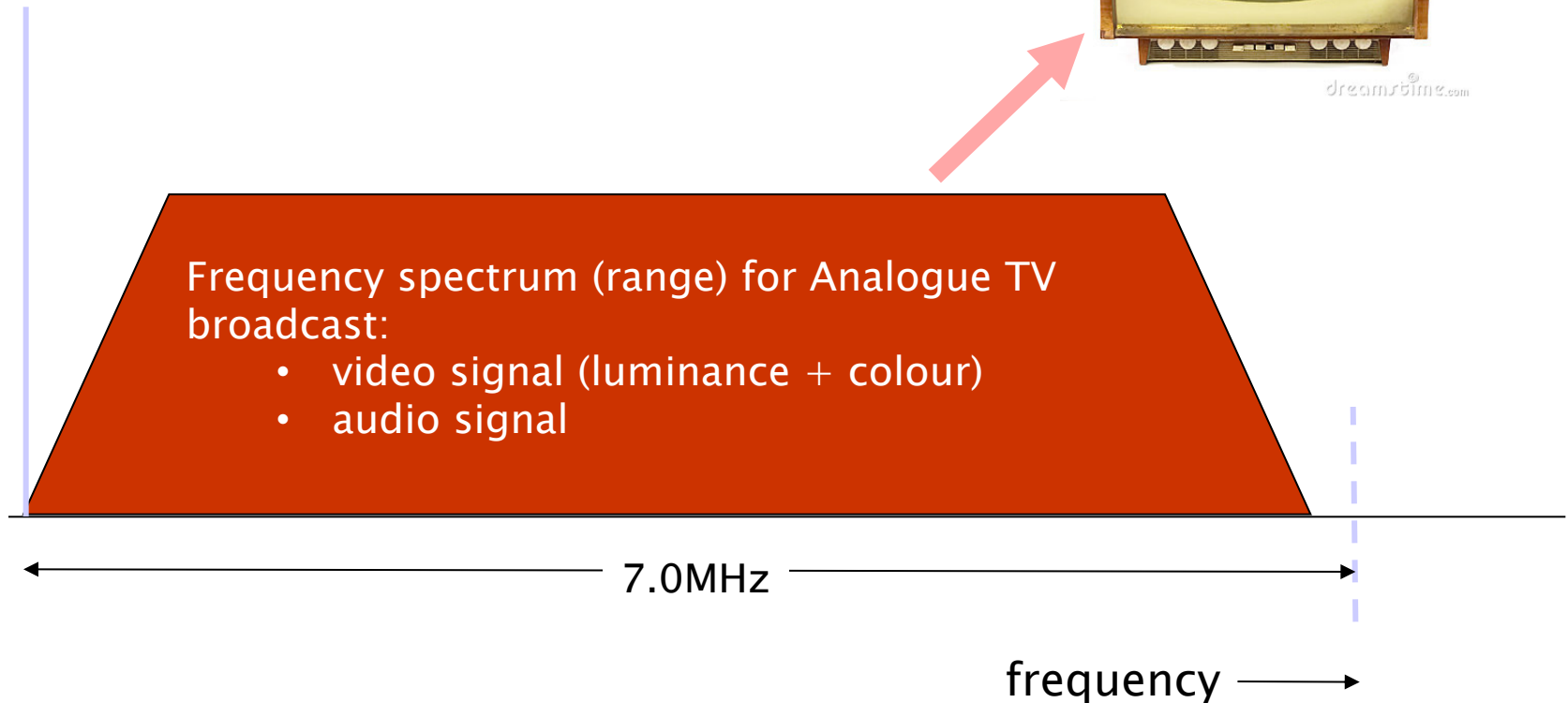
- so carry 1's and 0's on analogue signals:



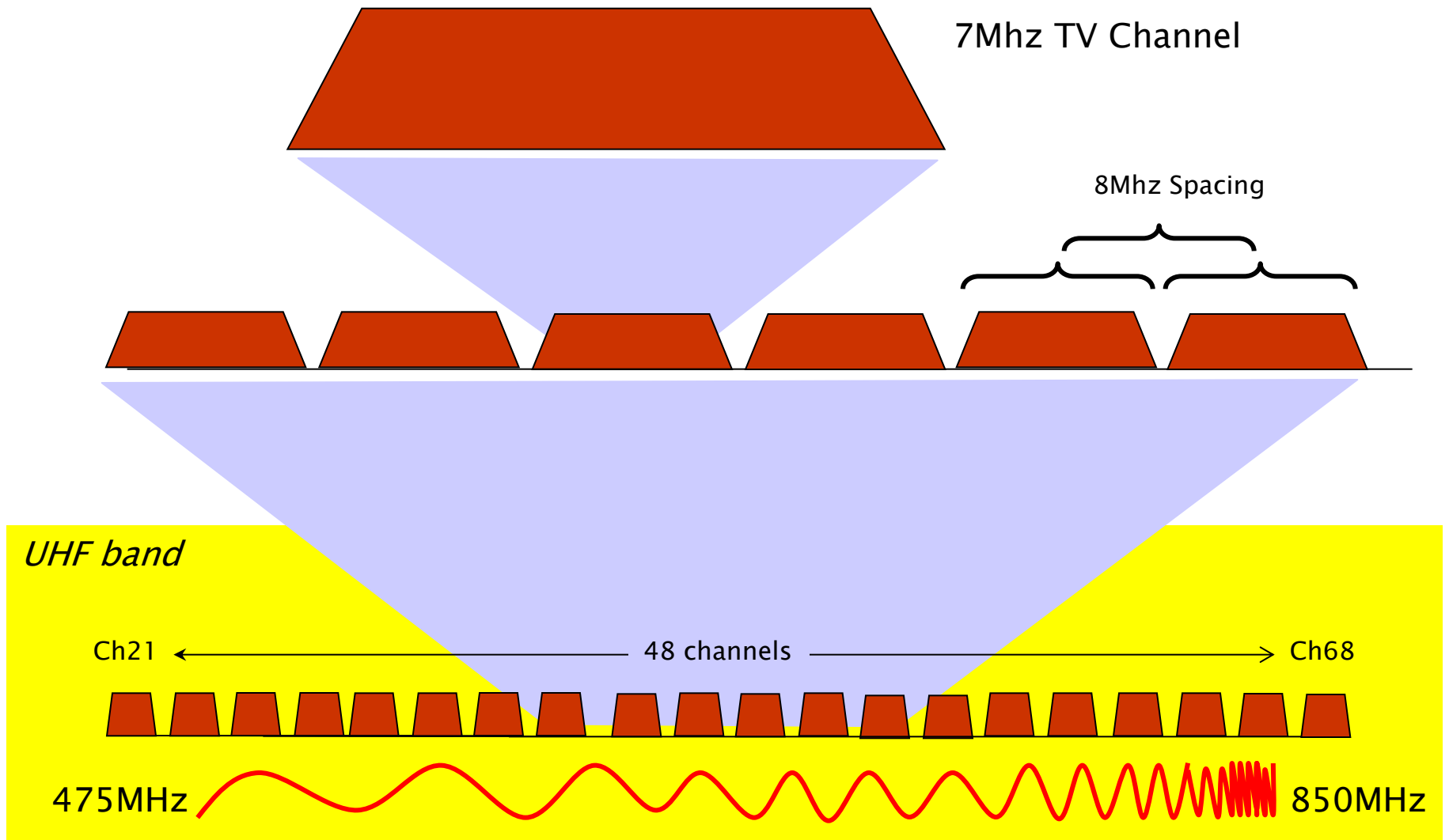
... often at hundreds of Mbits/s

e.g: Analogue TV ... to Digital TV

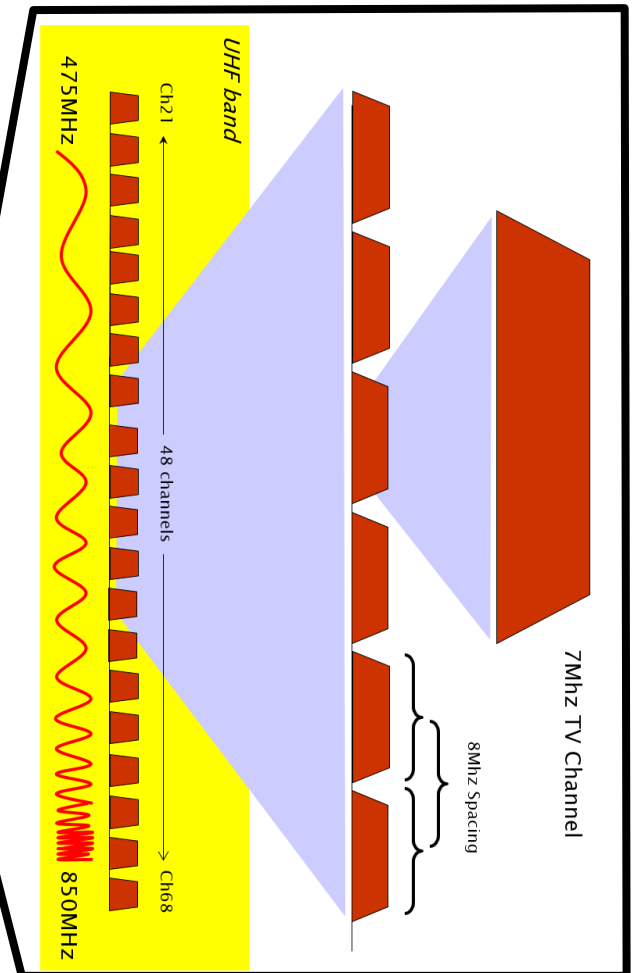
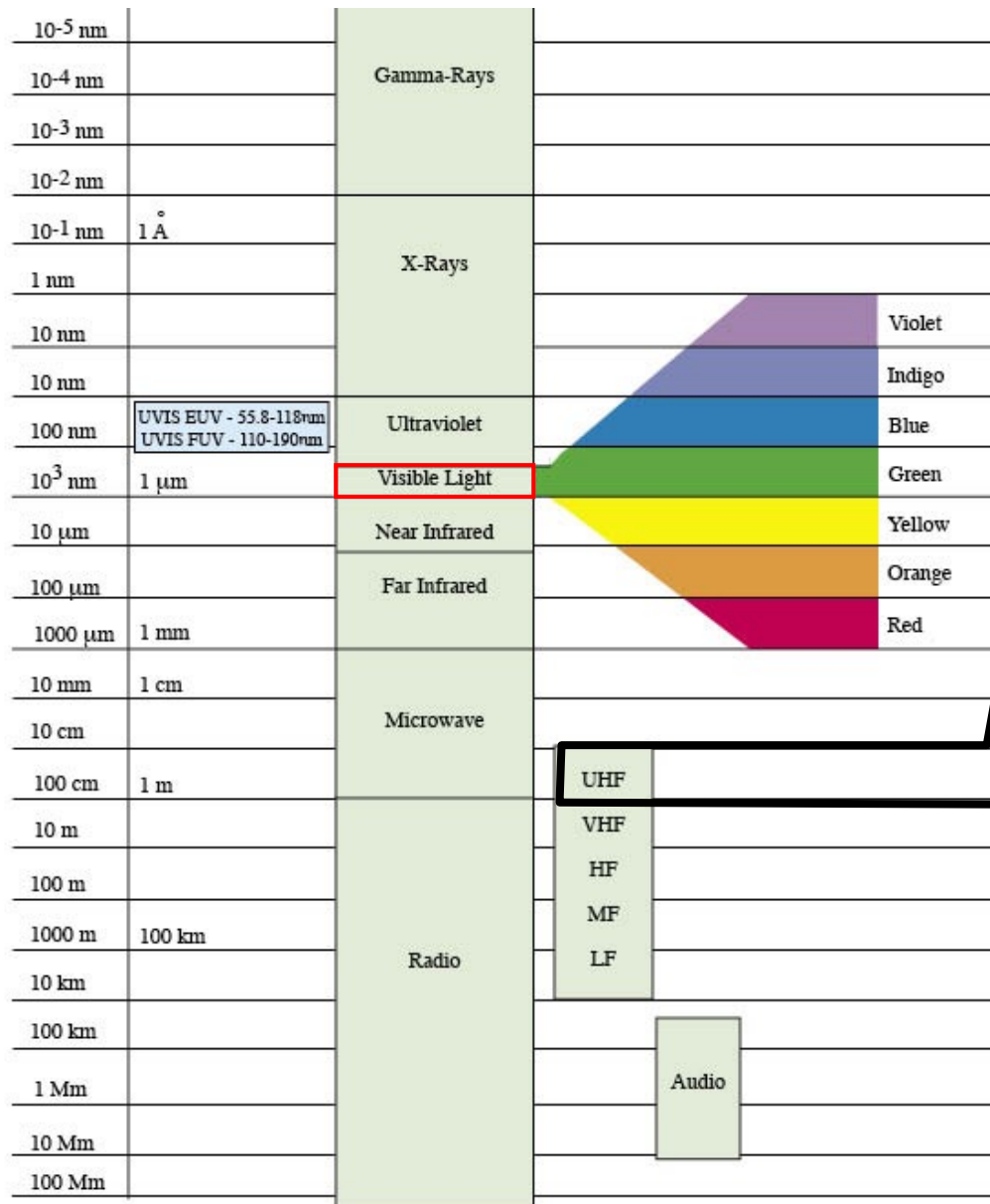
Analogue TV Signal Spectrum



Broadcast Channels Frequency Spacing



UHF TV in the Electromagnetic Spectrum



UK FREE-TV

Freeview: Sutton Coldfield Birmingham

Which Freeview channels does the Sutton Coldfield transmitter broadcast?

If you have any kind of Freeview fault, follow this [Freeview reset procedure](#) first.

Digital television services are broadcast on a multiplexes (or Mux) where many stations occupy a single broadcast frequency, as shown below.

Mux	Aerial position	Frequency	Height Mode	Watts
-----	-----------------	-----------	-------------	-------

PSB1 BBCA	horizontal max	C43 (650.0MHz)	433m 64QAM 8K 2/3 24.1Mb/s DVB-T MPEG2	200,000W
--------------	----------------	----------------	--	----------



1 BBC One West Midlands, 2 BBC Two England, 7 BBC Three, 9 BBC Four, 70 CBBC Channel, 71 CBeebies, 80 BBC News, 81 BBC Parliament, 301 BBC Red Button 1, plus 12 others

PSB2 D3+4	horizontal max	C46 (674.0MHz)	433m 64QAM 8K 2/3 24.1Mb/s DVB-T MPEG2	200,000W
--------------	----------------	----------------	--	----------



3 ITV (Central (West micro region)), 4 Channel 4 Midlands ads, 5 Channel 5 Part Network ads, 6 ITV 2, 13 Channel 4+1 Midlands ads, 14 More 4, 15 Film 4, 28 E4, 33 ITV+1 (Central west),

PSB3 BBCB	horizontal max	C40+ (626.2MHz)	433m 256QAM 32KE 2/3 40.2Mb/s DVB-T2 MPEG4	200,000W
--------------	----------------	-----------------	--	----------



73 CBBC HD, 101 BBC One HD (England no regional news), 102 BBC Two HD (England), 103 ITV HD (ITV Central West), 104 Channel 4 HD Midlands ads, 105 BBC Three HD, plus 1 others

COM4 SDN	horizontal max	C42 (642.0MHz)	433m 64QAM 8K 3/4 27.1Mb/s DVB-T MPEG2	200,000W
-------------	----------------	----------------	--	----------



ESP1 UK (not free), 10 ITV 3, 20 Drama, 25 Dave is vii, 27 ITV 2+1, 30 E, 31 SUBA, 38 Quest, 39 The Zone, 44 Channel 5 +1, 55 5 Later, 61 True Entertainment, 72 CITY, plus 23 others

COM5 ArqA	horizontal max	C45 (656.0MHz)	433m 64QAM 8K 3/4 27.1Mb/s DVB-T MPEG2	200,000W
--------------	----------------	----------------	--	----------



11 pick, 12 Dave, 17 Really, 29 E4+1, 32 Movie Mix, 41 Food Network, 42 Travel Channel, 46 Challenge, 82 Sky News, 87 Community Channel, plus 11 others

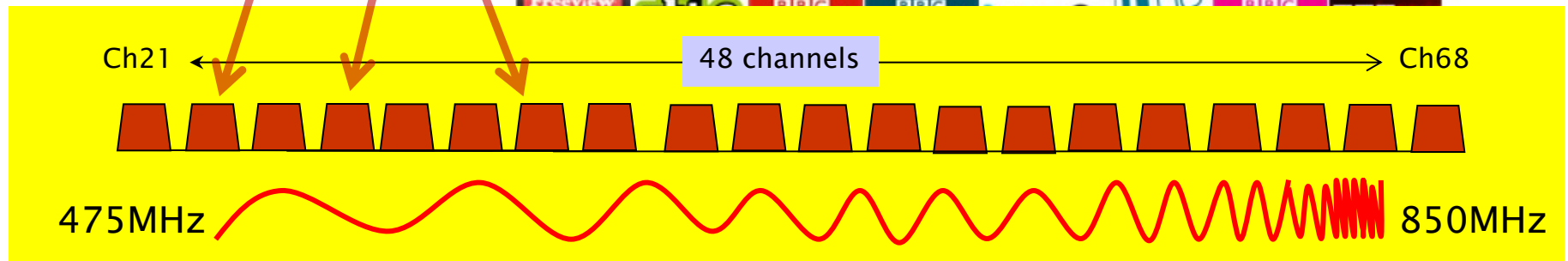
COM6 ArqB	horizontal max	C39+ (618.2MHz)	433m 64QAM 8K 3/4 27.1Mb/s DVB-T MPEG2	200,000W
--------------	----------------	-----------------	--	----------



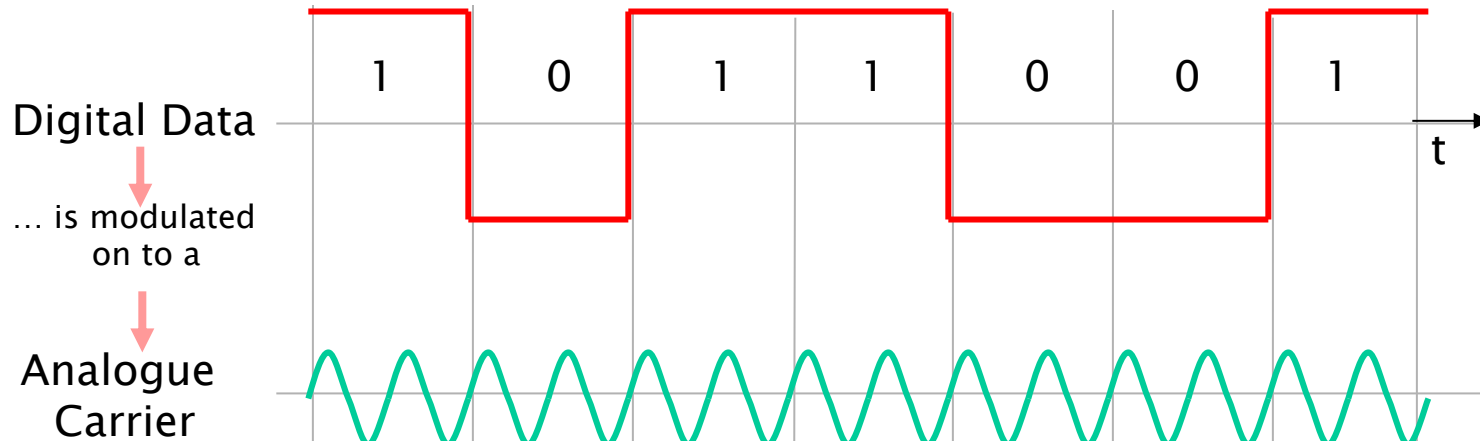
11 pick, 12 Dave, 17 Really, 29 E4+1, 32 Movie Mix, 41 Food Network, 42 Travel Channel, 46 Challenge, 82 Sky News, 87 Community Channel, plus 11 others

Mux	Aerial position	Frequency	Height Mode	Watts
PSB1 BBCA	horizontal max	C43 (650.0MHz)	433m 64QAM 8K 2/3 24.1Mb/s DVB-T MPEG2	200,000W
1 BBC One West Midlands, 2 BBC Two England, 7 BBC Three, 9 BBC Four, 70 CBBC Channel, 71 CBeebies, 80 BBC News, 81 BBC Parliament, 301 BBC Red Button 1, plus 12 others				
PSB2 D3+4	horizontal max	C46 (674.0MHz)	433m 64QAM 8K 2/3 24.1Mb/s DVB-T MPEG2	200,000W
3 ITV (Central (West micro region)), 4 Channel 4 Midlands ads, 5 Channel 5 Part Network ads, 6 ITV 2, 13 Channel 4+1 Midlands ads, 14 More 4, 15 Film 4, 28 E4, 33 ITV+1 (Central west),				
PSB3 BBCB	horizontal max	C40+ (626.2MHz)	433m 256QAM 32KE 2/3 40.2Mb/s DVB-T2 MPEG4	200,000W
73 CBBC HD, 101 BBC One HD (England no regional news), 102 BBC Two HD (England), 103 ITV HD (ITV Central West), 104 Channel 4 HD Midlands ads, 105 BBC Three HD, plus 1 others				
COM4 SDN	horizontal max	C42 (642.0MHz)	433m 64QAM 8K 3/4 27.1Mb/s DVB-T MPEG2	200,000W
ESP1 UK (not free), 10 ITV 3, 20 Drama, 25 Dave is vii, 27 ITV 2+1, 30 E, 31 SUBA, 38 Quest, 39 The Zone, 44 Channel 5 +1, 55 5 Later, 61 True Entertainment, 72 CITY, plus 23 others				
COM5 ArqA	horizontal max	C45 (656.0MHz)	433m 64QAM 8K 3/4 27.1Mb/s DVB-T MPEG2	200,000W
11 pick, 12 Dave, 17 Really, 29 E4+1, 32 Movie Mix, 41 Food Network, 42 Travel Channel, 46 Challenge, 82 Sky News, 87 Community Channel, plus 11 others				
COM6 ArqB	horizontal max	C39+ (618.2MHz)	433m 64QAM 8K 3/4 27.1Mb/s DVB-T MPEG2	200,000W
11 pick, 12 Dave, 17 Really, 29 E4+1, 32 Movie Mix, 41 Food Network, 42 Travel Channel, 46 Challenge, 82 Sky News, 87 Community Channel, plus 11 others				

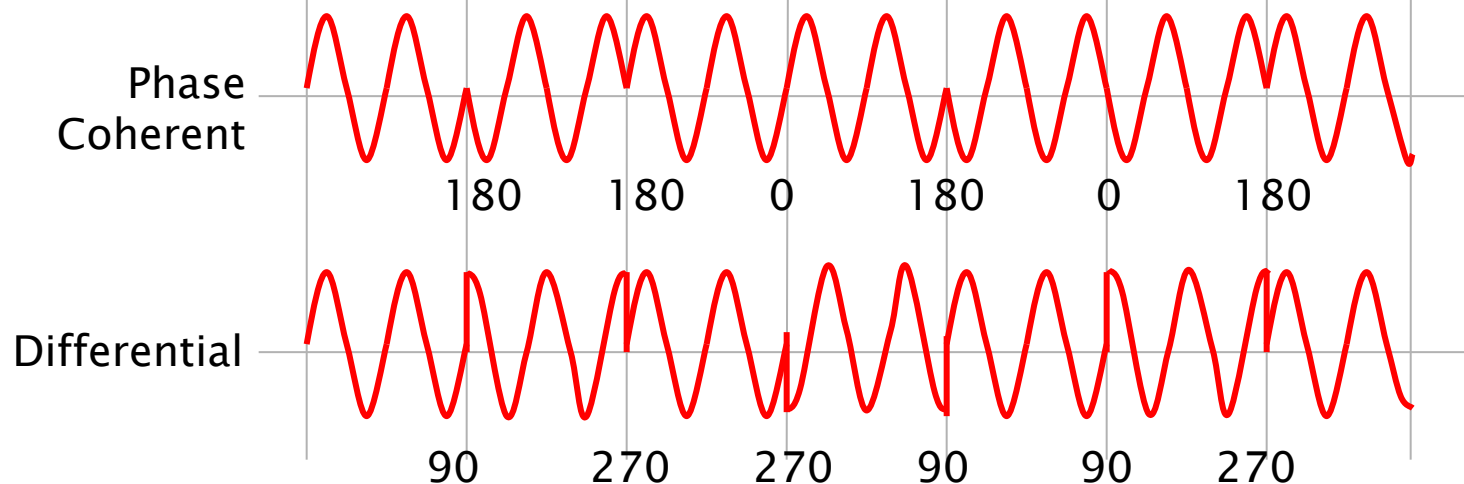
Mux	Aerial position	Frequency	Height	Mode	Watts
PSB1 BBCA	horizontal max	C43 (650.0MHz)	433m	64QAM 8K 2/3 24.1MB/s DVB-T MPEG2	200,000W
 1 BBC One West Midlands, 2 BBC Two England, 7 BBC Three, 9 BBC Four, 70 CBBC Channel, 71 CBeebies, 80 BBC News, 81 BBC Parliament, 301 BBC Red Button 1, plus 12 others					
PSB2 D3+4	horizontal max	C46 (674.0MHz)	433m	64QAM 8K 2/3 24.1MB/s DVB-T MPEG2	200,000W
 3 ITV (Central (West micro region)), 4 Channel 4 Midlands ads, 5 Channel 5 Part Network ads, 6 ITV 2, 13 Channel 4+1 Midlands ads, 14 More 4, 15 Film 4, 28 E4, 33 ITV +1 (Central west),					
PSB3 BBCB	horizontal max	C40+ (626.2MHz)	433m	256QAM 32KE 2/3 40.2MB/s DVB-T2 MPEG4	200,000W
					



Digital Data on Analogue Carrier *e.g. Phase Shift Keying (PSK)*



Binary PSK (there's various formats):



The Transmitted Signal

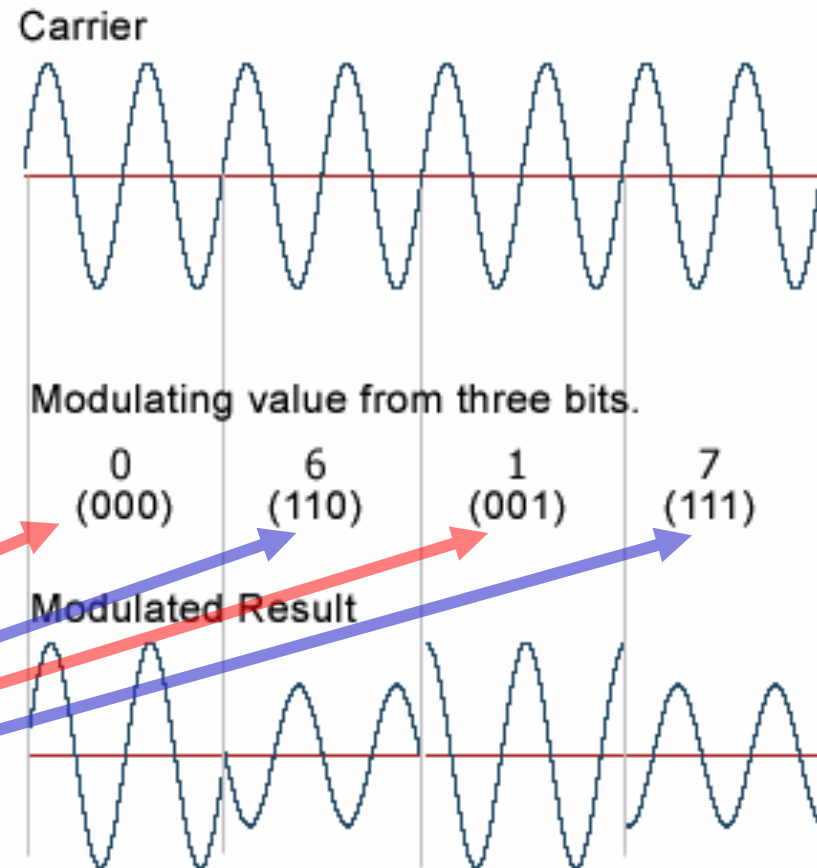
- Shown here – 8 QAM
- i.e. eight digital codes can be transmitted:
 - 000, 001, 010, 011,
100, 101, 110, 111

- So transmit the signal

000110001111:

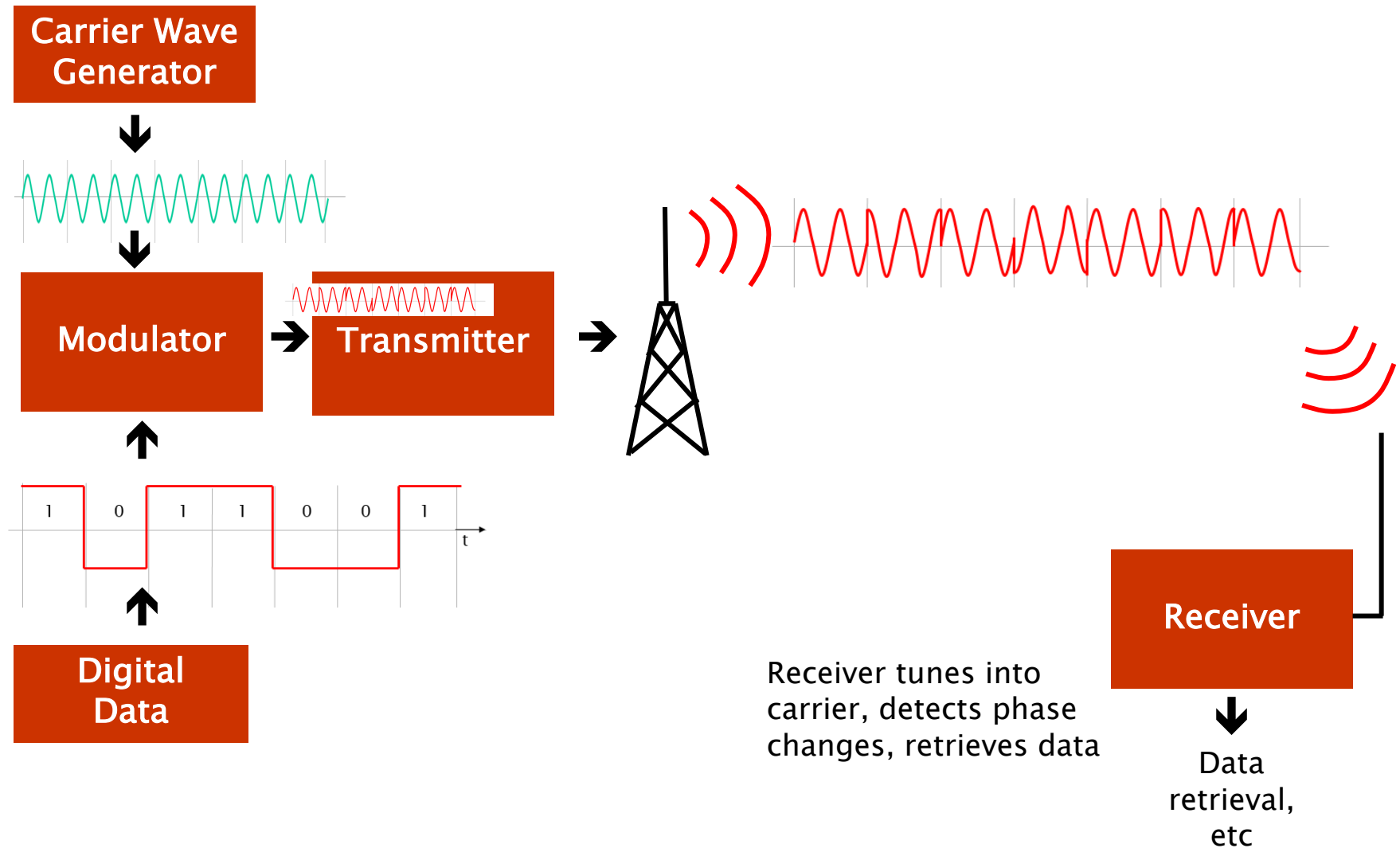
...break it into the
groups of codes
available

000 110 001 111



Note: Only four (0, 6, 1 and 7) out of the eight possible modulation states (0-7) are shown in this illustration.

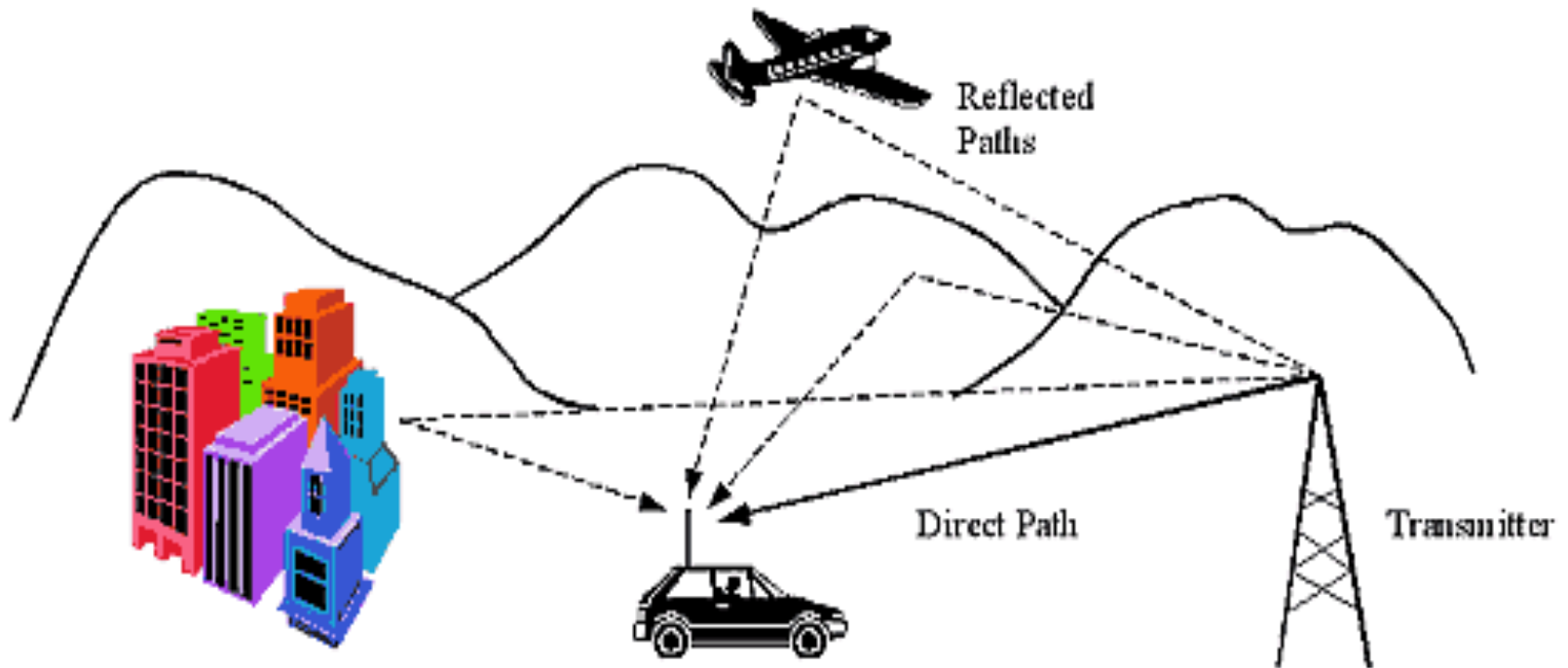
Digital Transmission



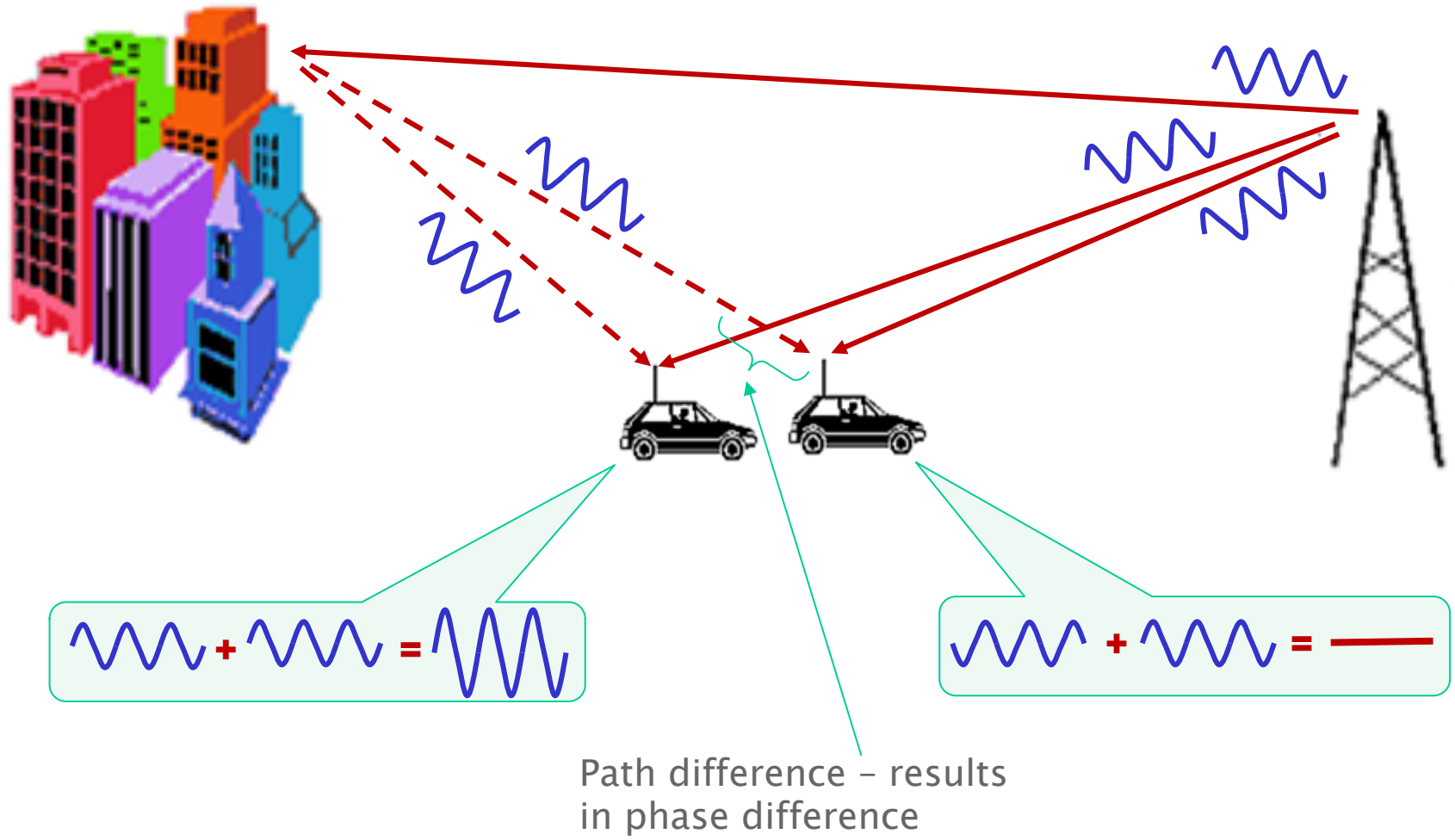
Inter-Symbol Interference for Modulated Data

Causes

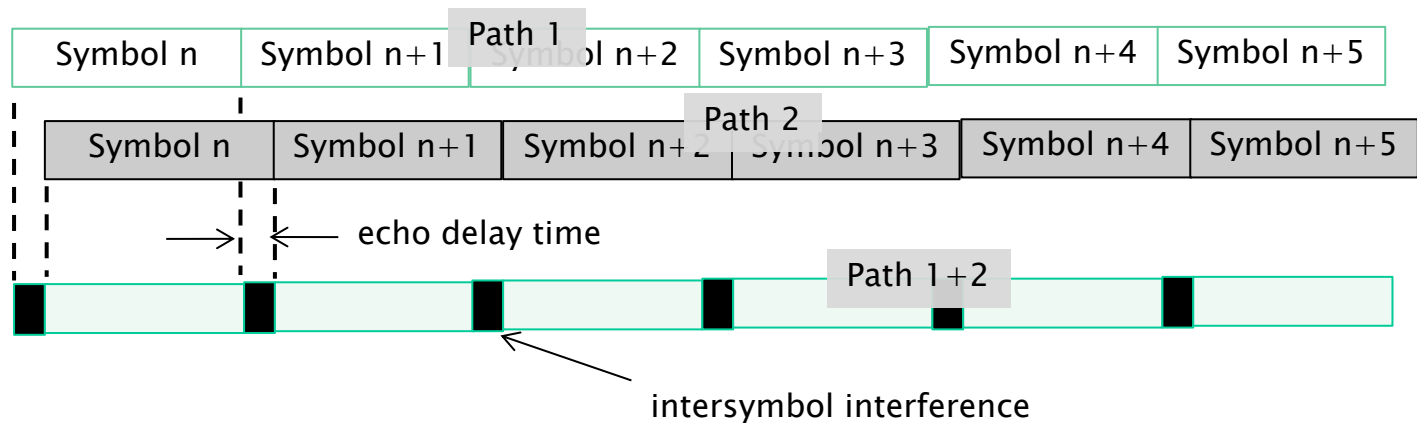
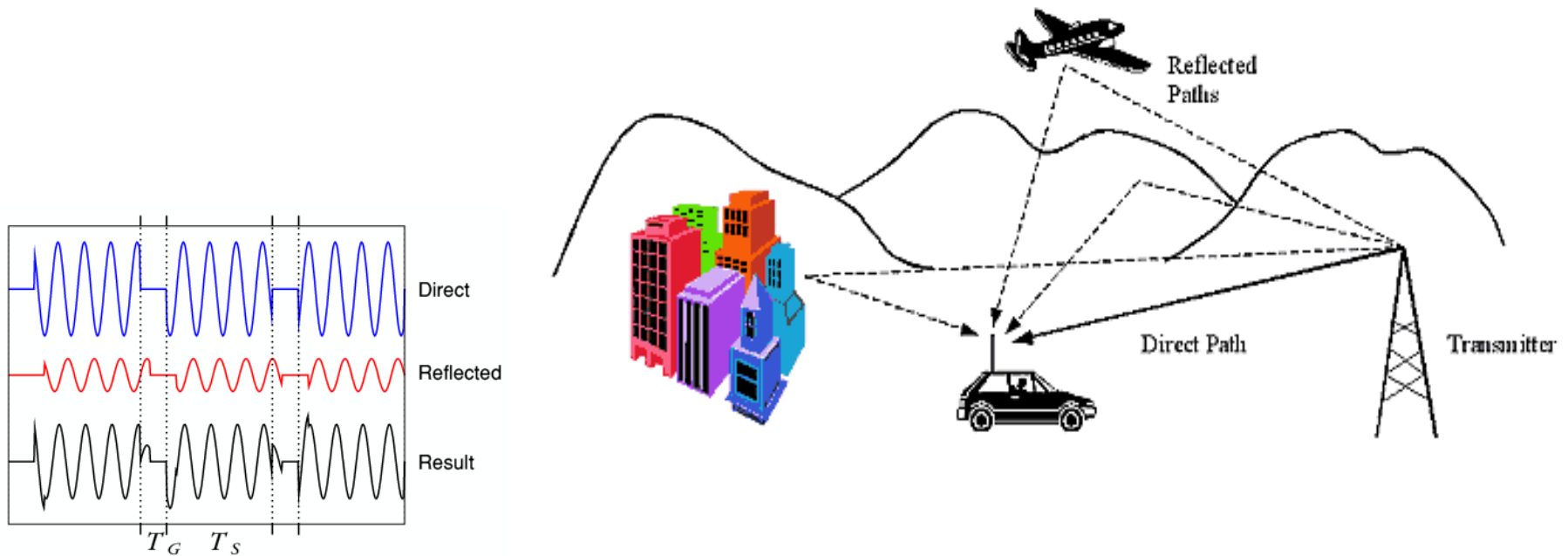
- multipath interference, including atmospherics
- channel gain variations
- signal spreading over a transmission line
- jitter and other timing issues

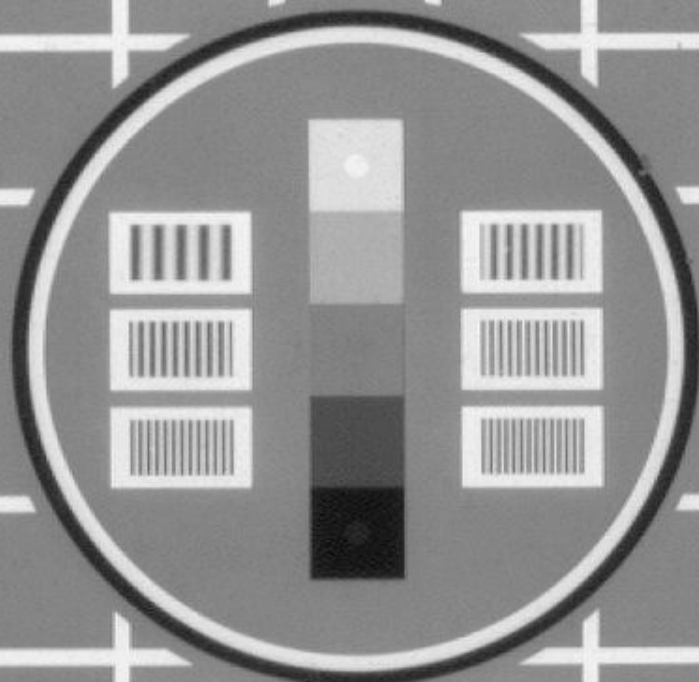


Inter-Symbol Interference for Modulated Data



Inter-Symbol Interference for Modulated Data



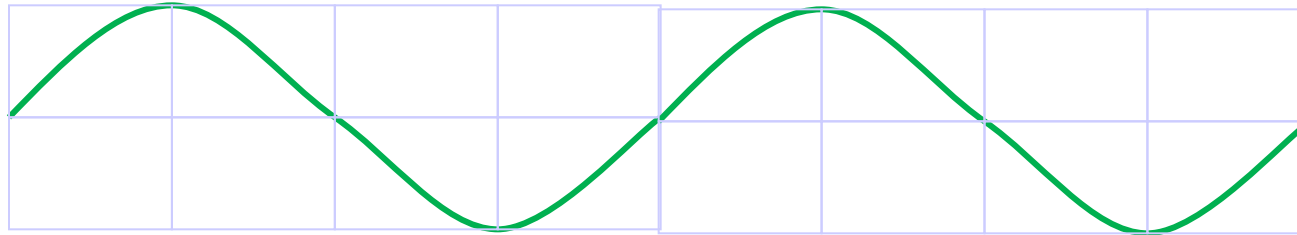


• | D | •

BBC 1 405 LINES

Advanced: QAM (later in the Course)

The Carrier for Digital – Still a Sinusoidal Wave



0°..... 90°.....180°.....270°...360°

Carrier wave

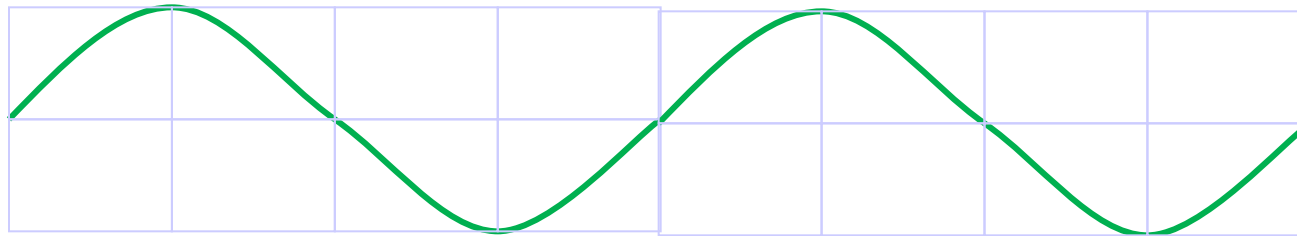
e.g. for Digital Terrestrial TV = 475 – 850 MHz

... for Bluetooth = 2.4GHz

... for Wifi = 2.4GHz and 5GHz

... for DSL broadband – many frequencies (OFDM)

Phase Changes In A Sinusoidal Wave

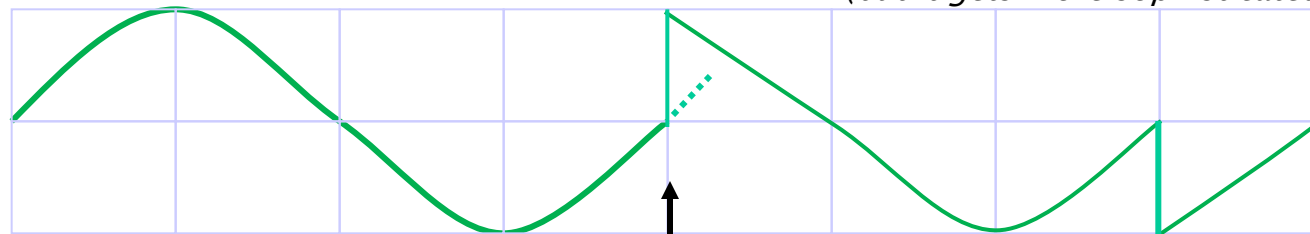


0°..... 90°.....180°.....270°...360°

Changes in phase (and amplitude) of a carrier can be used to denote digital code information

e.g. $+90^\circ = \text{"1"}; +270^\circ = \text{"0"}$

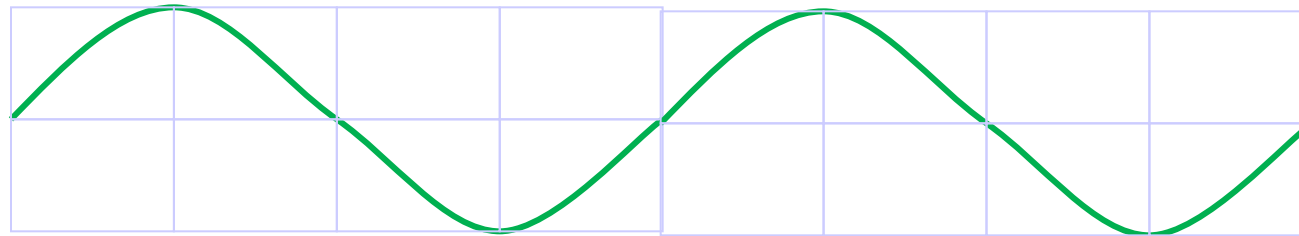
(but it gets more sophisticated than that!)



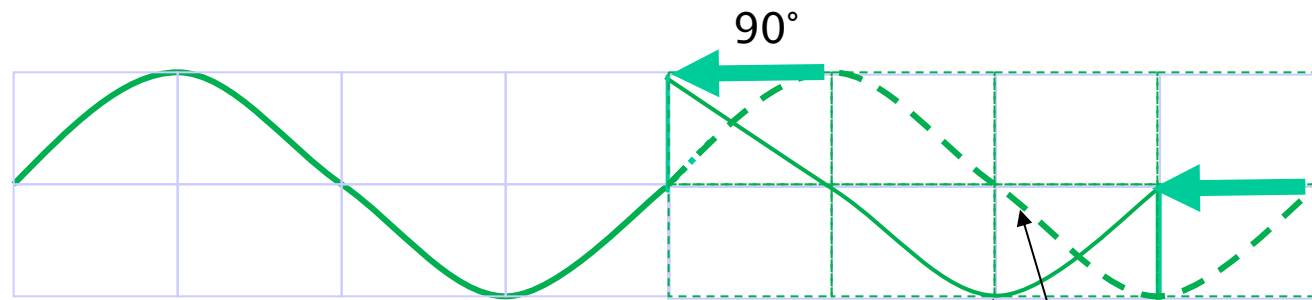
90° phase
change
(advanced)

270° phase
change
(advanced)

Phase Changes In A Sinusoidal Wave



0°..... 90°.....180°.....270°...360°

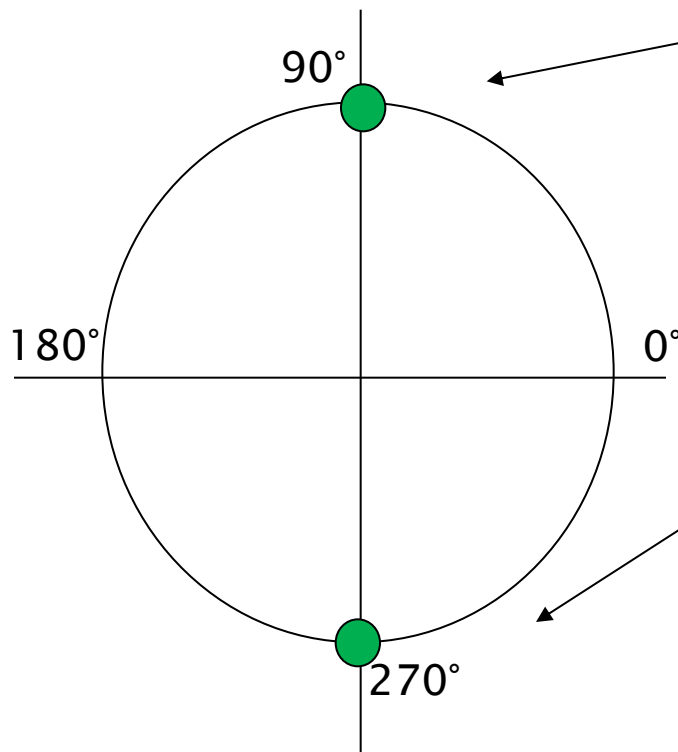
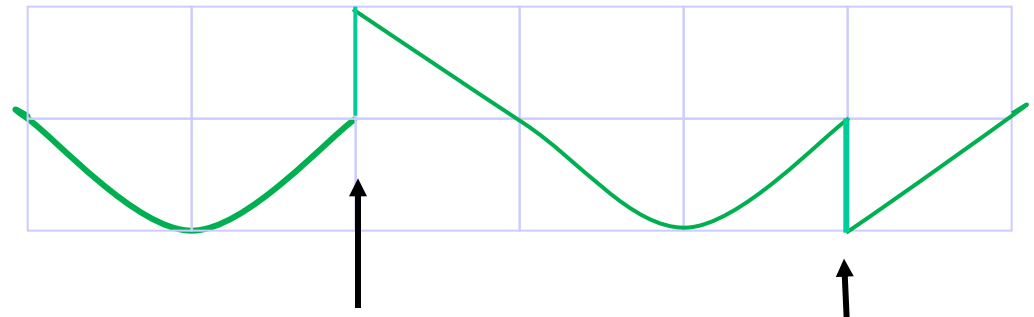


90° phase
change

- Signal was to be transmitted here ...
- But it has been moved forward (advanced) by 90°

Constellation Diagrams

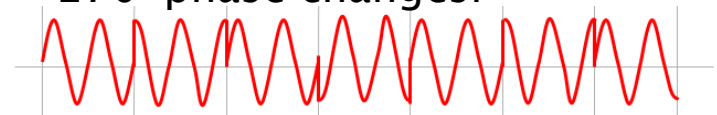
A graphical representation of digital codes corresponding to amplitude and phase-change in the carrier signal



90° phase
advance

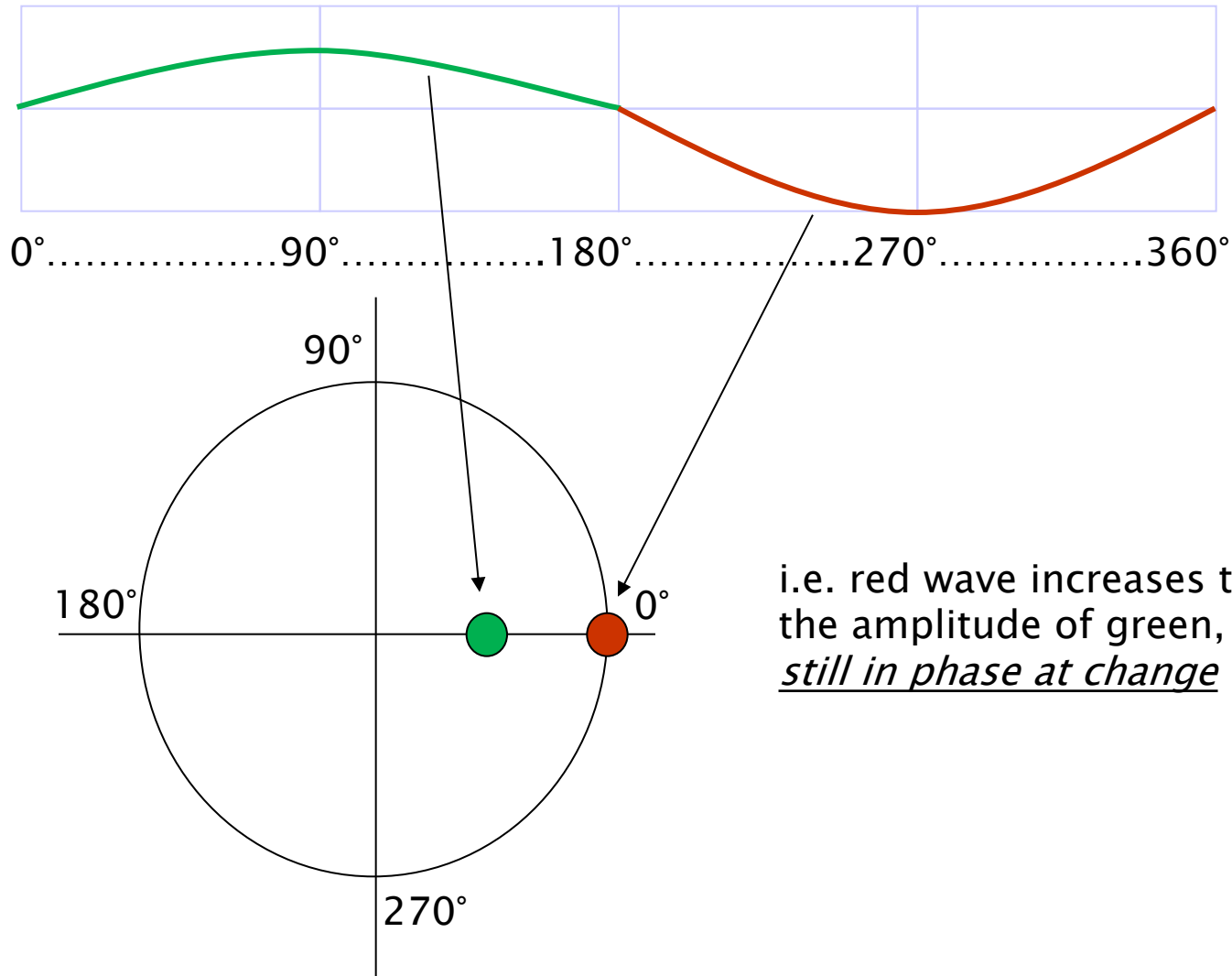
270° phase
advance

e.g. a signal with 90° and
270° phase changes:

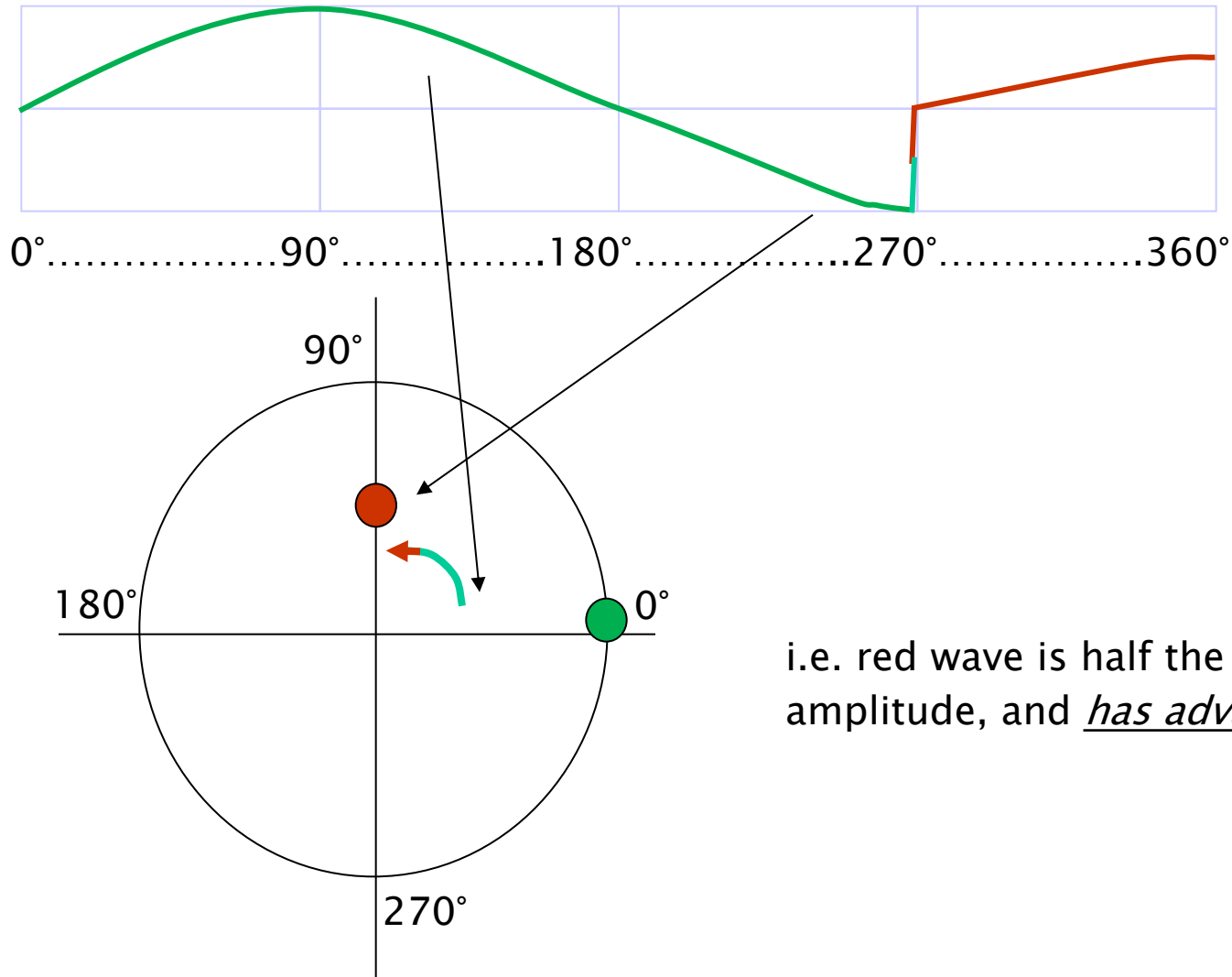


Amplitude Changes – Constellation Diagram

... an additional way to convey digital coded information

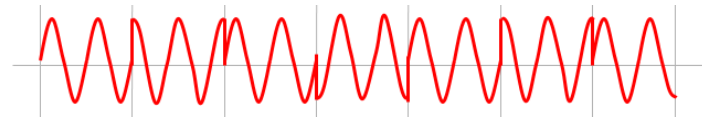
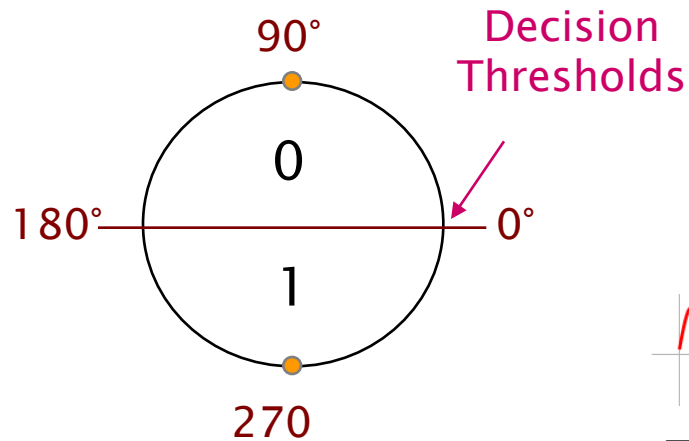


Phase and Amplitude Changes



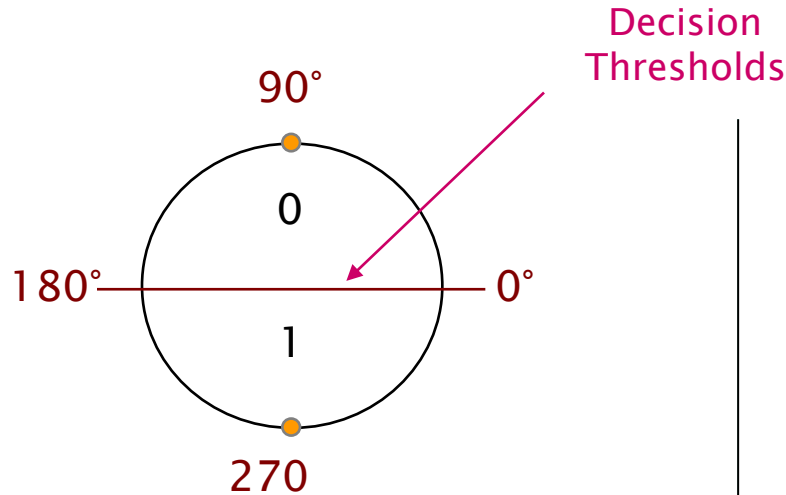
i.e. red wave is half the green amplitude, and has advanced by 90°

Constellation Diagrams

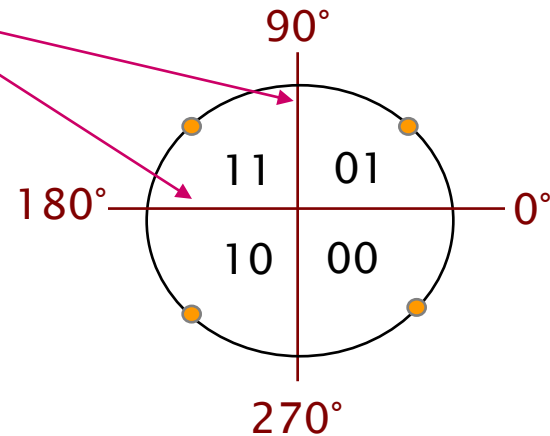
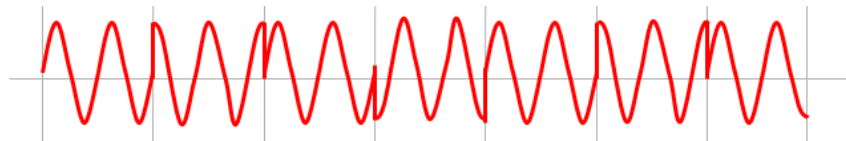


- Two phase changes – 90° & 270°
- allows intelligence – i.e. binary information – to be defined in these carrier changes
- two changes in state – allows two coded bits: 1 & 0

Constellation Diagrams



Two phase changes – 90° & 270°
i.e. two changes in state, allows
two coded bits: 1 & 0

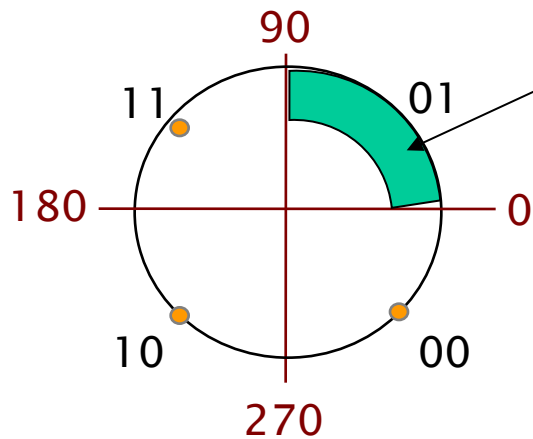


Four phase changes – 45°, 135°, 225°, 315°
i.e. four changes in state, allows two
coded bits *per symbol*.
00, 01, 10, 11

*Problem: The receiver has to detect
smaller phase changes*



Constellation Diagrams

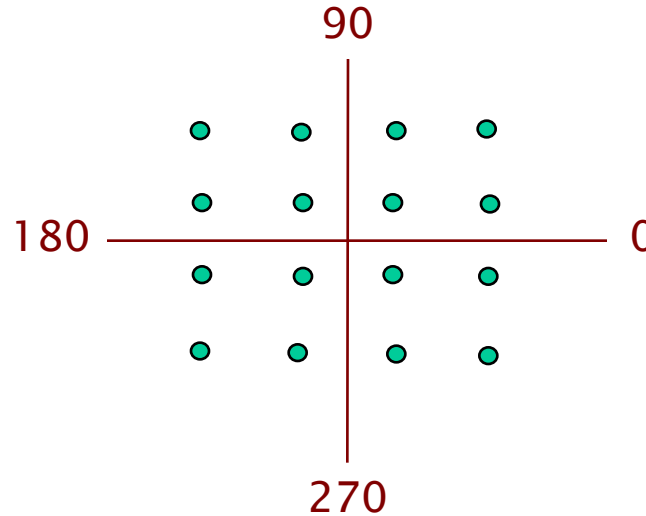


If the phase change falls within this range, it represents a signal "01" i.e. a fairly wide decision threshold

QAM – Quadrature Amplitude Modulation

16-QAM

- 16 decision points
- 4 coded bits per symbol



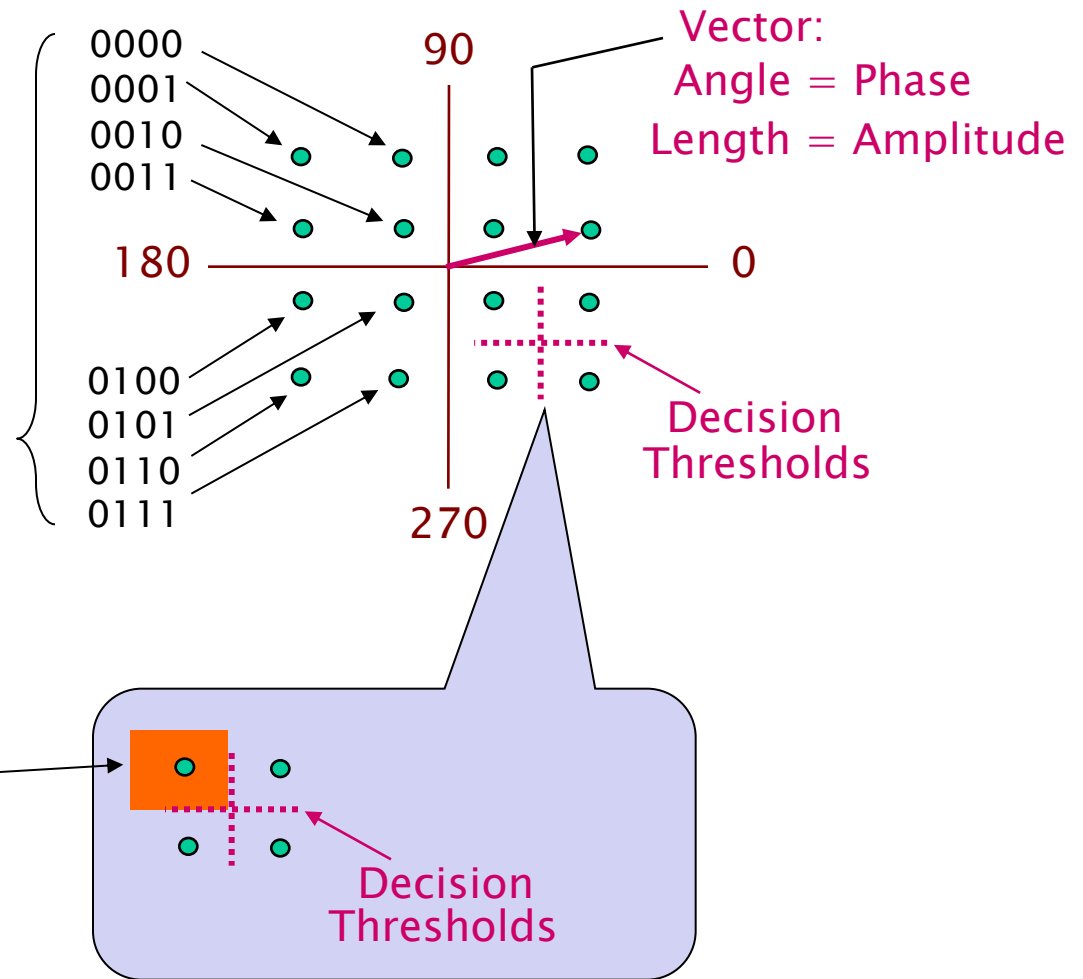
Used in Cable TV, IPTV, Satellite TV, WiFi
... most digital transmissions

16-QAM

16-QAM

- 16 decision points
- 4 coded bits per symbol

For example

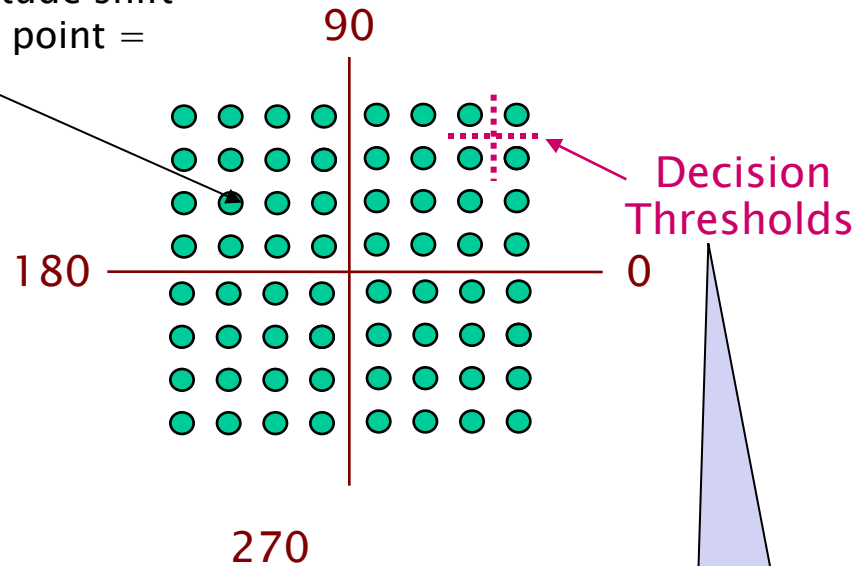


64-QAM

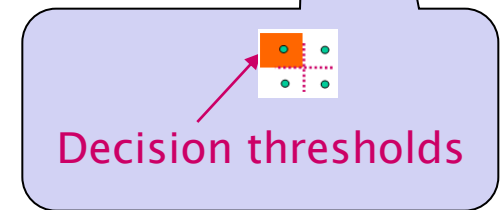
e.g. a phase/ amplitude shift
represented by this point =
(say...) "100101"

64-QAM

- 64 decision points
- 6 coded bits per symbol



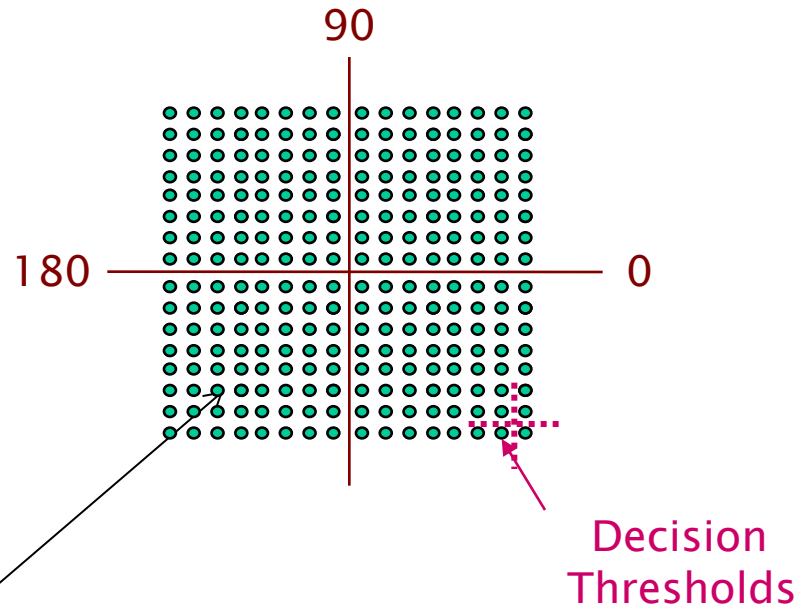
- More digital bits – more information transmitted
- More fragile signal (I.e. than 16QAM)–
- will suffer in a poor transmission
- results in more messages retransmitted



256-QAM

256-QAM

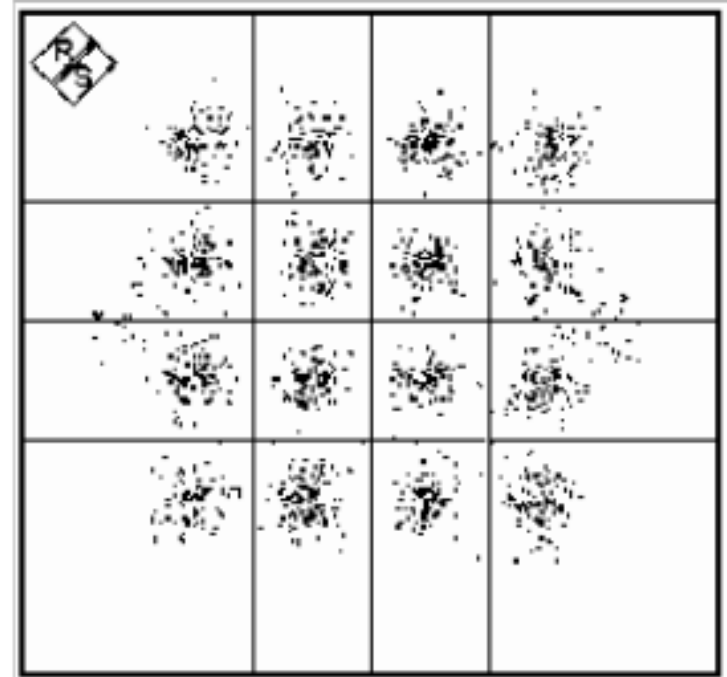
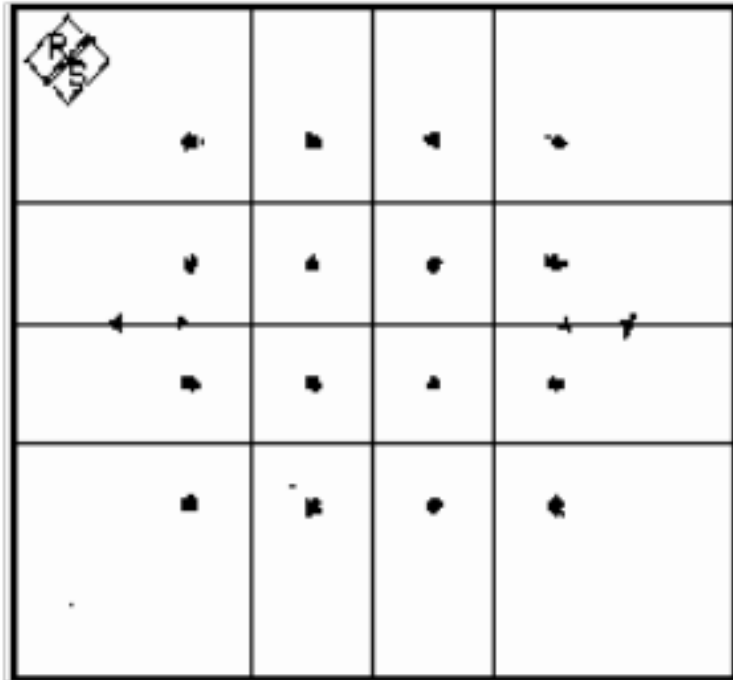
- 256 decision points
- 8 coded bits per symbol



e.g. a phase/ amplitude shift
represented by this point =
(say...) "10010111"

16-QAM Potential Errors

... due to phase and amplitude variations caused by transmission vagaries



Left – ideal reception of signal
Above – more practical reception

The Transmitted Signal

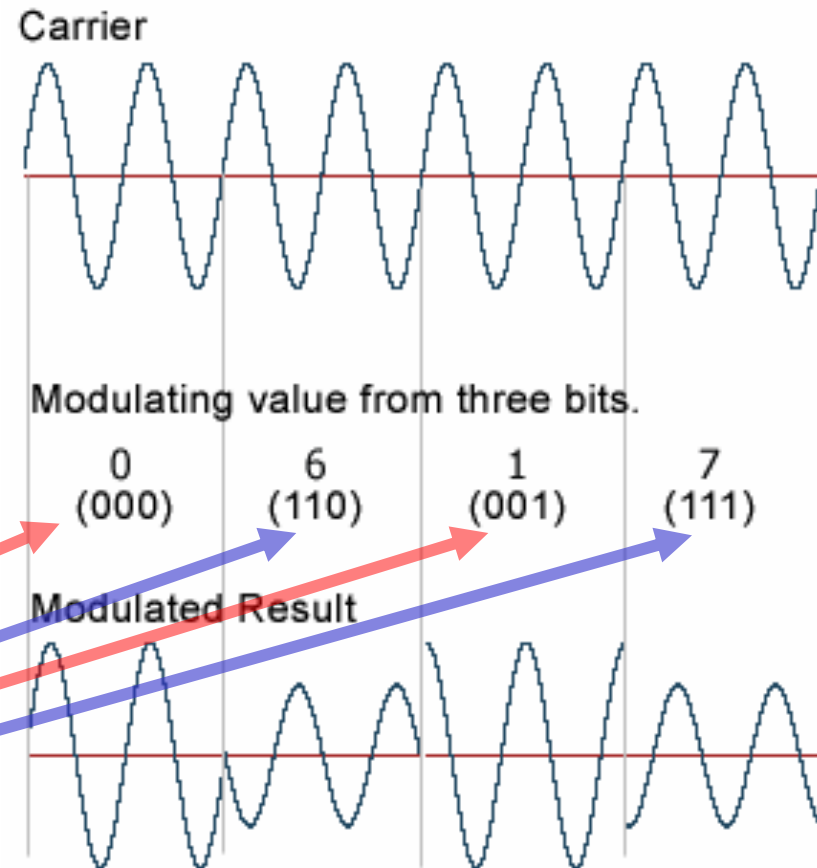
- Shown here – 8 QAM
- i.e. eight digital codes can be transmitted:
 - 000, 001, 010, 011,
100, 101, 110, 111

- So transmit the signal

000110001111:

...break it into the
groups of codes
available

000 110 001 111



Note: Only four (0, 6, 1 and 7) out of the eight possible modulation states (0-7) are shown in this illustration.



F
BBC 1

VHF



UHF



PULL-UP
VOLUME

VHF UHF

CONTRAST

BRIGHT

COLOUR

